

Online Supporting Information for: Weak-Instrument Inference and Residualized Concentration in Double Machine Learning for the Partially Linear Instrumental Variables Model

Airui Meng*

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This document is the Online Supporting Information for the article named above. The paper uses the partially linear IV setting and the CollegeDistance application of Çetin (2026a). Theorem 1 is established in Çetin (2026b); Theorems 2 and 3 and Propositions 4 and 5 are this paper’s. All numerical results are reproduced by the replication materials accompanying the paper.

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*Columbia University. Email: am6817@tc.columbia.edu.

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Appendix A. Formal results: sources and proofs

Notation is as in §2.1: the PLIV model is $Y = \tau_0 D + g_0(W) + \zeta$ with $\mathbb{E}[\zeta \mid Z, W] = 0$; the nuisances are $\mu_0 = \mathbb{E}[Y \mid W]$, $r_0 = \mathbb{E}[D \mid W]$, $\pi_0 = \mathbb{E}[Z \mid W]$; residuals at the truth are $\tilde{Z}_0 = Z - \pi_0$, $\tilde{D}_0 = D - r_0$, and at a general residualization $\tilde{Z} = Z - \pi$. The just-identified estimator solves the empirical analogue of $\mathbb{E}[(Z - \pi)\{(Y - \mu_0) - \tau(D - r_0)\}] = 0$; for $d_z > 1$ the Ω^{-1} -weighted (two-stage-least-squares-style) GMM root is $\hat{\tau} = (g'_{ZD}\Omega^{-1}g_{ZY})/(g'_{ZD}\Omega^{-1}g_{ZD})$ with $g_{ZD} = \mathbb{E}[\tilde{Z}\tilde{D}]$, $g_{ZY} = \mathbb{E}[\tilde{Z}\tilde{Y}]$, $\Omega = \mathbb{E}[\tilde{Z}\tilde{Z}']$. Each subsection records the statement of its result together with the source or the proof. A.1 covers Proposition 1, whose ingredients are available in the cited sources: its statement, sources, and a proof sketch. A.2 covers the joint weak-identification limit that Proposition 2 rests on (Lemma 1); the classical weak-instrument argument does not cover the cross-fitted statistic computed here, so that lemma and the exact limit of the feasible Wald statistic derived from it are given in full. Proposition 3 is, to our knowledge, new here, and A.3 gives its complete proof. Subsections A.4–A.6 hold the identification-robust distribution theory behind §3.2: A.4 the imported i.i.d. uniformity result (Theorem 1) and its clustered extension (Theorem 2); A.5 the cross-fitted score result (Theorem 3); A.6 the survey-weighted and generated-instrument extensions (Propositions 4 and 5). The boundary between the imported result and the results proved here is stated once, on the title page of this document.

A.1 Proposition 1: sources and proof sketch

Proposition 1 formalizes the estimand invariance implicit in the consistency theory for orthogonalized PLIV moments; the ingredients are those of Chernozhukov et al. (2018) and Çetin (2026a), so we record only the skeleton. We record it at all because the π -invariance of the slope is used repeatedly in the text. The argument is that the residualized instrument is a valid multiplier for *any* residualization. Because $\mathbb{E}[\zeta \mid Z, W] = 0$, for any W -measurable π_* we have $\mathbb{E}[(Z - \pi_*(W))\zeta] = \mathbb{E}[Z\mathbb{E}[\zeta \mid Z, W]] - \mathbb{E}[\pi_*(W)\mathbb{E}[\zeta \mid W]] = 0$, so the moment $\mathbb{E}[\psi(\tau_0, \eta)] = 0$ holds at the truth whatever (square-integrable) limit $\hat{\pi}$ takes. At $\tau = \tau_0$ the bracket of the moment collapses to the structural error: $\mu_0 - \tau_0 r_0 = \mathbb{E}[Y - \tau_0 D \mid W] = g_0$ by the tower property, so $m(\tau_0; \pi_*) = \mathbb{E}[(Z - \pi_*)\zeta]$. Instrument exogeneity kills both pieces of the population moment $m(\tau_0; \pi_*)$: $\mathbb{E}[Z\zeta] = \mathbb{E}[Z\mathbb{E}[\zeta \mid Z, W]] = 0$ (the **full** conditioning on Z is needed here, since Z is not W -measurable), and $\mathbb{E}[\pi_*(W)\zeta] = 0$ for any W -measurable π_* . The moment is affine in τ with slope $-g$, and $g = \mathbb{E}[(Z - \pi_*)\tilde{D}_0] = \mathbb{E}[\tilde{Z}_0\tilde{D}_0]$ is invariant to π_* because $\mathbb{E}[\tilde{D}_0 \mid W] = 0$; the root is therefore unique and equal to τ_0 for **every** admissible π_* . Cross-fitting transports the population root to the estimator under L^2 -consistent nuisances by the standard argument (Chernozhukov et al. 2018; Çetin 2026a), and for $d_z > 1$ the identity $g_{ZY} = \tau_0 g$ makes the Ω^{-1} -weighted GMM root equal to τ_0 for any positive-definite weight. Consistency uses only L^2 nuisance convergence; the $\sqrt{n}$ -asymptotic normality invoked in §2.1 and §3 additionally requires the product-rate condition of Çetin (2026a), which is not used for the present claim. The variance comparison $\text{Var}(\hat{\tau}) \propto 1/(nC)$ across residualizations is likewise a strong-identification statement established at a consistent $\hat{\pi}$; at a limit $\pi_* \neq \pi_0$ the score loses its orthogonality in the (μ, r) directions, and we do not develop distribution theory there. The proportionality is exact under conditional homoskedasticity of the score, where $\Sigma(\pi) \propto \Omega(\pi)$; under heteroskedasticity with conditional score variance bounded above

and below, C ranks weak-identification strength up to constants rather than tracking the sandwich variance itself.

A.2 Proposition 2: the weak-identification limit of the cross-fitted Wald layer

Part (i) is immediate: under $\tilde{Z} \mapsto c\tilde{Z}$ the score variance scales as $c^2\Sigma$ and the relevance as $c g_{ZD}$, so $V = \Sigma/g_{ZD}^2$ is unchanged. The correct reading concerns direction as well as scale. Holding Σ fixed, a *smaller* $|g_{ZD}|$ *raises* V ; under a pure rescaling both pieces move and V is unchanged. The absolute magnitude of the conventional standard error is therefore not a diagnostic of instrument strength, in either direction.

The rest of this subsection proves parts (ii) and (iii). Two features of the statistic computed in this paper lie outside the classical weak-instrument asymptotics of Staiger and Stock (1997) (Andrews, Stock, and Sun 2019 survey that theory): the residualized instrument is a cross-fitted, generated regressor, and the feasible variance evaluates residuals at the inconsistent $\hat{\tau}$. We therefore state and prove the joint limit we need (Lemma 1) and derive the exact limits of the null-normalized and feasible statistics from it by continuous mapping.

Notation, local to this subsection. All residuals are at the true nuisance functions, and we drop the 0 subscript that A.3 reserves for them: $\tilde{Z} = Z - \pi_0(W)$, $\tilde{D} = D - r_0(W)$, $\tilde{Y} = Y - \mu_0(W)$, where the nuisances are the P_n -conditional means of the triangular array introduced below. Cross-fitted objects carry hats: on outer fold k , $\hat{\tilde{Z}} = Z - \hat{\pi}_k(W)$, similarly $\hat{\tilde{D}}$ and $\hat{\tilde{Y}}$, with $\delta_{a,k} := \hat{a}_k - a_0$ for $a \in \{\mu, r, \pi\}$ and $\|\cdot\|$ the $L^2(P_n)$ norm treating the trained function as fixed. The structural score residual is $u = \tilde{Y} - \tau_0\tilde{D}$; since $\mu_0 - \tau_0 r_0 = g_0$ by the tower property, $u = \zeta$ and $\mathbb{E}[u | Z, W] = 0$; by construction $\mathbb{E}[\tilde{Z} | W] = 0$ and $\mathbb{E}[\tilde{D} | W] = 0$.

Statistics. With $\hat{u}_i(\tau) := \hat{\tilde{Y}}_i - \tau \hat{\tilde{D}}_i$, the just-identified estimator is $\hat{\tau} = \sum_i \hat{\tilde{Z}}_i \hat{\tilde{Y}}_i / \sum_i \hat{\tilde{Z}}_i \hat{\tilde{D}}_i$. Define the meat at τ , $\hat{\Sigma}(\tau) := n^{-1} \sum_i \hat{\tilde{Z}}_i^2 \hat{u}_i(\tau)^2$, the normalized sums

$$A_n := n^{-1/2} \sum_i \hat{\tilde{Z}}_i \hat{u}_i(\tau_0), \quad B_n := n^{-1/2} \sum_i \hat{\tilde{Z}}_i \hat{\tilde{D}}_i,$$

and the standard error at τ , $\hat{\text{se}}(\tau)^2 := \hat{\Sigma}(\tau)/B_n^2$. The *feasible* Wald statistic is $\hat{t}_n := (\hat{\tau} - \tau_0)/\hat{\text{se}}(\hat{\tau})$; the *null-normalized* statistic is $\tilde{t}_n(\tau_0) := (\hat{\tau} - \tau_0)/\hat{\text{se}}(\tau_0)$, feasible as a test of $H_0: \tau = \tau_0$ and infeasible as a companion of $\hat{\tau}$ only through its use of τ_0 .

Finite-sample identities. Because $\hat{\tau}$ solves $\sum_i \hat{\tilde{Z}}_i \hat{u}_i(\hat{\tau}) = 0$ and $\hat{u}(\tau_0) - \hat{u}(\hat{\tau}) = (\hat{\tau} - \tau_0)\hat{\tilde{D}}$ pointwise,

$$\hat{\tau} - \tau_0 = \frac{A_n}{B_n}, \quad \tilde{t}_n(\tau_0) = \frac{A_n \text{sign}(B_n)}{\sqrt{\hat{\Sigma}(\tau_0)}}, \quad \tilde{t}_n(\tau_0)^2 = \frac{A_n^2}{\hat{\Sigma}(\tau_0)} = \text{AR}_n(\tau_0), \quad (\text{A.1})$$

so the null-normalized Wald statistic squared *is* the cross-fitted AR statistic of §3, as an exact algebraic identity. The identity never touches the form of the meat: with a cluster-robust or survey-weighted $\hat{\Sigma}$ in both statistics, (A.1) holds verbatim. Squaring $\hat{u}(\hat{\tau}) = \hat{u}(\tau_0) - (\hat{\tau} - \tau_0)\hat{\tilde{D}}$ likewise gives the exact meat expansion

$$\hat{\Sigma}(\hat{\tau}) = \hat{\Sigma}(\tau_0) - 2(\hat{\tau} - \tau_0) S_{uD} + (\hat{\tau} - \tau_0)^2 S_{DD}, \quad (\text{A.2})$$

with $S_{uD} := n^{-1} \sum_i \hat{\tilde{Z}}_i^2 \hat{u}_i(\tau_0) \hat{\tilde{D}}_i$ and $S_{DD} := n^{-1} \sum_i \hat{\tilde{Z}}_i^2 \hat{\tilde{D}}_i^2$. The term the feasible residual adds to the sandwich is exactly this first-order perturbation, and it survives in the limit below.

Lemma 1 (joint weak-identification limit of the cross-fitted moment and relevance). *Assume: (L1) for each n the rows (Y_i, D_i, Z_i, W_i) are i.i.d. under P_n ; the PLIV model of §2.1 holds with fixed τ_0 ; the nuisances are the P_n -conditional means; and the relevance drifts locally to zero, $g_{ZD,n} := \mathbb{E}_{P_n}[\tilde{Z}\tilde{D}] = \gamma/\sqrt{n}$ for a fixed $\gamma \in \mathbb{R}$. (L2) for some $q > 2$ and $C < \infty$, uniformly in n , $\mathbb{E}[|\tilde{Z}u|^q + |\tilde{Z}\tilde{D}|^q] \leq C$; the second moments converge, $\Sigma_n := \mathbb{E}[\tilde{Z}^2 u^2] \rightarrow \Sigma > 0$, $\mathbb{E}[\tilde{Z}^2 \tilde{D}^2] \rightarrow \omega^2 > 0$, and $\kappa_n := \mathbb{E}[\tilde{Z}^2 u \tilde{D}] \rightarrow \kappa$, with $\rho := \kappa/(\sqrt{\Sigma}\omega) \in (-1, 1)$; and $\mathbb{E}[u^2 | Z, W] \leq C$, $\mathbb{E}[\tilde{D}^2 | W] \leq C$, $\mathbb{E}[\tilde{Z}^2 | W] \leq C$ almost surely. (L3) the sample is split into K fixed folds by a partition independent of the data, with n_k/n bounded away from zero and one; the fold- k nuisance estimates are measurable with respect to the out-of-fold observations (the protocol of §2.1); and the true nuisances and the fitted values take values in a fixed bounded set, enforced by truncating predictions if necessary, so that $\max_k \|\delta_{a,k}\|_\infty \leq C$ almost surely. (L4) $\max_k \|\delta_{a,k}\| = o_p(1)$ for each $a \in \{\mu, r, \pi\}$, and $\max_k \|\delta_{\pi,k}\| (\|\delta_{\mu,k}\| + \|\delta_{r,k}\|) = o_p(n^{-1/2})$. Then: (i) $A_n = n^{-1/2} \sum_i \tilde{Z}_i u_i + o_p(1)$ and $B_n = n^{-1/2} \sum_i \tilde{Z}_i \tilde{D}_i + o_p(1)$; (ii) $(A_n, B_n) \Rightarrow (\sqrt{\Sigma} \xi_1, \gamma + \omega \xi_2)$ with (ξ_1, ξ_2) bivariate standard normal of correlation ρ ; (iii) $\hat{\Sigma}(\tau_0) \rightarrow_p \Sigma$, $S_{uD} \rightarrow_p \kappa$, $S_{DD} \rightarrow_p \omega^2$, jointly with (ii).*

Two reading notes on the conditions. Along the drift, $\text{Var}_{P_n}(\tilde{Z}\tilde{D}) = \mathbb{E}[\tilde{Z}^2 \tilde{D}^2] - g_{ZD,n}^2$ shares the limit ω^2 , which is why a single ω serves both the CLT variance and the S_{DD} limit. And (L4)'s product display carries the $\pi \times r$ term without a $|\tau_0|$ factor because the relevance sum B_n has its own second-order term $\delta_\pi \delta_r$, which the moment expansion does not scale by τ_0 .

Proof. Throughout, conditioning on the fold- k training data $\mathcal{F}_k$ is licensed by (L3): the partition is independent of the data and the fold- k nuisances are out-of-fold measurable, so given $\mathcal{F}_k$ the fold- k rows remain i.i.d. P_n and each $\delta_{a,k}$ is a fixed function.

Step 1 (numerator). With $\delta u := \delta_\mu - \tau_0 \delta_r$, so that $\hat{u}(\tau_0) = u - \delta u$ pointwise on fold k , and $\hat{\tilde{Z}} = \tilde{Z} - \delta_\pi$,

$$\hat{\tilde{Z}} \hat{u}(\tau_0) - \tilde{Z} u = -\tilde{Z} \delta u - \delta_\pi u + \delta_\pi \delta u.$$

The first term is conditionally mean zero, $\mathbb{E}[\tilde{Z} \delta u(W) | \mathcal{F}_k] = \mathbb{E}[\mathbb{E}[\tilde{Z} | W] \delta u(W)] = 0$, with conditional variance $\leq \mathbb{E}[\mathbb{E}[\tilde{Z}^2 | W] \delta u^2] \leq C \|\delta u\|^2$; conditional Chebyshev gives, for every $\varepsilon > 0$, $P\left(\left|n^{-1/2} \sum_{i \in I_k} \tilde{Z}_i \delta u(W_i)\right| > \varepsilon\right) \leq \mathbb{E}[\min\{1, C \|\delta u\|^2 / \varepsilon^2\}] \rightarrow 0$ by (L4) and bounded convergence. The second term is handled identically using $\mathbb{E}[u | Z, W] = 0$ and $\mathbb{E}[u^2 | Z, W] \leq C$. For the third, Cauchy–Schwarz in the empirical norm gives $\left|n^{-1/2} \sum_{i \in I_k} \delta_\pi \delta u\right| \leq \sqrt{n} \|\delta_\pi\|_n \|\delta u\|_n$, and $\mathbb{E}[\|\delta\|_n^2 | \mathcal{F}_k] = \|\delta\|^2$ makes $\|\delta\|_n = O_p(\|\delta\|)$; since $\|\delta u\| \leq (1 \vee |\tau_0|) (\|\delta_\mu\| + \|\delta_r\|)$ and τ_0 is fixed, (L4) gives $\sqrt{n} \|\delta_\pi\| \|\delta u\| = o_p(1)$. Summing over the K folds proves the A_n half of (i).

Step 2 (relevance). $\hat{\tilde{Z}} \hat{\tilde{D}} - \tilde{Z} \tilde{D} = -\tilde{Z} \delta_r - \delta_\pi \tilde{D} + \delta_\pi \delta_r$. The first two terms are conditionally mean zero ($\mathbb{E}[\tilde{Z} | W] = 0$, $\mathbb{E}[\tilde{D} | W] = 0$) with conditional variances $\leq C \|\delta_r\|^2$ and $\leq C \|\delta_\pi\|^2$; the product term is $\leq \sqrt{n} \|\delta_\pi\|_n \|\delta_r\|_n = o_p(1)$ by (L4). The same fold-wise argument proves the B_n half of (i).

Step 3 (joint CLT). After Steps 1–2 the sums are over the full i.i.d. triangular array of oracle quantities, and the fold structure has dropped out. By Cramér–Wold, for $(a, b) \in \mathbb{R}^2$ the array $x_{n,i} := a \tilde{Z}_i u_i + b (\tilde{Z}_i \tilde{D}_i - g_{ZD,n})$ is i.i.d. mean zero ($\mathbb{E}[\tilde{Z}u] = \mathbb{E}[\tilde{Z} \mathbb{E}[u | Z, W]] = 0$) with variance $a^2 \Sigma_n + 2ab \kappa_n + b^2 (\mathbb{E}[\tilde{Z}^2 \tilde{D}^2] - g_{ZD,n}^2) \rightarrow a^2 \Sigma + 2ab \kappa + b^2 \omega^2$, the covariance being $\text{Cov}(\tilde{Z}u, \tilde{Z}\tilde{D}) = \kappa_n$ exactly; the limit matrix is positive definite by $\Sigma, \omega^2 > 0$ and $|\rho| < 1$. Lyapunov’s condition holds by (L2) with $q > 2$. The mean of the relevance sum is $n^{-1/2} \cdot n \cdot g_{ZD,n} = \gamma$. Together with (i) this gives (ii).

Step 4 (meat). The squares and cross products $(\tilde{Z}u)^2$, $(\tilde{Z}\tilde{D})^2$, $\tilde{Z}^2 u \tilde{D}$ are uniformly integrable: $q > 2$ bounds their $(1+\epsilon)$ -th moments, using $|\tilde{Z}^2 u \tilde{D}|^{1+\epsilon} \leq (|\tilde{Z}u|^{2+2\epsilon} + |\tilde{Z}\tilde{D}|^{2+2\epsilon})/2$ for the cross term. The weak law of large numbers for uniformly integrable triangular arrays then gives $n^{-1} \sum \tilde{Z}_i^2 u_i^2 \rightarrow_p \Sigma$, $n^{-1} \sum \tilde{Z}_i^2 u_i \tilde{D}_i \rightarrow_p \kappa$, and $n^{-1} \sum \tilde{Z}_i^2 \tilde{D}_i^2 \rightarrow_p \omega^2$. For the feasible-minus-oracle differences, expand as in Steps 1–2 and use (L3) boundedness with the conditional moment bounds: quadratic error terms such as $n^{-1} \sum \tilde{Z}^2 \delta u^2$ have conditional expectation $\leq C \|\delta u\|^2 = o_p(1)$; one-delta cross terms close by factor pairing, for example $|\tilde{Z}^2 u \delta_r| = |\tilde{Z}u| \cdot |\tilde{Z}\delta_r|$ with $\mathbb{E}[|\tilde{Z}u| |\tilde{Z}\delta_r|] \leq \Sigma_n^{1/2} (\mathbb{E}[\mathbb{E}[\tilde{Z}^2 | W] \delta_r^2])^{1/2} \leq C \|\delta_r\| = o_p(1)$, and likewise $|\tilde{Z}u \tilde{D} \delta_\pi| = |\tilde{Z}u| \cdot |\tilde{D}\delta_\pi|$ and $|\tilde{Z}^2 \tilde{D} \delta_r| = |\tilde{Z}\tilde{D}| \cdot |\tilde{Z}\delta_r|$; higher-order terms are absorbed by the (L3) sup bound. Hence $\hat{\Sigma}(\tau_0)$, S_{uD} , and S_{DD} share the oracle limits. Joint convergence with (ii) is Slutsky, the limits in (iii) being constants. ■

Limits of the three statistics. The map $(A, B) \mapsto A/B$ is continuous off $\{\gamma + \omega \xi_2 = 0\}$, a null set since $\omega > 0$, so continuous mapping applied to Lemma 1 yields

$$\hat{\tau} - \tau_0 \Rightarrow \Delta := \frac{\sqrt{\Sigma} \xi_1}{\gamma + \omega \xi_2} :$$

the estimator is inconsistent and $O_p(1)$, with no $\sqrt{n}$ rate. The standard errors satisfy

$$\widehat{\text{se}}(\tau_0) \Rightarrow \frac{\sqrt{\Sigma}}{|\gamma + \omega \xi_2|}, \quad \widehat{\text{se}}(\hat{\tau}) \Rightarrow \frac{\sqrt{\Sigma(1 - 2\rho h + h^2)}}{|\gamma + \omega \xi_2|}, \quad h := \frac{\omega \xi_1}{\gamma + \omega \xi_2},$$

using (A.2) with $\Delta\kappa = \rho \Sigma h$ and $\Delta^2 \omega^2 = \Sigma h^2$; both $\widehat{\text{se}}$ and $\widehat{\text{se}}^{-1}$ are $O_p(1)$, so the standard error neither vanishes nor explodes. The factor is strictly positive: $1 - 2\rho h + h^2 = (h - \rho)^2 + 1 - \rho^2 \geq 1 - \rho^2 > 0$. The null-normalized statistic obeys

$$\tilde{t}_n(\tau_0) \Rightarrow \xi_1 \cdot \text{sign}(\gamma + \omega \xi_2), \quad |\tilde{t}_n(\tau_0)| \Rightarrow |\xi_1| \sim |N(0, 1)|, \quad \text{AR}_n(\tau_0) \Rightarrow \chi_1^2.$$

One-sided use of $\tilde{t}_n$ is distorted when $\rho \neq 0$ (the sign factor is correlated with ξ_1), but the absolute statistic is exactly pivotal along the sequence, and by (A.1) its square is the AR statistic: inverting it over hypothesized values of τ is precisely the §3 confidence set. The feasible Wald statistic obeys

$$\hat{t}_n \Rightarrow L := \frac{\xi_1 \cdot \text{sign}(\gamma + \omega \xi_2)}{\sqrt{1 - 2\rho h + h^2}}. \quad (\text{A.3})$$

The law of L depends on γ/ω and ρ : the feasible statistic is non-pivotal for *every* ρ , including $\rho = 0$. As $\gamma/\omega \rightarrow \pm\infty$ the family of limit laws converges to $N(0, 1)$ (for $\gamma \rightarrow -\infty$ via $-\xi_1$), recovering the strong-identification Wald theory.

Whether a given configuration $(\gamma/\omega, \rho)$ over- or under-rejects is a property of the tails of L , not of its moments; at $\gamma = 0$ the moments of h do not exist. Three facts pin down the behavior. First, at $\rho = 0$ the factor $(1 + h^2)^{-1/2} \leq 1$ shrinks $|L|$ below $|\xi_1|$ pointwise, so the two-sided Wald test is asymptotically *conservative* along the sequence. Second, at $\gamma = 0$ the limit law has the closed form

$$L = \frac{\xi_1 \xi_2}{\sqrt{(1 - \rho^2)(\varepsilon^2 + \xi_2^2)}}, \quad \xi_1 = \rho \xi_2 + \sqrt{1 - \rho^2} \varepsilon, \quad \varepsilon \perp \xi_2,$$

obtained by substituting $h = \xi_1/\xi_2$ and completing the square in the denominator; its quantiles diverge as $|\rho| \rightarrow 1$, so for any fixed critical value there are configurations whose rejection rate is arbitrarily close to one. No fixed critical value therefore controls size uniformly over strength and score-correlation configurations, while a distribution-specific critical value can attain nominal size for a known configuration. Third, standard-normal critical values are exact only in the strong-identification limit and are not generically valid along the sequence: conservative at $\rho = 0$, under-covering at low strength and large $|\rho|$. The nominal $1 - \alpha$ Wald interval accordingly does not attain $1 - \alpha$ coverage in general. The failure is the *shape* of (A.3), not the magnitude of the standard error.

Remark (scope). Lemma 1 is stated for i.i.d. rows, $d_z = 1$, and a fixed local-to-zero drift, and Proposition 2 uses nothing more; the lemma is an i.i.d. statement and remains one. The identity (A.1) is free of the meat's form, so the §3 reporting layer carries to the clustered and survey-weighted cells at the identity level; the χ_1^2 limit itself is proved here only for the i.i.d. case. The clustered analogue of the χ^2 reference at the tested value is established by Theorem 2 (A.4), and the survey-weighted analogue by Proposition 4 (A.6). The stacked score underlying the LM statistic is treated by Theorem 3 (A.5).

Proposition 2(ii)–(iii) concern the local-to-zero *sequence*: at any fixed first-stage strength the feasible t -ratio is asymptotically $N(0, 1)$, and the §4.2 Monte-Carlo coverage evaluation measures this sequence's shadow at small effective- F .

A.3 Proof of Proposition 3

This is the paper's central theoretical claim, so we give it in full. **Setup.** Assume $\mathbb{E}\|Z\|^2 + \mathbb{E}[D^2] < \infty$, so that every second moment below is finite; $\Omega(\pi_0) \succ 0$, used at the inverse-monotonicity step; and $g \neq 0$, used in the equality discussion (with $g = 0$ the concentration is zero for every π and the equality classification is vacuous). The treatment nuisance is correctly specified, $r = r_0$, so $\tilde{D} = D - r_0$ is conditional-mean-zero: $\mathbb{E}[\tilde{D} \mid W] = 0$, the property the rest of the argument rests on. For a square-integrable, W -measurable residualization π (with $\pi_0 = \mathbb{E}[Z \mid W]$ the truth) write $\tilde{Z} = Z - \pi$, $\tilde{Z}_0 = Z - \pi_0$, and define the residualized concentration

$$C(\pi) = g_{ZD}(\pi)' \Omega(\pi)^{-1} g_{ZD}(\pi), \quad g_{ZD}(\pi) = \mathbb{E}[\tilde{Z}\tilde{D}] \in \mathbb{R}^{d_z}, \quad \Omega(\pi) = \mathbb{E}[\tilde{Z}\tilde{Z}'] \in \mathbb{R}^{d_z \times d_z}.$$

Strategy. $C(\pi)$ is a ratio: a numerator built from the relevance vector $g_{ZD}(\pi)$ and a denominator built from the instrument's second-moment matrix $\Omega(\pi)$. We show the numerator is *constant* in π (Lemma A) while the denominator is *Loewner-minimized* exactly at π_0 (Lemma B); combining the two through monotonicity of the matrix inverse forces $C(\pi) \leq C(\pi_0)$.

For how this result relates to, and differs from, the classical optimal-instruments theory (Amemiya 1974; Chamberlain 1987; Newey 1990), see the delineation in §5.

Under $r = r_0$ the relevance $g_{ZD}(\pi) = \mathbb{E}[\tilde{Z}_0 \tilde{D}_0]$ is independent of π : the W -predictable part of Z carries *zero* residualized covariance with $\tilde{D}$, because $\tilde{D}$'s W -structure is already removed by r_0 . Meanwhile $\Omega(\pi) = \Omega(\pi_0) + \mathbb{E}[(\pi_0 - \pi)(\pi_0 - \pi)']$ is minimized at π_0 . Recovering π_0 shrinks the residualized instrument toward its signal, raising C .

Lemma A (the relevance vector does not depend on π). For every admissible π ,

$$g_{ZD}(\pi) = \mathbb{E}[(Z - \pi)\tilde{D}] = \mathbb{E}[Z\tilde{D}] - \mathbb{E}[\pi(W)\tilde{D}] = \mathbb{E}[Z\tilde{D}] - \mathbb{E}[\pi(W) \underbrace{\mathbb{E}[\tilde{D} | W]}_{=0}] = \mathbb{E}[Z\tilde{D}] =: g,$$

where the third equality is the tower property applied to the W -measurable factor $\pi(W)$, and the fourth uses $\mathbb{E}[\tilde{D} | W] = 0$. Thus $g_{ZD}(\pi) \equiv g$ for all π ; in particular, taking $\pi = \pi_0$, $g = \mathbb{E}[\tilde{Z}_0 \tilde{D}]$. *Intuition.* Correct treatment residualization strips every W -predictable feature out of $\tilde{D}$, so whatever W -function $\pi(W)$ one subtracts from Z is uncorrelated with $\tilde{D}$: the identifying covariance lives entirely in the part of Z that *no* residualization can touch. Changing π moves only the noise in $\tilde{Z}$, never its signal.

Lemma B (the residualized second moment is Loewner-minimized at π_0 ; a standard best-prediction fact, recorded for completeness). Decompose the residualized instrument into the true residual plus a deterministic, W -measurable shift,

$$\tilde{Z} = Z - \pi = (Z - \pi_0) + (\pi_0 - \pi) = \tilde{Z}_0 + \Delta, \quad \Delta := \pi_0 - \pi \text{ (} W\text{-measurable)}.$$

Because $\pi_0 = \mathbb{E}[Z | W]$ is the L^2 -projection of Z onto W , its residual is orthogonal to every W -measurable function; concretely $\mathbb{E}[\tilde{Z}_0 | W] = \mathbb{E}[Z - \pi_0 | W] = 0$, so the cross term vanishes:

$$\mathbb{E}[\tilde{Z}_0 \Delta'] = \mathbb{E}[\mathbb{E}[\tilde{Z}_0 | W] \Delta'] = \mathbb{E}[0 \cdot \Delta'] = 0.$$

Expanding $\Omega(\pi) = \mathbb{E}[(\tilde{Z}_0 + \Delta)(\tilde{Z}_0 + \Delta)']$ and dropping the two (transpose-equal) cross terms gives the exact Pythagorean decomposition

$$\Omega(\pi) = \mathbb{E}[\tilde{Z}_0 \tilde{Z}_0'] + \mathbb{E}[\Delta \Delta'] = \Omega(\pi_0) + \mathbb{E}[(\pi_0 - \pi)(\pi_0 - \pi)'] \succeq \Omega(\pi_0), \quad (\text{A.4})$$

since $\mathbb{E}[\Delta \Delta']$ is a second-moment matrix, hence positive semidefinite, with $\Omega(\pi) = \Omega(\pi_0)$ iff $\mathbb{E}[\Delta \Delta'] = 0$, i.e., $\Delta = 0$ a.s., i.e., $\pi = \pi_0$ a.s. *Intuition.* Residualizing with the true π_0 leaves the minimum-variance residual $\tilde{Z}_0$; replacing π_0 by any other π injects a deterministic W -measurable error Δ that, being orthogonal to $\tilde{Z}_0$, can only *add* spread to the residualized instrument, never remove it.

Step (monotonicity of the matrix inverse). If $A \succeq B \succ 0$ then $A^{-1} \preceq B^{-1}$. Indeed $A \succeq B \succ 0$ is equivalent to $B^{-1/2} A B^{-1/2} \succeq I$, so all eigenvalues of $B^{-1/2} A B^{-1/2}$ are ≥ 1 ; the eigenvalues of its inverse $B^{1/2} A^{-1} B^{1/2}$ are then ≤ 1 , i.e., $B^{1/2} A^{-1} B^{1/2} \preceq I$, which is $A^{-1} \preceq B^{-1}$.

Conclusion. Take first the just-identified case $d_z = 1$, where everything is scalar: g is a number, $\Omega(\pi) = \mathbb{E}[\tilde{Z}^2] \geq \mathbb{E}[\tilde{Z}_0^2] = \Omega(\pi_0)$ by (A.4), and

$$C(\pi) = \frac{g^2}{\Omega(\pi)} \leq \frac{g^2}{\Omega(\pi_0)} = C(\pi_0),$$

because a fixed positive numerator over a larger denominator is smaller; this scalar case is the clearest form of the result. For general d_z , the relevance g is fixed (Lemma A) and $\Omega(\pi) \succeq \Omega(\pi_0) \succ 0$ (Lemma B), so by inverse monotonicity $\Omega(\pi)^{-1} \preceq \Omega(\pi_0)^{-1}$, and therefore

$$C(\pi) = g' \Omega(\pi)^{-1} g \leq g' \Omega(\pi_0)^{-1} g = C(\pi_0), \quad (\text{A.5})$$

the inequality being the statement that $g'(\Omega(\pi_0)^{-1} - \Omega(\pi)^{-1})g \geq 0$. The prediction-optimal residualization π_0 thus **maximizes** the residualized concentration over all square-integrable W -measurable π and every d_z , using nothing but $\mathbb{E}[\tilde{D} | W] = 0$ and orthogonal projection. A consistent flexible learner that recovers a nonlinear π_0 therefore *attains* the maximum concentration: it cannot, in population, lower it.

Equality. When is the bound (A.5) tight? Write $A = \Omega(\pi_0) \succ 0$ and $M = \mathbb{E}[\Delta \Delta'] = \Omega(\pi) - A \succeq 0$, and factor $M = LL'$ (e.g., $L = M^{1/2}$). The Woodbury identity gives the difference of inverses in manifestly positive-semidefinite form,

$$A^{-1} - (A + M)^{-1} = A^{-1} L (I + L' A^{-1} L)^{-1} L' A^{-1},$$

so the concentration gap is a squared length,

$$C(\pi_0) - C(\pi) = g' (A^{-1} - (A + M)^{-1}) g = \|(I + L' A^{-1} L)^{-1/2} L' A^{-1} g\|^2 \geq 0,$$

vanishing iff $L' A^{-1} g = 0$, equivalently $M A^{-1} g = 0$, i.e.

$$M \Omega(\pi_0)^{-1} g = 0. \quad (\text{A.6})$$

For $d_z = 1$, $M = \mathbb{E}[\Delta^2] \geq 0$ is a scalar and $\Omega(\pi_0)^{-1} g \neq 0$ (relevance), so (A.6) holds iff $M = 0$, i.e., **iff** $\pi = \pi_0$ **a.s.**: any nonzero perturbation strictly lowers concentration. For $d_z > 1$ the condition is genuinely weaker: $\Omega(\pi_0)^{-1} g$ is the population Ω^{-1} -weighted slope direction, and (A.6) says the added-variance matrix M *annihilates* that direction. A misspecified $\pi \neq \pi_0$ can then be **concentration-neutral**, $C(\pi) = C(\pi_0)$, precisely when the extra residual variance it injects lies entirely in directions orthogonal to the slope (variation that carries no identifying information). In every case, recovering π_0 attains the maximum. ■

Remark (apparent “absorption” requires a misspecified r). The entire argument used $\mathbb{E}[\tilde{D} | W] = 0$, i.e., $r = r_0$. If instead $\hat{r} \not\rightarrow r_0$ (for instance a linear $\hat{r}$ when $r_0 = \mathbb{E}[D | W]$ contains a nonlinear component $f_s \pi_0$, as in the §4.8 design), then $\mathbb{E}[\tilde{D} | W] \neq 0$, **Lemma A fails**, and $g_{ZD}(\pi)$ becomes π -dependent: a flexible π that strips the W -predictable part of Z now also removes variation correlated with the unmodeled first-stage structure left in $\tilde{D}$, **lowering** C . Yet $\hat{r}$ remains consistent when the instrument residualization is itself consistent: with $\hat{\pi} \rightarrow \pi_0$ the residualized instrument is conditional-mean-zero, so $\mathbb{E}[(Z - \pi_0)(r^* - r_0)] = 0$ for the limit r^* of $\hat{r}$ (and likewise for any outcome-nuisance limit), and the moment still has its root at τ_0 (Proposition 1’s blanket π -invariance, by contrast, requires $(\mu, r) \rightarrow (\mu_0, r_0)$). This nuisance-misspecification interaction, *not* a property of flexible π , is exactly the regime isolated by the §4.8 Monte Carlo. With estimated but correctly specified nuisances, the realized effective- F can fall short of the strength that $C(\pi_0)$ implies only through finite-sample estimation noise (the overfit-dilution channel, signed by $R_{\text{of}}^2(\hat{\pi}; Z) \leq 0$ together with $\text{Var}(\tilde{Z})/\text{Var}(Z) > 1$), which vanishes as $n \rightarrow \infty$. There is no population “genuine absorption” channel under correct nuisances; Proposition 3 closes it off.

A.4 Uniform validity of the cross-fitted Anderson–Rubin set

This subsection and the two that follow give the distribution theory behind the §3.2 inference layer. The i.i.d. baseline (Theorem 1) is imported from Çetin (2026b) and is stated with a remark mapping the implemented protocol to its assumptions; the clustered result (Theorem 2) is this paper’s and is proved in full. Throughout, τ_0 denotes the tested value, the PLIV model of §2.1 holds at τ_0 under every law considered, and $\chi_{d,1-\alpha}^2$ is the $1 - \alpha$ quantile of the χ_d^2 distribution.

Orthogonality at every τ (§3.2). The PLIV moment $\mathbb{E}[\tilde{Z}(\tilde{Y} - \tau\tilde{D})]$ is first-order Neyman-orthogonal in all three nuisance directions at every τ because the multiplier $\tilde{Z}$ and, at the true (μ_0, τ_0) , the bracket $\tilde{Y} - \tau\tilde{D}$ are both conditional-mean-zero given W .

The i.i.d. statistic. With out-of-fold scores $\hat{\psi}_i := \psi(\tau_0, \hat{\eta}_{k(i)})$, where $k(i)$ is observation i ’s inference fold and $\hat{\eta}_k$ is selected and estimated by the three-way protocol of §2.1, the d_z -dimensional cross-fitted AR statistic is the self-normalized quadratic form

$$\text{AR}_n(\tau_0) := \left(\sum_i \hat{\psi}_i \right)' \left(\sum_i \hat{\psi}_i \hat{\psi}_i' \right)^{-1} \left(\sum_i \hat{\psi}_i \right),$$

which at $d_z = 1$ coincides with the AR_n of the identity (A.1).

Theorem 1 (uniform size of the cross-fitted AR statistic under adaptive selection; imported). (Çetin 2026b, Theorem 3.) *Let the rows (Y_i, D_i, Z_i, W_i) be i.i.d. draws from P in a triangular-array class $\mathcal{P}_n$, let the nuisances be selected and estimated by the leakage-free nested protocol of §2.1, and let Assumptions 1–7 of Çetin (2026b), labeled A1–A7 here, hold. Then*

$$\sup_{P \in \mathcal{P}_n} \left| P\left(\text{AR}_n(\tau_0) > \chi_{d_z,1-\alpha}^2\right) - \alpha \right| \rightarrow 0,$$

and the inverted set $\{t : \text{AR}_n(t) \leq \chi_{d_z,1-\alpha}^2\}$ has uniform asymptotic coverage $1 - \alpha$.

The result is established in Çetin (2026b), not here; we restate it, in this paper’s notation and without proof, because the §3.2 reporting layer and the weighted extension (Proposition 4) rely on it.

Implementation of the inverted set (§3.2). At $d_z = 1$ the reported AR set is inverted exactly (analytic quadratic inversion); at $d_z > 1$ it is inverted by an adaptive expanding-window inversion with an analytic tail check and explicit truncation labels.

Remark (mapping the implemented protocol to A1–A7). The labels A1–A7 below name Assumptions 1–7 of Çetin (2026b) in order. The implemented pipeline satisfies the hypotheses of the import as follows. A1 (i.i.d. triangular-array sampling) holds for the i.i.d. cells (Card, and the i.i.d. Monte-Carlo arms of §4.2); the clustered cells are covered by Theorem 2, not by the import. A2 (a $q > 4$ moment bound on the score and eigenvalue bounds on its covariance) is a primitive condition on each application’s data distribution; the rank and conditioning diagnostics of §3.2 flag near-failure of the eigenvalue floor but do not replace the assumption. A3 is the Neyman orthogonality of the PLIV score (§2.1). A4 (leakage-free adaptivity) is the three-way protocol itself: every selected pipeline and nuisance fit is measurable with respect to data disjoint from the fold it scores. A5 (selected-nuisance rates and the product rate) is carried by the curated library discipline together with Çetin (2026b, Theorem 2 and Corollary 2), by which the additive predictive screen preserves the rate and product-rate

conditions across the library; fitted values take values in a fixed bounded set, enforced by truncation where necessary (condition (L3) of Lemma 1 is the same device). A6 holds trivially: the library is finite with $|\Gamma_n| = 6$, so $\log |\Gamma_n| = o(n_{\text{sel}})$. A7 fixes d_z and contains both strong-identification laws and Staiger–Stock local-to-zero sequences; the import is used here at the just-identified i.i.d. cells, and the $d_z = 30$ cell is read with the rank-guard caveat of §3.2 and the finite- G remark below (the imported framework also offers a dimension-agnostic ridge AR under stronger conditions, Çetin 2026b, Theorem 4).

Cluster-level objects. Data arrive as cluster bundles $V_c = (n_c, \{(Y_{c,i}, D_{c,i}, Z_{c,i}, W_{c,i})\}_{i \leq n_c})$, $c = 1, \dots, G$, i.i.d. draws from P in a triangular-array class $\mathcal{P}_G$ indexed by the number of clusters G , with $n = \sum_c n_c$. The cluster score is $S_c(\tau, \eta) := \sum_{i \in c} \psi_{c,i}(\tau, \eta)$, the sum of the observation scores of cluster c . The K outer folds partition *clusters*; $k(c)$ is cluster c 's fold, and $\hat{S}_c := S_c(\tau_0, \hat{\eta}_{k(c)})$. Write $\mathbf{n}_G := (n_1, \dots, n_G)$ for the size vector and $\mathcal{U}$ for the fold randomization variable (degenerate when the partition rule is deterministic); the fold-assignment vector is $\mathbf{k}_G := (k(1), \dots, k(G))$. The fold- k training field is $\mathcal{F}_k := \sigma(\mathbf{n}_G, \mathcal{U}) \vee \sigma(V_c \mathbf{1}\{k(c) \neq k\} : c \leq G)$; it contains the size vector, the randomization, the fold assignment, and every training bundle. The clustered statistic is the normalization-free quadratic form

$$\text{AR}_G(\tau_0) := \left(\sum_c \hat{S}_c \right)' \left(\sum_c \hat{S}_c \hat{S}_c' \right)^{-1} \left(\sum_c \hat{S}_c \right) = \bar{m}_G' \bar{\Sigma}_G^{-1} \bar{m}_G, \quad (\text{A.7})$$

with $\bar{m}_G := G^{-1/2} \sum_c \hat{S}_c$ and $\bar{\Sigma}_G := G^{-1} \sum_c \hat{S}_c \hat{S}_c'$; being self-normalized, the statistic is invariant to the n -versus- G normalization, so everything is stated at cluster level with $G \rightarrow \infty$. Fold- k nuisance errors are $\delta_{a,k} := \hat{a}_k - a_0$ for $a \in \{\mu, r, \pi\}$, and $\zeta = \tilde{Y} - \tau_0 \tilde{D}$ is the structural error, as in A.2. Norms $\|\cdot\|$ on nuisance errors are $L^2(\bar{P})$ norms treating the trained function as fixed, where $\bar{P}$ is the size-biased observation marginal $\bar{P}(A) := \mathbb{E}_P[\sum_{i \leq n_c} \mathbf{1}\{W_{c,i} \in A\}] / \mathbb{E}_P[n_c]$: with random cluster sizes, sizes may covary with the law of the controls, so $\bar{P}$, not the per-cluster mixture, is the marginal that observation-level sums follow; with nonrandom sizes $\bar{P}$ is the observation marginal of P . For s in the support of n_c , let $\bar{P}_s$ denote the per-observation control marginal given size, $\bar{P}_s(B) := s^{-1} \mathbb{E}_P[\sum_{i \leq n_c} \mathbf{1}\{W_{c,i} \in B\} \mid n_c = s]$ for Borel sets B ; averaging $\bar{P}_s$ over the size-biased size law recovers $\bar{P}$.

Assumptions C1–C8.

(C1) *sampling.* The bundles V_c are i.i.d. across c under every $P \in \mathcal{P}_G$, with $G \rightarrow \infty$; cluster sizes n_c are i.i.d. with $\sup_P \mathbb{E}_P[n_c^{2q}] \leq C$, for the q of C2. The leading case, used in the lemma proofs, is bounded sizes $n_c \leq \bar{n}$; Lemma 3 records the relaxation. Within-cluster dependence is unrestricted.

(C2) *moments.* For some $q > 4$ and $C < \infty$, $\sup_P \mathbb{E}_P \|S_c(\tau_0, \eta_0)\|^q \leq C$; the eigenvalues of $\Sigma_P(\tau) := \mathbb{E}_P[S_c(\tau, \eta_0) S_c(\tau, \eta_0)']$ lie in $[c_\Omega, C_\Omega] \subset (0, \infty)$ for τ near τ_0 , uniformly over $\mathcal{P}_G$; and the conditional second moments are bounded: $\mathbb{E}_P[\zeta^2 \mid Z, W] \leq C$, $\mathbb{E}_P[\|\tilde{Z}\|^2 \mid W] \leq C$, and $\mathbb{E}_P[\tilde{D}^2 \mid W] \leq C$ almost surely, uniformly over $\mathcal{P}_G$. (The conditional bounds are the cluster-level counterpart of (L2). They are a genuine addition relative to the imported package, whose A5 assumes the resulting $o_p(1)$ displays directly; Lemmas 5 and 8 prove those displays from these primitives.)

(C3) *orthogonality.* The observation-level score is Neyman-orthogonal at (τ_0, η_0) , as in §2.1; cluster sums inherit it.

(C4) *cluster-level leakage-free adaptivity with size-measurable folds.* (i) The K outer folds, and within each outer-training sample the selection and estimation subsamples, partition clusters. (ii) The fold-assignment vector satisfies $\mathbf{k}_G = \phi_G^{\text{fold}}(\mathbf{n}_G, \mathcal{U})$, where ϕ_G^{fold} is a fixed measurable function of the size vector and $\mathcal{U}$ is a randomization variable independent of the bundles $(V_1, \dots, V_G)$. Both degenerate cases are allowed: a deterministic function of $\mathbf{n}_G$ alone, which is the implementation's grouped K -fold rule (§3.1), and a randomized partition that ignores $\mathbf{n}_G$, which is the data-independent partition of (L3). Under the LF-NCF protocol the selection/estimation subsplit is a fixed rotation of the outer folds and is therefore of the same form; under the nested-CV protocol the inner grouped split of the training clusters influences only the selected pipeline, which (iii) already constrains. (iii) The selected pipeline and nuisance fits $(\hat{\gamma}_k, \hat{\eta}_k)$ are $\mathcal{F}_k$ -measurable, with $\mathcal{F}_k$ as defined above. Assignment rules that depend on the bundles through anything beyond the size vector, for example through realized values of Y , D , Z , or W , are outside C4.

(C5) *rates.* Uniformly over $P \in \mathcal{P}_G$ and folds k : (i) $\max_k \|\delta_{a,k}\| = o_p(n^{-1/4})$ for each $a \in \{\mu, r, \pi\}$, the norms in $L^2(\bar{P})$; (ii) the product rate $\max_k \|\delta_{\pi,k}\|(\|\delta_{\mu,k}\| + |\tau_0| \|\delta_{r,k}\|) = o_p(n^{-1/2})$; (iii) the score-difference display $\varepsilon_n := n^{-1} \sum_i \|\hat{\psi}_i - \psi_i(\eta_0)\|^2 = o_p(1)$, strengthened to $\varepsilon_n = o_p(n^{-1/2})$ when cluster sizes are random (the rates in (i) deliver the stronger form, every term of the squared expansion being a product of two rates or a rate times a bounded conditional moment); (iv) bounded realizations: the fitted nuisances take values in a fixed bounded set, $\max_k \|\hat{\eta}_k - \eta_0\|_\infty \leq C$ almost surely, enforced by truncation as in (L3) and standard for curated, clipped libraries (alternatively, the rates in (i) may be stated in L^4). The sup-over-class form of these displays (for every $\varepsilon > 0$, $\sup_P P(n^{1/4} \max_k \|\delta_{a,k}\| > \varepsilon) \rightarrow 0$, and similarly for (ii)–(iii)) is the uniform-rate, equicontinuity-type condition that the uniform argument below requires.

(C6) *library.* The candidate library Γ_n is finite with $\log |\Gamma_n| = o(G_{\text{sel}})$, where G_{sel} is the number of selection-subsample clusters.

(C7) *uniformity class.* $\mathcal{P}_G$ contains strong-identification laws and cluster-level weak-identification drifts (relevance sequences proportional to $G^{-1/2}$); d_z is fixed.

(C8) *size-conditional design.* Under every $P \in \mathcal{P}_G$, with n_c the size of the observation's own cluster: (i) *exogeneity given size*: $\mathbb{E}_P[\zeta \mid Z, W, n_c] = 0$. (ii) *nuisance stability*: $\mathbb{E}_P[Z \mid W, n_c] = \pi_0(W)$ and $\mathbb{E}_P[D \mid W, n_c] = r_0(W)$; the residuals $\tilde{Z}$ and $\tilde{D}$ then remain conditionally centered given (W, n_c) , and $\mathbb{E}_P[Y \mid W, n_c] = \mu_0(W)$ follows from (i) and (ii). (iii) *moments given size*: $\mathbb{E}_P[\zeta^2 \mid Z, W, n_c] \leq C$, $\mathbb{E}_P[\|\tilde{Z}\|^2 \mid W, n_c] \leq C$, and $\mathbb{E}_P[\tilde{D}^2 \mid W, n_c] \leq C$ almost surely, uniformly over $\mathcal{P}_G$. These strengthen the conditional bounds of C2 by adding n_c to the conditioning; the C2 forms follow by the tower property. (iv) *size-conditional design domination*: there is $C_s < \infty$ with $\bar{P}_s \ll \bar{P}$ and $d\bar{P}_s/d\bar{P} \leq C_s$, uniformly over s in the size support and over $\mathcal{P}_G$.

Remark (content of C8). Say sizes are *independent of the within-cluster variables* when the bundle is generated by truncating an infinite within-cluster array: $V_c = (n_c, (X_{c,i})_{i \leq n_c})$ with $(X_{c,i})_{i \geq 1}$ independent of n_c . (Without such an array representation the phrase has no content, since the bundle carries only n_c coordinates.) Three leading cases. (a) *Sizes independent of the array*: conditioning on n_c then changes no (Z, W) -conditional statement, so (i)–(iii) reduce to the maintained model conditions. Part (iv) does *not* follow from independence alone: $\bar{P}_s$ averages the first s position marginals, which C1 leaves free to

differ across positions. If, in addition, the array positions are identically distributed in W (within-cluster exchangeability suffices), then $\bar{P}_s = \bar{P}$ for every s and (iv) holds with $C_s = 1$. (b) *Nonrandom sizes*: all size conditioning is trivial, the size support is a single point, and (i)–(iv) hold with $C_s = 1$. (c) *Cluster size among the controls*, n_c a component of W : (i)–(iii) are again the maintained conditions, because conditioning on (W, n_c) is conditioning on W ; only (iv) retains content. Regardless of any independence structure, (iv) holds automatically whenever the size support is finite with $\min_s P(n_c = s) \geq p_0 > 0$, with $C_s \leq \mathbb{E}[n_c]/p_0$, since $\bar{P}(B) \geq s P(n_c = s) \bar{P}_s(B)/\mathbb{E}[n_c]$ for every s and B ; the condition has bite only under large or thin-tailed size support. The substantive requirement is *instrument exogeneity conditional on cluster size*: how large a cluster is must carry no information about the structural error or the instrument assignment beyond the controls. In the Oregon cell the lottery is randomized conditional on list household size, which the controls include as household-size indicators, so (i)–(ii) are close to the design’s own statement; the residual content is that the realized number of responding members adds no further information. In the examiner cell the condition asks that shift caseload be uninformative about case severity given the controls; a violation is repaired by adding caseload to W , at which point (i)–(iii) hold by construction.

Theorem 2 (clustered cross-fitted AR statistic, uniform size). *Under C1–C8,*

$$\sup_{P \in \mathcal{P}_G} \left| P\left(\text{AR}_G(\tau_0) > \chi_{d_z, 1-\alpha}^2\right) - \alpha \right| \rightarrow 0,$$

and the inverted set $\{t : \text{AR}_G(t) \leq \chi_{d_z, 1-\alpha}^2\}$ covers τ_0 with probability tending to $1 - \alpha$ uniformly over $\mathcal{P}_G$.

Remark (no relevance condition). At τ_0 , $\tilde{Y} - \tau_0 \tilde{D} = \zeta$ with $\mathbb{E}[\zeta \mid Z, W] = 0$, so $\mathbb{E}[\psi(\tau_0, \eta_0)] = \mathbb{E}[\tilde{Z}\zeta] = 0$ for *any* strength of the instrument: identification strength enters only through the distribution of S_c , which the class $\mathcal{P}_G$ leaves free (C7 includes the local-to-zero drifts). This is the clustered analogue of the observation that the AR statistic needs no Jacobian (relevance) condition.

Remark (rank deficiency and the pseudo-inverse convention). The theorem is stated under C2’s eigenvalue bounds: $\bar{\Sigma}_G$ is invertible with probability tending to one (Weyl’s inequality), and the plain inverse is meant in (A.7). The implementation’s Moore–Penrose pseudo-inverse with a numerical rank guard (§3.2) is a diagnostic layer outside the theorem’s assumptions: when the guard flags numerical rank $r < d_z$, the reported χ_r^2 reference is a flagged diagnostic convention, not a claim covered by Theorem 2. The statistic’s pseudo-inverse and its degrees of freedom are computed from the same relative truncation, a single retained eigenspace, so the reported pair does not depend on the implementation of the cutoff. (A dimension-agnostic ridge-regularized AR with its own guarantee is available in the imported framework: Çetin 2026b, Theorem 4.)

The proof runs through four lemmas that restate, at cluster level, the four steps of the imported argument, one generic Gaussian-approximation lemma that supplies the central limit steps, and two supplementary lemmas that complete the random-size and empirical-norm steps. The supplementary pair is stated immediately after Lemma 2, ahead of the proofs that use it.

Lemma 2 (conditionally independent clusters given sizes). *Under C1 and C4, for each fold k : conditional on $\mathcal{F}_k$, the inference-fold bundles $\{V_c : k(c) = k\}$ are mutually independent, the conditional law of V_c is the size-conditional law $P(\cdot | n_c)$, and $(\hat{\gamma}_k, \hat{\eta}_k)$ is $\mathcal{F}_k$ -measurable, hence fixed given $\mathcal{F}_k$. Explicitly, for bounded measurable $f_1, \dots, f_G$,*

$$\mathbb{E} \left[\prod_{c:k(c)=k} f_c(V_c) \mid \mathcal{F}_k \right] = \prod_{c:k(c)=k} \mathbb{E} [f_c(V_c) \mid n_c] \quad \text{almost surely.}$$

If ϕ_G^{fold} does not depend on $\mathbf{n}_G$, then with $\mathcal{F}_k = \sigma(\mathcal{U}) \vee \sigma(V_c : k(c) \neq k)$ the inference-fold bundles are conditionally i.i.d. from P , the statement of the data-independent case.

Proof. Step 0. The size is a coordinate of the bundle, so $\{n_c = s\}$ is $\sigma(V_c)$ -measurable, and $\mathbf{n}_G$ takes values in the countable set $\mathbb{N}^G$, so all conditioning on size events is elementary.

Step 1 (conditioning on sizes preserves independence). Fix $\mathbf{s} \in \mathbb{N}^G$ with $P(\mathbf{n}_G = \mathbf{s}) > 0$. The event $\{\mathbf{n}_G = \mathbf{s}\} = \bigcap_c \{n_c = s_c\}$ is an intersection of events that are independent across c by C1, so for bounded measurable $h_1, \dots, h_G$,

$$\mathbb{E} \left[\prod_c h_c(V_c) \mid \mathbf{n}_G = \mathbf{s} \right] = \frac{\prod_c \mathbb{E} [h_c(V_c) \mathbf{1}_{\{n_c = s_c\}}]}{\prod_c P(n_c = s_c)} = \prod_c \mathbb{E} [h_c(V_c) \mid n_c = s_c].$$

Conditional on $\mathbf{n}_G$, the bundles are therefore independent, each with its size-conditional law.

Step 2 (test-function identity). Fix k . For $t \subseteq \{1, \dots, G\}$ and $\mathbf{s}$ as above, let $B_{t,\mathbf{s}} := \{u : \{c : \phi_G^{\text{fold}}(\mathbf{s}, u)_c = k\} = t\}$, a measurable set; given $\{\mathbf{n}_G = \mathbf{s}\}$, the events $\{\mathcal{U} \in B_{t,\mathbf{s}}\}$ partition the space over the finitely many t , and on $E_{t,\mathbf{s}} := \{\mathbf{n}_G = \mathbf{s}\} \cap \{\mathcal{U} \in B_{t,\mathbf{s}}\}$ the inference index set is the fixed set t . Take bounded measurable g, h_c, f_c . Using first the independence of $\mathcal{U}$ from the bundles (C4(ii)) and then the factorization across c from C1,

$$\begin{aligned} & \mathbb{E} \left[g(\mathbf{n}_G, \mathcal{U}) \prod_{c \notin t} h_c(V_c) \prod_{c \in t} f_c(V_c) ; E_{t,\mathbf{s}} \right] \\ &= \mathbb{E} \left[g(\mathbf{s}, \mathcal{U}) \mathbf{1}_{B_{t,\mathbf{s}}}(\mathcal{U}) \right] \prod_{c \notin t} \mathbb{E} [h_c(V_c) \mathbf{1}_{\{n_c = s_c\}}] \prod_{c \in t} \mathbb{E} [f_c(V_c) \mathbf{1}_{\{n_c = s_c\}}]. \end{aligned}$$

The factors over $c \in t$ equal $\mathbb{E}[f_c(V_c) \mid n_c = s_c] P(n_c = s_c)$. Running the same computation with $f_c(V_c)$ replaced by the constant $\mathbb{E}[f_c(V_c) \mid n_c = s_c]$ produces the identical right side. Hence

$$\mathbb{E} \left[g \prod_{c \notin t} h_c \prod_{c \in t} f_c ; E_{t,\mathbf{s}} \right] = \mathbb{E} \left[g \prod_{c \notin t} h_c \prod_{c \in t} \mathbb{E}[f_c(V_c) \mid n_c] ; E_{t,\mathbf{s}} \right],$$

and summing over the countably many $(t, \mathbf{s})$ extends the identity to the whole space.

Step 3 (monotone-class conclusion). Let $\mathcal{C}$ be the class of random variables $g(\mathbf{n}_G, \mathcal{U}) \prod_{c \leq G} h_c(V_c \mathbf{1}_{\{k(c) \neq k\}})$ with g, h_c bounded measurable. Since $\{k(c) = k\}$ is $\sigma(\mathbf{n}_G, \mathcal{U})$ -measurable, on $E_{t,\mathbf{s}}$ each factor $h_c(V_c \mathbf{1}_{\{k(c) \neq k\}})$ equals $h_c(V_c)$ for $c \notin t$ and the constant $h_c(0)$ for $c \in t$; absorbing the constants into g , every element of $\mathcal{C}$ restricted to $E_{t,\mathbf{s}}$ is of the Step-2 form, and summing over the partition $\{E_{t,\mathbf{s}}\}$ gives, for every $\Xi \in \mathcal{C}$,

$$\mathbb{E} \left[\Xi \cdot \prod_{c:k(c)=k} f_c(V_c) \right] = \mathbb{E} \left[\Xi \cdot \prod_{c:k(c)=k} \mathbb{E}[f_c(V_c) \mid n_c] \right].$$

$\mathcal{C}$ is closed under multiplication, contains the constants, and generates $\mathcal{F}_k$ by the definition of $\mathcal{F}_k$. The set of bounded $\mathcal{F}_k$ -measurable Ξ satisfying the display is a vector space containing

$\mathcal{C}$ and closed under bounded monotone limits, so by the functional monotone class theorem it contains all bounded $\mathcal{F}_k$ -measurable variables. The right-hand product is $\mathcal{F}_k$ -measurable (the index set $\{c : k(c) = k\}$ is $\sigma(\mathbf{n}_G, \mathcal{U})$ -measurable and each $\mathbb{E}[f_c(V_c) \mid n_c]$ is $\sigma(\mathbf{n}_G)$ -measurable), so the display characterizes the conditional expectation: $\mathbb{E}[\prod_{c:k(c)=k} f_c(V_c) \mid \mathcal{F}_k] = \prod_{c:k(c)=k} \mathbb{E}[f_c(V_c) \mid n_c]$ almost surely. Taking all but one $f_c \equiv 1$ identifies the conditional marginals as $P(\cdot \mid n_c)$; the product form is the claimed conditional independence. $(\hat{\gamma}_k, \hat{\eta}_k)$ is $\mathcal{F}_k$ -measurable by C4(iii). For the final claim, when ϕ_G^{fold} ignores $\mathbf{n}_G$ the partition $E_t = \{\mathcal{U} \in B_t\}$ does not involve the sizes; Step 2 then runs with the unconditional factors $\mathbb{E}[f_c(V_c)]$, giving conditionally i.i.d. draws from P . ■

Lemma 3 (random cluster sizes). *Let C5(iii) hold in its $\varepsilon_n = o_p(n^{-1/2})$ form and let cluster sizes be i.i.d. with $\sup_P \mathbb{E}_P[n_c^{2q}] \leq C$ ($q > 4$ as in C2). Then*

$$\begin{aligned} G^{-1} \sum_c \left\| \hat{S}_c - S_c(\eta_0) \right\|^2 &\leq \left(\max_{c \leq G} n_c \right) (n/G) \varepsilon_n = O_p(G^{1/(2q)}) \cdot O_p(1) \cdot o_p(n^{-1/2}) \\ &= o_p(G^{1/(2q)-1/2}) = o_p(1), \end{aligned}$$

and in Term C of the proof of Lemma 5 below, the factor $G^{-1/2} N_k \leq \sqrt{n} (n/G)^{1/2} = \sqrt{n} \cdot O_p(1)$, so the bounded-size steps go through with $\bar{n}$ replaced by $O_p(1)$ size statistics.

Proof. $\max_{c \leq G} n_c = O_p(G^{1/(2q)})$ under $\sup_P \mathbb{E}_P[n_c^{2q}] \leq C$: a union bound at level $t G^{1/(2q)}$ gives probability at most $t^{-2q} C$, small for large t (an o_p version would need uniform integrability, which we do not assume). $n/G \rightarrow_p \mathbb{E}[n_c] < \infty$ by the law of large numbers, and the final step uses $n \asymp G$ (equality of orders, not of values); the exponent $1/(2q) - 1/2$ is negative for every $q > 1$. Bounded sizes ($n_c \leq \bar{n}$) remain the clean leading case in the lemma proofs; this lemma is the stated relaxation. ■

Lemma 4 (empirical-to-population norm switch, size-conditional form). *Let δ be a scalar fold- k nuisance error ($\mathcal{F}_k$ -measurable); a vector-valued error such as δ_π enters componentwise, summing the displays over its components. Under C1, C4, and C8(iv),*

$$\mathbb{E} \left[\sum_{i \in \text{fold } k} \delta(W_i)^2 \mid \mathcal{F}_k \right] = \sum_{c:k(c)=k} n_c \|\delta\|_{L^2(\bar{P}_{n_c})}^2 \leq C_s N_k \|\delta\|_{L^2(\bar{P})}^2,$$

where $N_k = \sum_{c:k(c)=k} n_c$ is $\mathcal{F}_k$ -measurable. Conditional Markov then yields $\|\delta\|_{N_k}^2 = O_p(\|\delta\|_{L^2(\bar{P})}^2)$, which is all Term C of the proof of Lemma 5 needs, provided the C5 rates are stated in $L^2(\bar{P})$, as they are.

Proof. By Lemma 2, conditional on $\mathcal{F}_k$ the inference bundles are independent with laws $P(\cdot \mid n_c)$, and δ is a fixed function. Hence

$$\mathbb{E} \left[\sum_{i \leq n_c} \delta(W_{c,i})^2 \mid \mathcal{F}_k \right] = \mathbb{E} \left[\sum_{i \leq n_c} \delta(W_{c,i})^2 \mid n_c \right] = n_c \|\delta\|_{L^2(\bar{P}_{n_c})}^2 \leq C_s n_c \|\delta\|_{L^2(\bar{P})}^2,$$

the middle equality by the definition of $\bar{P}_s$ and the inequality by C8(iv). Summing over the inference clusters gives the display. Since N_k is $\mathcal{F}_k$ -measurable, dividing by N_k gives $\mathbb{E}[\|\delta\|_{N_k}^2 \mid \mathcal{F}_k] \leq C_s \|\delta\|_{L^2(\bar{P})}^2$, and conditional Markov delivers the O_p bound. ■

No concentration inequality is required. In cases (a)-with-exchangeability and (b) of the C8 remark, $\bar{P}_{n_c} = \bar{P}$ for every realized size, so $C_s = 1$ and the inequality is an identity, recovering the data-independent case.

Lemma 5 (oracle equivalence of the cluster-score mean). *Under C1–C5 and C8, for every $\varepsilon > 0$, $\sup_{P \in \mathcal{P}_G} P\left(\|\bar{m}_G - \bar{m}_G^0\| > \varepsilon\right) \rightarrow 0$, where $\bar{m}_G^0 := G^{-1/2} \sum_c S_c(\tau_0, \eta_0)$.*

Proof. Work fold by fold (K is fixed); fix fold k and condition on $\mathcal{F}_k$. For an inference-fold observation, write $\delta_\pi := \delta_{\pi,k}$, $\delta_\mu := \delta_{\mu,k}$, $\delta_r := \delta_{r,k}$ (all functions of W); using $\tilde{Y} - \tau_0 \tilde{D} = \zeta$ and $\tilde{Z} = Z - \pi_0(W)$,

$$\psi(\tau_0, \hat{\eta}_k) - \psi(\tau_0, \eta_0) = -\delta_\pi(W) \zeta - \tilde{Z} (\delta_\mu - \tau_0 \delta_r)(W) + \delta_\pi(W) (\delta_\mu - \tau_0 \delta_r)(W) =: A_i + B_i + C_i. \quad (\text{A.8})$$

Term A (conditionally mean zero). By Lemma 2, conditional on $\mathcal{F}_k$ the law of an inference bundle is the size-conditional law $P(\cdot \mid n_c)$ and δ_π is a fixed function, so

$$\mathbb{E}[A_i \mid \mathcal{F}_k] = -\mathbb{E}[\delta_\pi(W_i) \mathbb{E}[\zeta_i \mid Z_i, W_i, n_c] \mid n_c] = 0$$

by C8(i). The cluster sums $\sum_{i \in c} A_i$ are, again by Lemma 2, conditionally independent mean zero across inference clusters. They are no longer identically distributed, but the Chebyshev step below never uses identical distributions, only independence, mean zero, and a uniform variance bound. That bound is

$$\begin{aligned} \mathbb{E}\left[\left\|\sum_{i \in c} A_i\right\|^2 \mid \mathcal{F}_k\right] &\leq n_c \mathbb{E}\left[\sum_{i \leq n_c} \|\delta_\pi(W_i)\|^2 \zeta_i^2 \mid n_c\right] \\ &\leq C n_c \mathbb{E}\left[\sum_{i \leq n_c} \|\delta_\pi(W_i)\|^2 \mid n_c\right] \leq C C_s n_c^2 \|\delta_{\pi,k}\|^2, \end{aligned}$$

using within-cluster Cauchy–Schwarz, then C8(iii), then C8(iv). Under the leading bounded-size case $n_c \leq \bar{n}$ this is $C C_s \bar{n}^2 \|\delta_{\pi,k}\|^2$, the bound of the data-independent case with an extra factor C_s . Hence $\mathbb{E}\left[\|G^{-1/2} \sum_{c:k(c)=k} \sum_{i \in c} A_i\|^2 \mid \mathcal{F}_k\right] \leq C C_s \bar{n}^2 \|\delta_{\pi,k}\|^2 = o_p(1)$ by C5(i) (indeed $\|\delta_{\pi,k}\| = o_p(1)$ suffices for this term). Conditional Chebyshev and bounded convergence remove the conditioning, exactly as in the proof of Lemma 1; the C5 displays are uniform over $\mathcal{P}_G$, so the bound is too. With random sizes, replace $\bar{n}^2$ by $G^{-1} \sum_{c \leq G} n_c^2 = O_p(1)$ (Markov with $\mathbb{E}[n_c^{2q}] \leq C$), the Lemma 3 machinery.

Term B (conditionally mean zero). Symmetric: $\mathbb{E}[\tilde{Z} \mid W, n_c] = 0$ at the true π_0 by C8(ii), and $\delta_\mu - \tau_0 \delta_r$ is fixed given $\mathcal{F}_k$, so B_i is conditionally mean zero under the size-conditional law; the same variance bound holds with $\|\delta_{\mu,k} - \tau_0 \delta_{r,k}\|^2$, by C8(iii)–(iv), and the term is $o_p(1)$ as above.

Term C (the nuisance product). No conditional centering is available; we bound the term directly. Let N_k be the number of inference-fold observations. Empirical Cauchy–Schwarz gives

$$\left\|G^{-1/2} \sum_{i \in \text{fold } k} C_i\right\| \leq G^{-1/2} N_k \|\delta_{\pi,k}\|_{N_k} \|\delta_{\mu,k} - \tau_0 \delta_{r,k}\|_{N_k},$$

with $\|\cdot\|_{N_k}$ the empirical L^2 norm on the inference fold. Since $N_k \leq n \leq \bar{n}G$ by C1, $G^{-1/2} N_k \leq G^{-1/2} n = n^{1/2} (n/G)^{1/2} \leq n^{1/2} \bar{n}^{1/2}$. Empirical norms are O_p of the $L^2(\bar{P})$ norms on the held-out fold (Lemma 4; the constant C_s of C8(iv) is absorbed into the $O_p(1)$ factor), so the product rate C5(ii) gives

$$\left\|G^{-1/2} \sum_i C_i\right\| \leq \bar{n}^{1/2} n^{1/2} \|\delta_{\pi,k}\| \|\delta_{\mu,k} - \tau_0 \delta_{r,k}\| \cdot O_p(1) = n^{1/2} \cdot o_p(n^{-1/2}) \cdot O_p(1) = o_p(1).$$

The product-rate display is therefore unchanged from the i.i.d. case: bounded cluster sizes absorb the clustering entirely, and with random sizes $\mathbb{E}[n_c^{2q}] < \infty$ and Minkowski's inequality give the same conclusion with $\bar{n}$ replaced by a moment constant (Lemma 3). Summing the three terms over the K folds completes the proof. ■

Lemma 6 (uniform Gaussian approximation on convex sets). *Let d be fixed. For each P in a class $\mathcal{P}$, let $\xi_1, \dots, \xi_G$ be i.i.d. mean-zero random vectors in $\mathbb{R}^d$ under P , with $\sup_P \mathbb{E}_P \|\xi_1\|^3 \leq C_3 < \infty$ and $\lambda_{\min}(\Lambda_P) \geq c_0 > 0$, where $\Lambda_P := \mathbb{E}_P[\xi_1 \xi_1']$. Let $\mathcal{A}_d$ be the class of convex Borel subsets of $\mathbb{R}^d$. Then*

$$\sup_{P \in \mathcal{P}} \sup_{A \in \mathcal{A}_d} \left| P\left(G^{-1/2} \sum_{g \leq G} \xi_g \in A\right) - N(0, \Lambda_P)(A) \right| \leq c_B d^{1/4} c_0^{-3/2} C_3 G^{-1/2},$$

with c_B an absolute constant.

Proof. Fix P . The summands $X_g := G^{-1/2} \xi_g$ are independent and mean zero, and their sum has covariance Λ_P , invertible since $c_0 > 0$. The convex-set Berry–Esseen bound of Bentkus (2005) gives

$$\sup_{A \in \mathcal{A}_d} \left| P\left(\sum_{g \leq G} X_g \in A\right) - N(0, \Lambda_P)(A) \right| \leq c_B d^{1/4} \sum_{g \leq G} \mathbb{E}_P \left\| \Lambda_P^{-1/2} X_g \right\|^3.$$

The third-moment sum is at most $G \cdot G^{-3/2} \lambda_{\min}(\Lambda_P)^{-3/2} \mathbb{E}_P \|\xi_1\|^3 \leq c_0^{-3/2} C_3 G^{-1/2}$. The resulting bound does not depend on P , so the supremum over the class obeys it too. ■

Lemma 7 (uniform cluster-level central limit theorem). *Under C1–C3, C7, and C8(i), uniformly over $\mathcal{P}_G$, $G^{-1/2} \sum_c S_c(\tau_0, \eta_0) \Rightarrow N(0, \Sigma_P)$, where $\Sigma_P := \mathbb{E}_P[S_c S_c']$ (per-cluster normalization). Uniformity is meant in the convex-set sense: the approximation error against $N(0, \Sigma_P)$, taken over all convex Borel sets, is at most a constant multiple of $G^{-1/2}$, with the constant built from C2's bounds, the fixed d_z of C7, and an absolute constant.*

Proof. Write $S_c := S_c(\tau_0, \eta_0)$. The cluster scores are i.i.d. across clusters under every P (C1) and mean zero cluster-wise: conditioning on n_c and applying C8(i) through the tower property gives $\mathbb{E}_P[S_c] = \mathbb{E}_P\left[\sum_{i \leq n_c} \tilde{Z}_{c,i} \mathbb{E}_P[\zeta_{c,i} \mid Z_{c,i}, W_{c,i}, n_c]\right] = 0$ term by term; with nonrandom sizes this reduces to the remark following Theorem 2. By C2 and Lyapunov's moment inequality with $q > 4 > 3$, $\mathbb{E}_P \|S_c\|^3 \leq (\mathbb{E}_P \|S_c\|^q)^{3/q} \leq C^{3/q}$, and $\lambda_{\min}(\Sigma_P) \geq c_\Omega$. Lemma 6, applied with the cluster scores as the ξ_g , $d = d_z$ (fixed by C7), $C_3 = C^{3/q}$, and $c_0 = c_\Omega$, gives

$$\sup_{P \in \mathcal{P}_G} \sup_{A \in \mathcal{A}_{d_z}} \left| P\left(G^{-1/2} \sum_c S_c \in A\right) - N(0, \Sigma_P)(A) \right| \leq C_* G^{-1/2} \rightarrow 0,$$

with $C_* := c_B d_z^{1/4} c_\Omega^{-3/2} C^{3/q}$ built from C2's constants, the fixed d_z of C7, and the absolute constant c_B of Lemma 6. This display is the uniform statement: the approximation error against the matched Gaussian $N(0, \Sigma_P)$ vanishes at rate $G^{-1/2}$ simultaneously over the class and over all convex sets, so weak convergence holds along every sequence $P_G \in \mathcal{P}_G$ (half-spaces are convex, so the Cramér–Wold directions are a special case). It is also the form used downstream: for every $t \geq 0$ the set $\{x : x' \Sigma_P^{-1} x \leq t\}$ is convex, and $N(0, \Sigma_P)$ assigns it exactly $F_{\chi_{d_z}^2}(t)$, the $\chi_{d_z}^2$ distribution function at t (substitute $x = \Sigma_P^{1/2} u$), which is what Step 4 of the theorem's proof uses. ■

Lemma 8 (variance consistency). *Under C1–C5 and C8(iii)–(iv), for every $\varepsilon > 0$, $\sup_{P \in \mathcal{P}_G} P(\|\bar{\Sigma}_G - \Sigma_P\| > \varepsilon) \rightarrow 0$, and $\bar{\Sigma}_G^{-1} - \Sigma_P^{-1} = o_p(1)$ uniformly over $\mathcal{P}_G$.*

Proof. Split $\bar{\Sigma}_G - \Sigma_P$ into $\bar{\Sigma}_G - \bar{\Sigma}_G^0$ and $\bar{\Sigma}_G^0 - \Sigma_P$, with $\bar{\Sigma}_G^0 := G^{-1} \sum_c S_c(\eta_0) S_c(\eta_0)'$ (the τ_0 argument suppressed). (i) With $\Delta_c := \hat{S}_c - S_c(\eta_0)$,

$$\begin{aligned} \|\bar{\Sigma}_G - \bar{\Sigma}_G^0\| &\leq G^{-1} \sum_c \left(2\|S_c(\eta_0)\| \|\Delta_c\| + \|\Delta_c\|^2 \right) \\ &\leq 2 \left(G^{-1} \sum_c \|S_c\|^2 \right)^{1/2} \left(G^{-1} \sum_c \|\Delta_c\|^2 \right)^{1/2} + G^{-1} \sum_c \|\Delta_c\|^2. \end{aligned}$$

By within-cluster Cauchy–Schwarz, $\|\Delta_c\|^2 \leq n_c \sum_{i \in c} \|\hat{\psi}_i - \psi_i(\eta_0)\|^2$, so $G^{-1} \sum_c \|\Delta_c\|^2 \leq \bar{n} (n/G) \varepsilon_n = \bar{n}^2 \cdot o_p(1)$ by C5(iii); the display itself follows from the decomposition (A.8) with squared terms, each of $\|\delta_\pi\|^2$, $\|\delta_\mu - \tau_0 \delta_r\|^2$, and their products being $o_p(1)$ under the C5 rates, the size-conditional moment bounds C8(iii) alongside C2 (the conditional expectations are taken under the size-conditional laws of Lemma 2), the norm domination C8(iv) wherever an empirical squared norm is replaced by its population counterpart (every constant multiplied by C_s), and the bounded realizations C5(iv). The first factor is $O_p(1)$ by C2 and Markov. (ii) $\bar{\Sigma}_G^0 - \Sigma_P$ is an i.i.d. average of mean- Σ_P outer products with $q/2 > 2$ moments (C2): the law of large numbers for uniformly integrable triangular arrays applies uniformly over the class. Finally, C2’s eigenvalue floor makes $M \mapsto M^{-1}$ uniformly continuous on the relevant set, giving the inverse statement. ■

Proof of Theorem 2. The argument is direct and uniform, following at cluster level the four-step proof of the imported result (Çetin 2026b, proof of Theorem 3); clustering enters only through Lemmas 2 and 7, the size-conditional conditions C4 and C8 they require, and the cluster outer products in Lemma 8.

Step 1 (uniform central limit theorem for the oracle cluster score). Lemma 7: $\bar{m}_G^0 \Rightarrow N(0, \Sigma_P)$ uniformly, with a $G^{-1/2}$ convex-set approximation rate from Lemma 6 (third moments from C2’s $q > 4$, the eigenvalue floor c_Ω , and d_z fixed).

Step 2a (score equivalence). Lemma 5: $\bar{m}_G - \bar{m}_G^0 = o_p(1)$ uniformly. Where the imported argument invokes its A5 display directly, the display is here proved constructively from the observation-level C5 rates via the three-term decomposition (A.8).

Step 2b (covariance equivalence). Lemma 8: $\bar{\Sigma}_G - \Sigma_P = o_p(1)$ and $\bar{\Sigma}_G^{-1} - \Sigma_P^{-1} = o_p(1)$ uniformly.

Step 3 (statistic equivalence). By Weyl’s inequality and C2’s eigenvalue floor, $\bar{\Sigma}_G$ is invertible with probability tending to one uniformly; the resolvent identity together with $\|\bar{m}_G^0\| = O_p(1)$ (Step 1) converts Steps 2a–2b into $\text{AR}_G(\tau_0) = \bar{m}_G^{0'} \Sigma_P^{-1} \bar{m}_G^0 + o_p(1)$ uniformly over $\mathcal{P}_G$.

Step 4 (size and coverage). For every $t \geq 0$ the set $\{x : x' \Sigma_P^{-1} x \leq t\}$ is convex and carries $N(0, \Sigma_P)$ -probability exactly $F_{\chi_{d_z}^2}(t)$ (the proof of Lemma 7), so Step 1’s convex-set bound gives

$$\sup_{P \in \mathcal{P}_G} \sup_{t \geq 0} \left| P(\bar{m}_G^{0'} \Sigma_P^{-1} \bar{m}_G^0 \leq t) - F_{\chi_{d_z}^2}(t) \right| \leq C_* G^{-1/2},$$

with C_* the constant of that proof; the approximating law is $\chi_{d_z}^2$, free of P , so no continuous-mapping step under drifting laws is needed. Slutsky and the continuity of the $\chi_{d_z}^2$ distribution

function at its $1 - \alpha$ quantile give the size statement, and inverting the family of tests gives the uniform coverage of the set. Every o_p and O_p above is quantified with sup-over- $\mathcal{P}_G$ thresholds: the C5 displays are assumed in sup form and the constants in C1–C2 are uniform, so the conclusion is the uniform one. ■

Remark (what clustering changes). Clustering changes four things relative to the imported i.i.d. argument: the sampling unit in Lemma 2 (fold assignment must be at cluster level, assumption C4, the cluster-aware folds of §3.1); the fold-assignment law (the assignment is a function of the cluster sizes, so the conditional sampling law inside a fold is the size-conditional law, assumptions C4 and C8); the carrier of the central limit theorem in Lemma 7 (cluster scores replace observation scores); and the variance estimator (cluster-summed outer products in (A.7)). The nuisance-rate and product-rate conditions remain observation-level displays, identical to the i.i.d. case.

Remark (asymptotics in the number of clusters). Theorem 2 is a $G \rightarrow \infty$ statement with bounded (or $2q$ -th-moment) cluster sizes. The examiner cell (1,831 shift clusters) and the OHIE cell (20,747 household clusters) sit squarely inside this regime. The Angrist–Krueger cell’s $G = 51$ state-of-birth clusters sit at its boundary, and we do not read Theorem 2 as finite- G evidence there; its fixed- G complement (wild-cluster bootstrap AR inference) lies outside this paper’s theory.

Remark (uniformity, and what was added to the imported package). Every step above is stated sup-over- $\mathcal{P}_G$, and no step required more than C1–C8. The only conditions beyond the cluster-level transcription of the imported package are the conditional second-moment bounds recorded in C2, which are genuine additions (the imported A5 assumes the resulting displays directly, where Lemmas 5 and 8 prove them constructively), the bounded-realization condition C5(iv), the size-measurable fold clause C4(ii), which restores in weakened form the partition-independence clause that (L3) carries, and the size-conditional design conditions C8, which supply the size-conditional centering, moment, and domination bounds that the fold-conditional sampling law of Lemma 2 requires. The weighted and generated-instrument variants are Propositions 4 and 5 (A.6): bounded weights multiply the score at observation level and pass through every lemma under the weighted target law and the conditions stated there (cluster scores become $\sum_{i \in c} \omega_i \psi_i$, with no within-cluster weight constancy assumed or needed), and the fold-nested generated instrument enters as a fourth cross-fitted nuisance whose estimation error enters every display of the AR chain exactly where δ_π already enters.

One further lemma serves a single application. The Angrist–Krueger cell of §4.9 runs the protocol after sweeping the state-of-birth effects by a within transformation: the outer folds partition states, and both the training rows and the held-out rows of a state are demeaned by that state’s own sample means. The lemma below records that this demeaning is a deterministic transformation of the sampling unit, not an estimated nuisance evaluated out of fold, and it states the price at which the theory above covers the transformed procedure.

Lemma 9 (within-transformation bridge). *Let the bundles V_c satisfy C1, and let the controls contain a cluster-constant block (here, state-of-birth dummies) plus a varying block $W_{c,i}^y$ (here, year-of-birth dummies). Let the outer folds satisfy C4(i)–(ii): the folds partition clusters, and the assignment is a fixed measurable function of the size vector $\mathbf{n}_G$ and a randomization $\mathcal{U}$ independent of the bundles. Define the within map T on bundles by $A_{c,i}^w := A_{c,i} - n_c^{-1} \sum_{j \leq n_c} A_{c,j}$ for $A \in \{Y, D, Z, W^y\}$, and $V_c^w := T(V_c)$; write $\bar{P}^w$ and $\bar{P}_s^w$*

for the size-biased observation marginal and the per-observation marginal given size of the transformed bundles, defined exactly as $\bar{P}$ and $\bar{P}_s$ are for the raw bundles. Assume: (W1) $Y_{c,i} = \tau_0 D_{c,i} + \alpha_c + W_{c,i}^y \gamma_0 + \varepsilon_{c,i}$, with α_c an arbitrary cluster-level random intercept; (W2) bundle exogeneity of the structural error: $\mathbb{E}[\varepsilon_{c,i} \mid Z_c, W_c^y, n_c, \alpha_c] = 0$ for every $i \leq n_c$, where Z_c and W_c^y collect all rows of the bundle; (W3) an additive first stage for the instrument, $Z_{c,i} = \Pi_0' W_{c,i}^y + \lambda_c + v_{c,i}$ with $\mathbb{E}[v_{c,i} \mid W_c^y, n_c, \lambda_c] = 0$ and λ_c a cluster-level random intercept vector; (W4) C2 read on the transformed problem, in all three parts. Write $\tilde{Z}^w := Z^w - \Pi_0' W^{y,w}$, $\zeta^w := (Y^w - \mu_0^w) - \tau_0(D^w - r_0^w)$, $\tilde{D}^w := D^w - r_0^w$, and $S_c^w := \sum_{i \leq n_c} \tilde{Z}_{c,i}^w \zeta_{c,i}^w$, where η_0^w collects the $L^2(\bar{P}^w)$ linear projections on $W^{y,w}$. Then (W4) requires: (a) $\sup_P \mathbb{E}_P \|S_c^w\|^q \leq C$ for the $q > 4$ of C2; (b) the eigenvalues of $\Sigma_P^w := \mathbb{E}[S_c^w S_c^{w'}]$ lie in $[c_\Omega, C_\Omega]$; (c) the conditional second moments are bounded at bundle level, $\mathbb{E}[(\zeta_{c,i}^w)^2 \mid Z_c, W_c^y, n_c] \leq C$, $\mathbb{E}[\|\tilde{Z}_{c,i}^w\|^2 \mid W_c^y, n_c] \leq C$, and $\mathbb{E}[(\tilde{D}_{c,i}^w)^2 \mid W_c^y, n_c] \leq C$ almost surely; the bundle-level conditioning is what the transformed Terms A and B of (A.8) require, because the transformed nuisance errors are measurable with respect to the whole bundle. Assume further: (W5) the eigenvalues of $\mathbb{E}_{\bar{P}^w}[W^{y,w} W^{y,w'}]$ are bounded below; and C8(iv) restated on the transformed marginals: there is $C_s < \infty$ with $\bar{P}_s^w \ll \bar{P}^w$ and $d\bar{P}_s^w/d\bar{P}^w \leq C_s$, uniformly over the size support and over $\mathcal{P}_G$ (T preserves the size support, and the finite-support sufficient condition of the C8 remark applies verbatim); and (W6) an additive treatment equation parallel to (W3), $D_{c,i} = \Pi_0^D W_{c,i}^y + \lambda_c^D + v_{c,i}^D$ with $\mathbb{E}[v_{c,i}^D \mid W_c^y, n_c, \lambda_c^D] = 0$ and λ_c^D a cluster-level random intercept. The remaining parts of C8, read on the transformed objects, are supplied by (W2), (W3), (W6), and (W4)(c), whose conditioning sets all contain n_c ; in particular the Z equation of C8(ii) comes from (W3) and the D equation from (W6). Then: (i) the V_c^w are i.i.d. across c with the same sizes, so C1 holds for the transformed bundles; (ii) the state-fold procedure of §4.9 is the protocol of this appendix applied to $\{V_c^w\}$: the training-side and held-out demeanings are the same bundle-internal map T ; T preserves n_c , so the fold assignment computed from the transformed bundles coincides with the one computed from the raw bundles and C4 holds for the transformed problem; the training coefficients are measurable with respect to the transformed training field; each held-out cluster's score is a measurable function of the cluster's transformed bundle and that field; and Lemma 2 applies to the transformed bundles, so cross-cluster conditional independence is preserved and Lemma 8 makes the clustered variance estimator consistent for Σ_P^w ; (iii) $\zeta_{c,i}^w = \varepsilon_{c,i} - \bar{\varepsilon}_c$ with $\bar{\varepsilon}_c := n_c^{-1} \sum_{j \leq n_c} \varepsilon_{c,j}$, and $\mathbb{E}[S_c^w(\tau_0, \eta_0^w) \mid n_c] = 0$ under (W1)–(W3), cluster by cluster; (iv) C3 and C5–C7 hold for the transformed problem: the Gateaux derivatives of the score in the μ , r , and π directions are the cross moments $-\mathbb{E}[\tilde{Z}^w W^{y,w'}]$, $+\tau_0 \mathbb{E}[\tilde{Z}^w W^{y,w'}]$, and $-\mathbb{E}[W^{y,w} \zeta^w]$, the first two zero by the normal equations of the π -projection, hence by (W3), and the third zero by (W2); C5 holds at the parametric $n^{-1/2}$ rate by (W5), with C5(iv) satisfied in its L^4 -rate alternative form. Consequently C1–C8 hold for the transformed bundles and Theorem 2 applies to them as stated; with K1 and K3 imposed on the transformed stacked score, so does Theorem 3.

Proof. (i). T is a fixed Borel map acting on each bundle alone: it depends only on n_c and the bundle's own rows, not on any other cluster, any estimate, or the fold partition. Measurable images of i.i.d. bundles are i.i.d., and T leaves n_c unchanged, so C1's size moments carry over.

(ii). The outer folds partition states, so every row of a state lies in one fold. The training-side per-state means and the held-out own-state means are therefore both the bundle's own

sample means, and the implemented procedure coincides with the appendix protocol run on $\{V_c^w\}$. The fold assignment is $\phi_G^{\text{fold}}(\mathbf{n}_G, \mathcal{U})$ by C4(ii), and T preserves $\mathbf{n}_G$, so the assignment is identical before and after the transformation and C4(i)–(ii) hold for the transformed bundles with the same ϕ_G^{fold} and $\mathcal{U}$; the implementation’s grouped rule of §3.1 is the deterministic case. The pooled OLS coefficients of fold k are functions of the transformed training bundles, hence measurable with respect to $\mathcal{F}_k^w := \sigma(\mathbf{n}_G, \mathcal{U}) \vee \sigma(V_c^w \mathbf{1}\{k(c) \neq k\} : c \leq G)$, which is C4(iii) for the transformed problem. Lemmas 2, 8, 3, and 4 then apply with V_c^w in place of V_c and $\bar{P}^w$ in place of $\bar{P}$, the variance bounds drawing on (W4)(c) and the norm switch on the transformed C8(iv).

(iii). T annihilates cluster-constant blocks, so the state dummies, α_c , and λ_c vanish, and (W1) becomes $Y^w = \tau_0 D^w + W^{y,w} \gamma_0 + \varepsilon^w$ with $\varepsilon_{c,i}^w = \varepsilon_{c,i} - \bar{\varepsilon}_c$. Each transformed variable is a measurable function of the whole bundle, and averaging (W2) over the rows gives $\mathbb{E}[\varepsilon_{c,i}^w \mid Z_c, W_c^y, n_c, \alpha_c] = 0$, hence $\mathbb{E}[W^{y,w} \varepsilon^w] = 0$. Linearity of the $L^2(\bar{P}^w)$ projection then gives $\mu_0^w - \tau_0 r_0^w = W^{y,w} \gamma_0$, hence $\zeta^w = \varepsilon^w$. Under (W3), $\tilde{Z}_{c,i}^w = v_{c,i}^w$ is a measurable function of (Z_c, W_c^y, n_c) , and the conditioning set of (W2) contains n_c , so each term of S_c^w is conditionally mean zero given n_c by the tower property. The treatment equation runs in parallel: T annihilates λ_c^D , so (W6) becomes $D^w = \Pi_0^{D'} W^{y,w} + v^{D,w}$; averaging (W6)’s centering over the rows and the tower property over λ_c^D give $\mathbb{E}[v_{c,i}^{D,w} \mid W_c^y, n_c] = 0$, hence $\mathbb{E}[W^{y,w} v^{D,w}] = 0$, the linear projection gives $r_0^w = \Pi_0^{D'} W^{y,w}$, and $\tilde{D}^w = v^{D,w}$ is conditionally centered given (W_c^y, n_c) . This is the D equation of C8(ii) read on the transformed problem, at the bundle-level conditioning that (W4)(c) uses.

(iv). For C3, the three Gateaux derivatives are the displayed cross moments; the μ - and r -direction moments vanish by the normal equations of the projection of Z^w on $W^{y,w}$, which (W3) implies, and the π -direction moment vanishes by (W2). Term A of Lemma 5 centers with (W2), Term B with (W3), and Term C is unchanged; the variance sides of Terms A and B use the bundle-level bounds of (W4)(c). The nuisance is a fixed-dimension regression on bounded regressors (demeaned dummies take values in $[-1, 1]$), so (W5) delivers the C5 rates with room to spare, in L^4 as well as L^2 , which settles C5(iv) through its stated alternative. The step from (W5)’s population floor to the per-fold Gram matrices is a matrix law of large numbers: with K fixed, each training complement contains a constant fraction of the G i.i.d. clusters, the dimension is fixed, and the entries of $W^{y,w} W^{y,w'}$ are bounded with cluster sums controlled by C1’s size moments, so each fold’s Gram matrix concentrates entrywise on $\mathbb{E}_{\bar{P}^w}[W^{y,w} W^{y,w'}]$ uniformly over $\mathcal{P}_G$, and Weyl’s inequality keeps its smallest eigenvalue above half of (W5)’s floor with probability tending to one; on that event the pooled OLS coefficient error is $O_p(G^{-1/2})$, which closes the C5 rate claim. C6 is trivial for the single linear learner, and C7 is read on the transformed class. ■

Remark (what the bridge does and does not buy). The lemma reclassifies own-state demeaning: it is not an estimated nuisance evaluated out of fold but a deterministic transformation of the sampling unit, applied identically on both sides of every fold boundary. The price is condition (W2), the within-estimator’s strict exogeneity in this setting: each individual’s structural error must be mean-independent of the whole state bundle of instruments and year-of-birth controls, not only of the individual’s own row. Under random within-state sampling and the quarter-of-birth exclusion, another individual’s quarter of birth carries no information about one’s own wage error, so (W2) is the natural reading of the design; it

still rules out cross-individual interference within a state. The eigenvalue floor in (W4) is a genuinely new primitive: the within transformation keeps only within-state variation, so the floor for the untransformed score does not imply it. The conditions also stratify across statistics: the AR chain uses only C8(i), the Z equation of C8(ii), C8(iii), and C8(iv), because $\zeta^w = \varepsilon^w$ carries the projection cancellation through Terms A and B ; (W6) is needed only for the score (LM) extension, whose q -block invokes the D equation of C8(ii) in the proof of Lemma 10. No new condition on unbalanced state sizes is needed: C1's size moments already bound the largest state share, $\max_c n_c/n = O_p(G^{1/(2q)-1}) = o_p(1)$, by the argument of Lemma 3. The finite- G caveat of §4.9 is unchanged.

A.5 The cross-fitted score statistic

The score (LM) statistic of Kleibergen (2005) directs power at the τ direction and carries a χ_1^2 reference whatever d_z (§3.2). This subsection states and proves its cross-fitted version under the leakage-free protocol, in both the i.i.d. and the clustered sampling frames; τ is scalar throughout, so Kleibergen's parameter dimension is $m = 1$.

Objects. The score and its τ -derivative (the Jacobian contribution) are $\psi_i(\eta) = \tilde{Z}_i(\tilde{Y}_i - \tau_0 \tilde{D}_i)$ and $q_i(\eta) := \partial_\tau \psi_i(\eta) = -\tilde{Z}_i \tilde{D}_i$, with $\tilde{Z} = Z - \pi(W)$, $\tilde{Y} = Y - \mu(W)$, $\tilde{D} = D - r(W)$; the stacked vector is $\varphi_i(\eta) := (\psi_i(\eta)', q_i(\eta)')' \in \mathbb{R}^{2d_z}$. Cross-fitted versions carry hats and use $\hat{\eta}_{k(i)}$. Note $\mathbb{E}_P[q_i(\eta_0)] = -g_{ZD}$, the relevance, which the identification classes allow to drift to zero: recentering is at the drifting truth, exactly the unconditional recentering of Kleibergen (2005, Assumption 1). The sample objects are, in the i.i.d. case (the clustered case replaces every sum over observations by the corresponding sum of cluster-summed scores, with G in place of n):

$$\hat{m}_n := n^{-1} \sum_i \hat{\psi}_i, \quad \hat{g}_D := n^{-1} \sum_i \hat{q}_i, \quad \hat{V}_{ff} := n^{-1} \sum_i \hat{\psi}_i \hat{\psi}_i', \quad \hat{V}_{\theta f} := n^{-1} \sum_i (\hat{q}_i - \hat{g}_D) \hat{\psi}_i', \quad (\text{A.9})$$

where $\hat{V}_{ff}$ and $\hat{V}_{\theta f}$ share one aggregation operator (condition K3 below), and, following Kleibergen (2005, eq. (16)) at $m = 1$,

$$\hat{D}_n := \hat{g}_D - \hat{V}_{\theta f} \hat{V}_{ff}^{-1} \hat{m}_n, \quad K_n(\tau_0) := \frac{n \left(\hat{m}_n' \hat{V}_{ff}^{-1} \hat{D}_n \right)^2}{\hat{D}_n' \hat{V}_{ff}^{-1} \hat{D}_n}, \quad (\text{A.10})$$

the statistic of his Definition 1 (eq. (21)) at $m = 1$.

(K1) *stacked moments and non-degeneracy.* $\sup_P \mathbb{E}_P \|\varphi(\tau_0, \eta_0)\|^q < \infty$ for the $q > 4$ of A2/C2 (for the PLIV score this follows from $2q$ -th moments of the residuals via Cauchy-Schwarz: $\mathbb{E}|\tilde{Z}_j \tilde{D}_j|^q \leq (\mathbb{E}\tilde{Z}_j^{2q} \mathbb{E}\tilde{D}_j^{2q})^{1/2}$), and the eigenvalues of $V_{ff} := \mathbb{E}_P[\psi(\eta_0)\psi(\eta_0)']$ lie in $[c, C] \subset (0, \infty)$, as in A2/C2. In the i.i.d. case K1 also carries the conditional second-moment bounds $\mathbb{E}_P[\zeta^2 \mid Z, W] \leq C$, $\mathbb{E}_P[\|\tilde{Z}\|^2 \mid W] \leq C$, and $\mathbb{E}_P[\tilde{D}^2 \mid W] \leq C$ almost surely, uniformly over the class: these are the (L2) bounds of Lemma 1, restated here because the conditional-variance steps for the Jacobian block of Lemma 10 use them, and the imported A5, which assumes the score-block displays directly, does not cover that block. In the clustered case C2 and C8(iii) already carry these bounds. The full joint covariance V_P of φ is not restricted directly: Kleibergen (2005, p. 1106) explicitly allows a positive semidefinite joint covariance, and only V_{ff} must be positive definite. Under K1 a floor is nonetheless automatic: the V_{ff}

bounds and the non-degeneracy display below force $\lambda_{\min}(V_P) \geq c' > 0$ over the class (derived in the proof of Lemma 10). In the clustered case the same bounds are imposed directly on the stacked cluster score $\Phi_c := \sum_{i \in c} \varphi_i$, exactly as C2 imposes the moment bound on S_c (under bounded cluster sizes $n_c \leq \bar{n}$ they follow from the observation-level bounds by Minkowski's inequality), with the eigenvalue condition on the cluster-level V_{ff} . Non-degeneracy: with $V_{\theta\theta.f} := V_{\theta\theta} - V_{\theta f} V_{ff}^{-1} V_{f\theta}$,

$$\lambda_{\min}(V_{\theta\theta.f}) \geq c > 0 \quad \text{over the class.}$$

This rules out the collinear corner $\zeta \propto \tilde{D}$ (score and Jacobian contributions collinear), in which the orthogonalized Jacobian's limiting direction is undefined; the conditioning step of the proof requires it (Lemma 11 turns the requirement into a quantitative small-ball bound).

(K3) *coherence*. $\hat{V}_{ff}$ and $\hat{V}_{\theta f}$ are built with the *same* aggregation operator: per-observation outer products in the i.i.d. case, per-cluster score sums in the clustered case. The published home of this condition is Kleibergen (2005, Assumption 2 and eq. (28)): the covariance estimator of the moment block must imply, through the same weighting, a consistent estimator of the cross block. The implementation satisfies it by construction (the moment and cross meats share one cluster-sum operator); the reference degrees of freedom are 1 whatever d_z , and the score set inversion reuses the adaptive-grid machinery of §3.2.

Remark (K2: the product display, recorded rather than assumed). The Jacobian block needs $\max_k \|\delta_{\pi,k}\| \|\delta_{r,k}\| = o_p(n^{-1/2})$. Under the blanket $o_p(n^{-1/4})$ individual rates of A5/C5 this holds automatically, so it carries no extra content as an assumption; we record it separately only because the corresponding moment-block display carries a $|\tau_0|$ factor that degenerates at $\tau_0 = 0$ on the inversion grid, and the τ_0 -free form is the one the score statistic needs. If the blanket rates were ever weakened to asymmetric per-nuisance rates, this display would regain independent content.

Lemma 10 (joint oracle equivalence and central limit theorem). *Assume A1–A7 (i.i.d.) or C1–C8 (clustered, with cluster sums throughout) together with K1, and write $\bar{\varphi}_i := \varphi_i(\eta_0) - \mathbb{E}_P \varphi_i(\eta_0)$. Then, uniformly over the class: (i) $n^{-1/2} \sum_i [\hat{\varphi}_i - \varphi_i(\eta_0)] = o_p(1)$ (both blocks; the Jacobian's drifting mean $\mathbb{E}_P q(\eta_0) = -g_{ZD}$ is a property of the oracle score, absorbed by the recentering in (ii), and cancels identically from this feasible-minus-oracle difference); (ii) $n^{-1/2} \sum_i \bar{\varphi}_i \Rightarrow N(0, V_P)$ (with $\lambda_{\min}(V_P) \geq c' > 0$ over the class, a consequence of K1 derived in the proof); (iii) $\hat{V}_{ff} - V_{ff} = o_p(1)$ and $\hat{V}_{\theta f} - V_{\theta f} = o_p(1)$.*

Proof. (ii): K1 forces a uniform eigenvalue floor on the stacked covariance. Write $x = (u, v)$; the Schur identity gives

$$x' V_P x = \left\| V_{ff}^{1/2} u + V_{ff}^{-1/2} V_{f\theta} v \right\|^2 + v' V_{\theta\theta.f} v,$$

and K1 supplies $\lambda_{\min}(V_{ff}) \geq c$, $\lambda_{\min}(V_{\theta\theta.f}) \geq c$, and $\|V_{f\theta}\| \leq C_V$ for a constant C_V (Cauchy–Schwarz on K1's moments). Fix a unit x and an $\varepsilon_0 \in (0, 1)$ small enough that $\sqrt{c}(1 - \varepsilon_0^2)^{1/2} - c^{-1/2} C_V \varepsilon_0 \geq \sqrt{c}/2$; such ε_0 depends only on c and C_V . If $\|v\| \geq \varepsilon_0$, the second term alone gives $x' V_P x \geq c \varepsilon_0^2$. If $\|v\| < \varepsilon_0$, then $\|u\|^2 > 1 - \varepsilon_0^2$ and the first term alone gives $x' V_P x \geq c/4$. Hence $\lambda_{\min}(V_P) \geq c' := \min(c \varepsilon_0^2, c/4) > 0$ over the class. Lemma 6 then applies to the stacked score exactly as in the proof of Lemma 7, with the sampling unit as

in the case at hand (observations, or cluster sums), $d = 2d_z$ fixed by C7, $c_0 = c'$, and third moments bounded by K1's $q > 4$ moments via Lyapunov's moment inequality. The convex-set approximation by $N(0, V_P)$ therefore holds at the root- G (or root- n) rate, uniformly over the class, which is the uniform form used in the proof of Theorem 3.

(i), ψ block: exactly Lemma 5 (Terms A, B, C ; in the i.i.d. case, the corresponding steps in the proof of Lemma 1, whose (L2) conditional-moment inputs K1 carries). Nothing new is needed.

(i), q block: for i in fold k 's inference set,

$$q_i(\hat{\eta}_k) - q_i(\eta_0) = -(\tilde{Z} - \delta_\pi)(\tilde{D} - \delta_r) + \tilde{Z}\tilde{D} = \tilde{Z}\delta_r(W) + \tilde{D}\delta_\pi(W) - \delta_\pi(W)\delta_r(W). \quad (\text{A.11})$$

For the first term, $\mathbb{E}[\tilde{Z} \mid W] = 0$ at π_0 and δ_r is $\mathcal{F}_k \vee \sigma(W)$ -measurable, so the term is conditionally mean zero, with conditional variance of the normalized sum at most $C\|\delta_{r,k}\|^2 = o_p(1)$, the constant supplied by K1's conditional second-moment bounds (clustered case: cluster sums and C1's size bounds, as in Lemma 5, with the centering under the size-conditional law given by $\mathbb{E}[\tilde{Z} \mid W, n_c] = 0$, C8(ii)). For the second term, $\mathbb{E}[\tilde{D} \mid W] = 0$ at r_0 (clustered case: $\mathbb{E}[\tilde{D} \mid W, n_c] = 0$, which is C8(ii); this is why C8(ii) includes the D equation and not only the Z equation), and the same argument runs with $\|\delta_{\pi,k}\|^2$. For the product term, $|n^{-1/2} \sum_i \delta_\pi \delta_r| \leq \sqrt{n} \|\delta_\pi\|_{N_k} \|\delta_r\|_{N_k} = \sqrt{n} \cdot o_p(n^{-1/2}) = o_p(1)$ by the product display of the remark above and the norm switch of Lemma 4. The q block thus needs no new mechanism: it is the Lemma 5 argument with the roles $(\zeta, \delta_\mu - \tau_0 \delta_r)$ replaced by $(\tilde{D}, \delta_r)$ and $(\tilde{Z}, \delta_\pi)$, a consequence of the score being linear in τ : the Jacobian is itself a doubly residualized cross moment, so the same orthogonality logic covers its nuisance drift.

(iii) is Lemma 8's argument applied to the stacked outer products: the $\Delta\varphi$ cross terms are controlled by the squared versions of the six displays above (each a product of two rates, or a rate times a bounded conditional moment), plus the law of large numbers for the oracle outer products under K1 ($q/2 > 2$ moments), uniformly over the class. The recentering of $\hat{V}_{\theta f}$ by $\hat{g}_D$ adds an $o_p(1) \cdot O_p(1)$ term via (i)–(ii). ■

Lemma 11 (Gaussian small ball and direction non-degeneracy). *Let c be K1's non-degeneracy floor $\lambda_{\min}(V_{\theta\theta.f}) \geq c$, and let $[c_f, C_f]$ denote the eigenvalue band of V_{ff} that K1 states as $[c, C]$; we rename it here because K1's letter c also names the non-degeneracy floor. Here $V_{\theta\theta} := \text{Var}_P(q(\eta_0))$, the $\theta\theta$ block of the joint covariance V_P of the centered score $\bar{\varphi}$. The clustered case replaces observation scores by cluster-summed scores and n by G throughout. Write $m_n^0 := n^{-1} \sum_i \psi_i(\eta_0)$, $g_D^0 := n^{-1} \sum_i q_i(\eta_0)$, $D_n^0 := g_D^0 - V_{\theta f} V_{ff}^{-1} m_n^0$, and $h_n := \sqrt{n} g_{ZD}(P)$. Then: (i) if $X \sim N(\mu, \Sigma)$ on $\mathbb{R}^d$ with $\lambda_{\min}(\Sigma) \geq \underline{\lambda} > 0$, then for every $\varepsilon > 0$,*

$$\sup_{\mu \in \mathbb{R}^d} P(\|X\| \leq \varepsilon) \leq \sqrt{\frac{2}{\pi \underline{\lambda}}} \varepsilon,$$

so in particular $P(X = 0) = 0$ and the direction $X/\|X\|$ is well defined almost surely, whatever the mean; (ii) under A1–A7 (or C1–C8) and K1, for every $\varepsilon > 0$ and every n ,

$$\sup_{P \in \mathcal{P}_n} P(\|\sqrt{n} D_n^0\| \leq \varepsilon) \leq \sqrt{\frac{2}{\pi c}} \varepsilon + C_{\text{BE}} n^{-1/2},$$

with C_{BE} depending only on d_z , K1's moment ceiling, c , and c_f ; consequently, for every $\varepsilon > 0$, $\limsup_n \sup_P P(\|\sqrt{n} \hat{D}_n\| \leq \varepsilon) \leq 2\sqrt{2/(\pi c)} \varepsilon$ and

$$\limsup_n \sup_P P(\hat{V}_{ff} \text{ singular, or } n \hat{D}'_n \hat{V}_{ff}^{-1} \hat{D}_n \leq \varepsilon^2/(2C_f)) \leq 2\sqrt{2/(\pi c)} \varepsilon.$$

In particular $\sup_P P(\hat{D}_n = 0 \text{ or } \hat{V}_{ff} \text{ singular}) \rightarrow 0$, so the direction of $\hat{D}_n$ is well defined, and K_n itself defined, with probability tending to one uniformly over the class and over all identification regimes at once; and for every $\delta > 0$ there is an $\varepsilon > 0$ keeping the scaled denominator $n \hat{D}'_n \hat{V}_{ff}^{-1} \hat{D}_n$ above $\varepsilon^2/(2C_f)$ with uniform asymptotic probability at least $1 - \delta$ (an ε -indexed small-ball family, not a fixed positive floor holding with probability tending to one); (iii) with $u_n := \hat{V}_{ff}^{-1/2} \hat{D}_n / \|\hat{V}_{ff}^{-1/2} \hat{D}_n\|$ and $u_n^0 := V_{ff}^{-1/2} D_n^0 / \|V_{ff}^{-1/2} D_n^0\|$, each defined on the event that its argument is nonzero (and, for u_n , that $\hat{V}_{ff}$ is positive definite), $u_n - u_n^0 = o_p(1)$ uniformly over the class.

Proof. (i) Fix any unit vector v . The event $\{\|X\| \leq \varepsilon\}$ is contained in $\{|v'X| \leq \varepsilon\}$, and $v'X \sim N(v'\mu, v'\Sigma v)$ with $v'\Sigma v \geq \underline{\lambda}$. A normal density is bounded by $(2\pi v'\Sigma v)^{-1/2}$, so the interval of length 2ε carries probability at most $2\varepsilon(2\pi\underline{\lambda})^{-1/2}$, a bound free of μ . Letting $\varepsilon \downarrow 0$ gives $P(X = 0) = 0$.

(ii) Write $\sqrt{n} D_n^0 = T_n^0 - h_n$ with $T_n^0 := n^{-1/2} \sum_i \xi_i$ and $\xi_i := (q_i(\eta_0) + g_{ZD}) - V_{\theta f} V_{ff}^{-1} \psi_i(\eta_0)$. The ξ_i are i.i.d. (clustered case: i.i.d. across clusters by C1) with $\mathbb{E}\xi_i = 0$, $\text{Var}(\xi_i) = V_{\theta\theta.f} \succeq cI$, and $\text{Cov}(\xi_i, \psi_i) = 0$: the orthogonalization removes the cross block exactly, so the argument runs on $V_{\theta\theta.f}$ alone and never on the full joint covariance V_P . Third absolute moments are bounded uniformly: $\|\xi_i\| \leq (1 + \|V_{\theta f}\|/c_f) \|\bar{\varphi}_i\|$ and K1 gives $q > 4$ moments for $\bar{\varphi}$, so $\sup_P \mathbb{E}_P \|\xi_i\|^3 \leq C_3 < \infty$ by Lyapunov's moment inequality. The ξ_i therefore satisfy every hypothesis of Lemma 6, with $d = d_z$, $c_0 = c$, and this C_3 ; since the closed ball $\bar{B}(h_n, \varepsilon)$ is convex, that lemma gives

$$P(T_n^0 \in \bar{B}(h_n, \varepsilon)) \leq P(Z_P \in \bar{B}(h_n, \varepsilon)) + C_{\text{BE}} n^{-1/2}, \quad Z_P \sim N(0, V_{\theta\theta.f}),$$

and part (i) applied to $X = Z_P - h_n$ bounds the first term by $\sqrt{2/(\pi c)} \varepsilon$. For the feasible version,

$$\sqrt{n}(\hat{D}_n - D_n^0) = \sqrt{n}(\hat{g}_D - g_D^0) - (\hat{V}_{\theta f} \hat{V}_{ff}^{-1} - V_{\theta f} V_{ff}^{-1}) \sqrt{n} \hat{m}_n - V_{\theta f} V_{ff}^{-1} \sqrt{n}(\hat{m}_n - m_n^0) = o_p(1)$$

uniformly: the first and third terms by Lemma 10(i) (the drifting mean cancels from the feasible-minus-oracle difference) with $\|V_{\theta f} V_{ff}^{-1}\|$ bounded by K1, the middle term by Lemma 10(iii) and the resolvent identity on K1's eigenvalue band, against $\sqrt{n} \hat{m}_n = O_p(1)$ uniformly (Lemma 10(i)–(ii) and Markov). Hence $\{\|\sqrt{n} \hat{D}_n\| \leq \varepsilon\} \subseteq \{\|\sqrt{n} D_n^0\| \leq 2\varepsilon\} \cup \{\|\sqrt{n}(\hat{D}_n - D_n^0)\| > \varepsilon\}$, which gives the first lim sup bound. By Lemma 10(iii) and Weyl's inequality, the eigenvalues of $\hat{V}_{ff}$ lie in $[c_f/2, 2C_f]$ with probability tending to one uniformly; on that event $\hat{V}_{ff}$ is nonsingular and $n \hat{D}'_n \hat{V}_{ff}^{-1} \hat{D}_n \geq \|\sqrt{n} \hat{D}_n\|^2 / \lambda_{\max}(\hat{V}_{ff}) \geq \|\sqrt{n} \hat{D}_n\|^2 / (2C_f)$, which gives the second lim sup bound with the exceptional event absorbed into it. The remaining statements follow by sending $\varepsilon \downarrow 0$ after $n \rightarrow \infty$.

(iii) For nonzero vectors, $\|x/\|x\| - y/\|y\|\| \leq 2\|x - y\|/\|y\|$. Apply it with $x = b_n := \hat{V}_{ff}^{-1/2} \sqrt{n} \hat{D}_n$ and $y = b_n^0 := V_{ff}^{-1/2} \sqrt{n} D_n^0$. On the event, of probability tending to one

uniformly by Lemma 10(iii) and Weyl, that the eigenvalues of $\hat{V}_{ff}$ lie in $[c_f/2, 2C_f]$, the map $A \mapsto A^{-1/2}$ is Lipschitz over this widened band with a constant depending only on c_f and C_f , so on that event

$$\|b_n - b_n^0\| \leq \|\hat{V}_{ff}^{-1/2} - V_{ff}^{-1/2}\| \|\sqrt{n} \hat{D}_n\| + \|V_{ff}^{-1/2}\| \|\sqrt{n}(\hat{D}_n - D_n^0)\| = o_p(1) \|\sqrt{n} D_n^0\| + o_p(1).$$

The absolute difference need not vanish: under strong identification $\|\sqrt{n} \hat{D}_n\|$ diverges with $\|h_n\|$. The relative bound does. Since $\|b_n^0\| \geq \|\sqrt{n} D_n^0\|/\sqrt{C_f}$, on the event $\{\|\sqrt{n} D_n^0\| > \varepsilon\}$,

$$\|u_n - u_n^0\| \leq 2\sqrt{C_f} \left(o_p(1) + \frac{o_p(1)}{\varepsilon} \right),$$

which is $o_p(1)$ uniformly for each fixed ε , while the exceptional event has $\limsup_n \sup_P$ probability at most $\sqrt{2/(\pi c)} \varepsilon$ by (ii). Sending $\varepsilon \downarrow 0$ after $n \rightarrow \infty$ completes the proof. $\blacksquare$

Theorem 3 (cross-fitted score statistic, uniform size). *Assume A1–A7 (i.i.d.) or C1–C8 (clustered, with cluster sums throughout), together with K1 and K3. Then*

$$\sup_{P \in \mathcal{P}_n} \left| P(K_n(\tau_0) > \chi_{1,1-\alpha}^2) - \alpha \right| \rightarrow 0,$$

with $\mathcal{P}_G$ in place of $\mathcal{P}_n$ and G in place of n in the clustered case.

Proof. Lemma 10 delivers precisely the two inputs of Kleibergen (2005). The joint central limit theorem for the (moment, recentered-Jacobian) vector is his Assumption 1 (unconditional recentering, pp. 1105–1106); his mixing conditions are offered there as sufficient conditions only, so the cross-fitted joint central limit theorem is an alternative sufficient route. K3 together with Lemma 10(iii) delivers his Assumption 2 (consistency of the covariance blocks, with the coherence property of eq. (14) and eq. (28)). Downstream of these inputs, his argument is distribution-level algebra that does not depend on how the moment was produced: his Lemma 1 (the block-triangular factorization, by which the limit of $\hat{D}_n$ is independent of the score limit regardless of the value or drift of the population Jacobian), his Theorem 1 (the normalized score is pivotal, $N(0, I_1)$), and the $\chi^2(1)$ statement recorded inside his Definition 1 (the χ^2 reference is stated there, not in a separately numbered theorem of that article).

Uniformity is obtained by rerunning the four-step argument of the proof of Theorem 2 on the stacked vector. Step 1 is Lemma 10(ii); Steps 2a–2b are Lemma 10(i) and (iii). Step 3, the statistic-equivalence step, closes through Lemma 11. Write $K_n(\tau_0) = (a_n' u_n)^2$ with $a_n := \hat{V}_{ff}^{-1/2} \sqrt{n} \hat{m}_n$ and u_n the unit vector of Lemma 11(iii); the rewriting is exact whenever $\hat{D}_n \neq 0$ and $\hat{V}_{ff}$ is nonsingular (both are needed for K_n to be defined at all), a joint event of probability tending to one uniformly by Lemma 11(ii), which also supplies the quantitative small-ball family keeping the scaled denominator $n \hat{D}_n' \hat{V}_{ff}^{-1} \hat{D}_n$ above any fixed threshold $\varepsilon^2/(2C_f)$ except on events of uniform asymptotic probability $O(\varepsilon)$. On that event $a_n - a_n^0 = o_p(1)$ uniformly, with $a_n^0 := V_{ff}^{-1/2} \sqrt{n} m_n^0$ the standardized oracle score (Lemma 10(i) and (iii); the score block carries no drifting mean), and $u_n - u_n^0 = o_p(1)$ uniformly (Lemma 11(iii)), so $K_n(\tau_0) = (a_n^0 u_n^0)^2 + o_p(1)$ uniformly over the class: the feasible statistic is the oracle statistic plus a uniform $o_p(1)$.

Step 4 is the conditioning step, run on the oracle pair along subsequences. Let $\{P_n\}$ be any sequence in the class and write $h_n := \sqrt{n} g_{ZD}(P_n)$, so that $\sqrt{n} D_n^0 = T_n^0 - h_n$ with T_n^0 the transformed oracle sum of the proof of Lemma 11(ii). Every subsequence has a further subsequence along which $V_{P_n} \rightarrow V_*$ (K1's moment ceiling and eigenvalue bounds are closed under such limits) and either (a) $h_n \rightarrow h \in \mathbb{R}^{d_z}$ or (b) $\|h_n\| \rightarrow \infty$ with $h_n/\|h_n\| \rightarrow u_0$ on the unit sphere. In case (a), note $(a_n^0, T_n^0) = n^{-1/2} \sum_i M_{P_n} \bar{\varphi}_i$ with $M_P := \begin{bmatrix} V_{ff}^{-1/2} & 0 \\ -V_{\theta f} V_{ff}^{-1} & I \end{bmatrix}$. Under each P_n the summands are i.i.d. and mean zero. Their covariance is exactly $M_{P_n} V_{P_n} M_{P_n}' = \text{diag}(I_{d_z}, V_{\theta\theta.f}(P_n))$, block diagonal because the orthogonalization removes the cross block identically, with eigenvalues at least $\min(1, c) > 0$ under K1. Third absolute moments are bounded uniformly over the class: $\|M_P\|$ is bounded by K1's moment ceiling and the c_f floor, and K1's $q > 4$ moments bound $\mathbb{E}\|\bar{\varphi}\|^3$ via Lyapunov's moment inequality. Lemma 6, applied with $d = 2d_z$ (fixed by C7), therefore bounds the convex-set distance between the law of (a_n^0, T_n^0) and $N(0, \text{diag}(I_{d_z}, V_{\theta\theta.f}(P_n)))$ by a constant multiple of $n^{-1/2}$, uniformly over the class. Along the subsequence the Schur complement is continuous on K1's eigenvalue band, so $V_{\theta\theta.f}(P_n) \rightarrow V_{*,\theta\theta.f}$ and the approximating Gaussians converge weakly to $N(0, \text{diag}(I_{d_z}, V_{*,\theta\theta.f}))$; the triangle inequality then gives $(a_n^0, T_n^0) \Rightarrow (a, T)$, jointly Gaussian with $a \sim N(0, I_{d_z})$ independent of $T \sim N(0, V_{*,\theta\theta.f})$ (a nonsingular Gaussian puts zero mass on the boundary of any convex set, so convex-set approximation suffices for weak convergence). By Lemma 11(i) the vector $T - h$ is nonzero almost surely, so $(a, x) \mapsto (a'x/\|x\|)^2$ is continuous almost everywhere under the limit law, and continuous mapping gives $(a_n^0 u_n^0)^2 \Rightarrow (a'u)^2$ with $u := V_{*,ff}^{-1/2}(T - h)/\|V_{*,ff}^{-1/2}(T - h)\|$ a function of T alone, hence independent of a . Conditional on u , $a'u \sim N(0, 1)$: the projection of a standard Gaussian vector onto an independent one-dimensional span is exactly $\chi^2(1)$, a distributional identity, so $(a_n^0 u_n^0)^2 \Rightarrow \chi_1^2$. In case (b) the direction is asymptotically deterministic: $T_n^0 = O_p(1)$ uniformly while $\|h_n\| \rightarrow \infty$, so the perturbation bound in the proof of Lemma 11(iii) gives $u_n^0 = -V_{*,ff}^{-1/2} u_0/\|V_{*,ff}^{-1/2} u_0\| + o_p(1)$, and Slutsky gives $(a_n^0 u_n^0)^2 \Rightarrow \chi_1^2$ directly. In both cases Step 3, Slutsky, and the continuity of the χ_1^2 distribution function at its quantile give $P_n(K_n(\tau_0) > \chi_{1,1-\alpha}^2) \rightarrow \alpha$ along the subsequence. A sequence $\{P_n\}$ attaining the sup-over-class deviation within n^{-1} therefore has every subsequence containing a further subsequence along which the deviation vanishes, so the supremum itself vanishes. Kleibergen handles the degenerate corner instead by regime bookkeeping (the $T^{-\nu}$ scaling in the published proof of his Theorem 1); here the small-ball bound of Lemma 11(i) is uniform in the mean, so the strong, weak, and zero-relevance regimes are covered by one bound, and the excluded corner ($\zeta \propto \bar{D}$, collinear score and Jacobian, $V_{\theta\theta.f}$ singular) is exactly the corner K1 removes. ■

Remark (relation to the published result). The uniform statement is a strengthening relative to Kleibergen (2005), whose identification-robustness is proved pointwise across regimes of the population Jacobian; no uniform-over-laws language appears in the published article. The clustered statement is likewise this paper's: the published article contains no panel or cluster asymptotics (a panel section in its 2001 working-paper predecessor was dropped in publication), and the clustered case above is obtained from the Theorem 2 machinery.

Remark (just-identified case, and the over-identified cell). At $d_z = 1$ the projection in K_n is the identity almost surely and $K_n = AR_n$ (on the null event $\hat{D}_n = 0$ we set $K_n := AR_n$, its continuous extension): Theorem 3 adds content only for the over-identified case, in this paper the Angrist–Krueger cell. There the clustered-score asymptotics inherit the $G = 51$ boundary caveat of the Theorem 2 remark, aggravated at $d_z = 30$ by a 30×30 cluster meat built from 51 cluster scores; the clean theoretical case for the score statistic is an over-identified design with many clusters.

Remark (the companion J statistic). Kleibergen (2005, eq. (29)) defines a companion J statistic with a $\chi^2_{d_z-1}$ limit, asymptotically independent of K_n , and his eq. (30) records the decomposition of the Stock–Wright S statistic as $S = K + J$; the same transfer argument applies to it. (The name “JKLM” for this statistic belongs to the later literature, not to the 2005 article.) The over-identification diagnostic reported in this paper is instead the grid-minimized objective $J_{CUE} = \min_{\tau} AR(\tau)$ of §3.2; its conservative χ^2 reading rests on the inequality $\min_{\tau} AR(\tau) \leq AR(\tau_0)$ together with Theorem 2 (Theorem 1 in the i.i.d. case) applied at τ_0 ; the grid minimum lies weakly above the exact minimum, so numerical minimization error can only enlarge the reported value, and a non-rejection read from it survives exact minimization.

Remark (the LM and conditional-QLR statistics as reported). In the over-identified cell’s reporting (§3.2), the Kleibergen (2005) score (LM) statistic directs power at the τ direction (with known power dips away from the null); the conditional QLR statistic, in Kleibergen’s (2005) GMM extension of Moreira’s (2003) CLR, adapts between the score and the full-vector AR given its conditioning statistic.

A.6 Weighted and generated-instrument extensions

The two extensions are deliberately proved by reduction: Proposition 4 reduces to Theorems 1–3 for the weighted score, and Proposition 5 reduces to Theorem 2 with an enlarged nuisance vector, its score clause being the just-identified identity.

Survey-weighted target laws. Known sampling weights $\omega_i = \omega(U_i) > 0$ are a measurable function of observables U_i , which may include but need not be limited to the controls W_i , with $0 < c_{\omega} \leq \omega \leq C_{\omega} < \infty$. The weighted target law is P_{ω} with $dP_{\omega}/dP = \omega/\mathbb{E}_P[\omega]$. Weighted nuisances are the P_{ω} -conditional means, $\mu_{\omega}(W) := \mathbb{E}_{P_{\omega}}[Y | W] = \mathbb{E}_P[\omega Y | W]/\mathbb{E}_P[\omega | W]$, and r_{ω}, π_{ω} analogously; weighted residuals are $\tilde{Y}^{\omega} := Y - \mu_{\omega}(W)$, $\tilde{D}^{\omega}$, $\tilde{Z}^{\omega}$, and the weighted score at a tested value t is $\psi^{\omega}(t) := \omega \tilde{Z}^{\omega}(\tilde{Y}^{\omega} - t \tilde{D}^{\omega})$. The null hypothesis is primitive and per tested value, because the AR framework must not presume a well-defined solved estimand under weak identification, where the weighted relevance can be zero or drifting:

$$\text{W-ID}(t) : \quad \mathbb{E}_P[\omega(\tilde{Y}^{\omega} - t \tilde{D}^{\omega}) | Z, W] = 0. \quad (\text{A.12})$$

Under strong identification the solved weighted estimand τ_{ω} , the unique root of $\mathbb{E}_P[\psi^{\omega}(\tau)] = 0$ (the τ_{ω} of §4.2 and §4.4), exists, and W-ID(τ_{ω}) is the usual weighted exclusion restriction; that interpretation is confined to this sentence and does not enter the hypotheses of Proposition 4.

Two regimes are distinguished. *Regime (a):* $\omega = \omega(W)$, weights are functions of the included controls. Then dP_{ω}/dP depends on W only, so the conditional law of (Y, D, Z) given W is identical under P and P_{ω} ; hence $\mu_{\omega} = \mu_0$, $r_{\omega} = r_0$, $\pi_{\omega} = \pi_0$, the weighted structural

error coincides with ζ , and W-ID(t) reduces to the unweighted exclusion restriction. In this regime unweighted nuisance estimation is consistent for the weighted nuisances, and a weighted-moment, unweighted-nuisance implementation is valid; regime (a) is the formal statement of when that invariance holds. *Regime (b): general ω , depending on variables beyond W .* Then $\mu_\omega \neq \mu_0$ in general, and the nuisances must be estimated under the weighted criterion (weighted empirical risk in the estimation and selection subsamples), with rates stated in $L^2(P_\omega)$, equivalent to $L^2(P)$ up to the constants c_ω, C_ω by boundedness.

Proposition 4 (survey-weighted target laws). *Assume bounded weights, W-ID(t) at the tested value t , and either (regime (b)) the package A1–A7 (i.i.d.) or C1–C8 (clustered) with every nuisance object replaced by its P_ω version and rates in $L^2(P_\omega)$, or (regime (a)) the original unweighted package plus $\omega = \omega(W)$. In the clustered regime (b), C8(i)–(ii) are replaced by their weighted analogues: W-ID(t) holds conditional on size, $\mathbb{E}_P[\omega(\tilde{Y}^\omega - t\tilde{D}^\omega) \mid Z, W, n_c] = 0$, and the weighted nuisances are stable under size conditioning, $\mathbb{E}_P[\omega Z \mid W, n_c] / \mathbb{E}_P[\omega \mid W, n_c] = \pi_\omega(W)$ and likewise for D and Y , equivalently $\mathbb{E}_P[\omega \tilde{Z}^\omega \mid W, n_c] = 0$ and $\mathbb{E}_P[\omega \tilde{D}^\omega \mid W, n_c] = 0$. The moment bounds of C8(iii) are imposed on the ω -weighted residuals; bounded weights transfer them from the unweighted forms up to C_ω^2 . C8(iv) is unchanged: it constrains only the control marginals $\bar{P}_s, \bar{P}$, and the $L^2(P_\omega)$ norms are equivalent to $L^2(P)$ norms up to c_ω, C_ω , so the norm switch holds with constant $C_s C_\omega / c_\omega$. In regime (a), $\omega = \omega(W)$ factors out of every (Z, W, n_c) -conditional expectation and C8 itself suffices; no weighted analogue is needed. In the clustered case, either regime, assume the weighted non-degeneracy condition (C2 $_\omega$): with $S_c^\omega(t) := \sum_{i \in c} \psi_i^\omega(t)$ the weighted cluster score (each ψ_i^ω carries its own weight ω_i), the eigenvalues of $\Sigma_P^\omega(t) := \mathbb{E}_P[S_c^\omega(t) S_c^\omega(t)']$ lie in $[c_\Omega, C_\Omega] \subset (0, \infty)$ at the tested value t , uniformly over $\mathcal{P}_G$, in the per-cluster normalization of C2. In regime (b) this is C2's eigenvalue clause read on the weighted problem; in regime (a) it is a genuine addition to the unweighted package, required only in the clustered case (the proof shows the i.i.d. floor transfers automatically). Then the conclusions of Theorem 1 (i.i.d.) and Theorem 2 (clustered) hold for the weighted statistics built from ψ^ω , and so does Theorem 3 under K1 and K3 imposed on the weighted stacked score and Jacobian, $\varphi^\omega := (\psi^{\omega'}, q^{\omega'})'$ with $q^\omega := -\omega \tilde{Z}^\omega \tilde{D}^\omega$, cluster-summed in the clustered case: the weighted AR statistic with weighted meat (cluster-summed weighted scores in the clustered case) has uniform asymptotic size α at every (P, t) pair in the class satisfying W-ID(t), and the inverted set covers any such t with probability tending to $1 - \alpha$.*

Proof. Boundedness transfers all moment upper bounds, $\mathbb{E}_P|\omega h|^q \leq C_\omega^q \mathbb{E}_P|h|^q$, and the squared norms satisfy $\|h\|_{L^2(P_\omega)}^2 \in [c_\omega / \mathbb{E}_P \omega, C_\omega / \mathbb{E}_P \omega] \cdot \|h\|_{L^2(P)}^2$. The three oracle-equivalence terms of Lemma 5 carry the factor ω . Term A becomes $-\delta_\pi \cdot \omega(\tilde{Y}^\omega - t\tilde{D}^\omega)$, conditionally mean zero by W-ID(t), in the clustered regime (b) in the size-conditional form stated in the premise (in regime (a), by $\mathbb{E}[\zeta \mid Z, W] = 0$, since $\omega(W)$ factors out of the conditional expectation, with C8(i) supplying the size-conditional form in the clustered frame). Term B becomes $-\omega \tilde{Z}^\omega(\delta_\mu - t\delta_r)$, conditionally mean zero because $\mathbb{E}_P[\omega \tilde{Z}^\omega \mid W] = \mathbb{E}_P[\omega \mid W] (\mathbb{E}_{P_\omega}[Z \mid W] - \pi_\omega(W)) = 0$ by construction of π_ω , in the clustered regime (b) by the size-conditional weighted stability of the premise, $\mathbb{E}_P[\omega \tilde{Z}^\omega \mid W, n_c] = 0$. Term C is bounded by C_ω times the unweighted product bound, with the norm switch of Lemma 4 carrying the constant $C_s C_\omega / c_\omega$ recorded in the premise. Lemmas 7 and 8 also need C2's eigenvalue bounds for the score they

are run on, and positive bounded weights transfer only the moment upper bounds, not the cluster-level floor: within-cluster weight heterogeneity can change the cancellation structure of the cluster sum, so the floor is not inherited from the unweighted score and is assumed separately as $C2_\omega$. In the i.i.d. case no separate floor is needed: writing $\psi^\omega = \omega \check{\psi}$ with $\check{\psi} := \tilde{Z}^\omega(\tilde{Y}^\omega - t\tilde{D}^\omega)$, the pointwise semidefinite order $\omega^2 \check{\psi} \check{\psi}' \succeq c_\omega^2 \check{\psi} \check{\psi}'$ transfers the eigenvalue floor of $\mathbb{E}_P[\check{\psi} \check{\psi}']$, A2/C2's clause read at the tested value, with constant c_ω^2 . Under these bounds the central limit theorem (Lemma 7) applies to the weighted scores, or their cluster sums, with the transferred moment bounds; variance consistency (Lemma 8) likewise; and the four-step closure is unchanged. For Theorem 3, K1 and K3 are read on the weighted stacked vector φ^ω , as the statement records, and the weighted Jacobian $q^\omega = -\omega \tilde{Z}^\omega \tilde{D}^\omega$ satisfies the three q -block arguments of Lemma 10 with the same factor. ■

Remark (OHIE placement). The OHIE weights are not household-constant: in the analysis sample, 92 of the 20,747 households show within-household weight variation. No within-cluster constancy is assumed or needed: weights enter at observation level, and the cluster score is $\sum_{i \in c} \omega_i \psi_i$, which is all the reduction uses. Whether OHIE sits in regime (a) is settled by the weight's construction: the design strata (household size) are in W , but the official weight is built with survey-wave and draw components that are not, so the weights are not a function of the controls alone. The §4.4 diagnostic (a regression of the weight on W , with $R^2 = 0.03$) is consistent with the construction. The primary OHIE specification accordingly takes the safe route, regime (b): the weighted target law is carried through every nuisance fit and selection loss (§4.4), with the regime-(a) regression reported as the diagnostic that would license the unweighted-nuisance shortcut if its residual variation were near zero.

Remark (estimand). τ_ω differs from the unweighted estimand whenever effects are heterogeneous across the weighted strata; this is the design of the §4.2 sixth arm, where the weighted AR statistic covers τ_ω and an unweighted statistic centered on the unweighted estimand does not.

Generated fold-nested cluster-jackknife instruments. The examiner design (§4.5) has court-by-date (bond-hearing shift) clusters c , a judge identifier J_i that is part of the conditioning data, and a leniency instrument that is constructed, not observed. Under the fold-nested implementation, for outer fold k the construction map $\hat{\ell}_k(\cdot)$, the fixed-effect residualization composed with the leave-one-cluster-out projection, is fitted on the training complement of fold k only. The construction involves no selection step, so it uses the full $(K-1)$ -fold training complement (the fixed-nuisance convention), with the leave-out applied within the training set during fitting; an estimation-subsample-only variant would apply if the construction itself were ever subject to selection. The instrument value for an inference observation is $\hat{Z}_i = \hat{\ell}_k(J_i)$: evaluation uses only the own-cluster conditioning variables (J_i, W_i) , never own- or other-inference-cluster treatment data. Either variant is $\mathcal{F}_k$ -measurable, which is all the proof uses. The population target is a leniency function ℓ_0 with $\pi_0(W) := \mathbb{E}_P[\ell_0(J) \mid W]$, so the instrument-side residual $\tilde{Z}^\ell := \ell_0(J) - \pi_0(W)$ is conditionally mean zero given W by construction.

(G-EX) exclusion. $\mathbb{E}_P[\zeta \mid J, W] = 0$: conditional quasi-random assignment and exclusion of the judge identity given the controls. This is the examiner design's identifying restriction, stated explicitly here because the generated instrument makes J an argument of the score. In the clustered frame the condition is imposed in its size-conditional form, $\mathbb{E}_P[\zeta \mid J, W, n_c] = 0$:

conditional quasi-random assignment and exclusion given the controls and the shift caseload. The instrument-side residual centering $\mathbb{E}_P[\tilde{Z}^\ell \mid W, n_c] = 0$ then follows from C8(ii) applied to $\ell_0(J)$ in the role of Z .

Structure. $\hat{Z}$ enters the score only through the residualized instrument $\hat{\tilde{Z}} = \hat{Z} - \hat{\pi}_k(W)$. Writing $\delta_{\ell,k} := \hat{\ell}_k - \ell_0$ and $\delta_{\pi,k} := \hat{\pi}_k - \pi_0$,

$$\hat{\tilde{Z}} = \tilde{Z}^\ell - (\delta_{\pi,k} - \delta_{\ell,k}), \quad (\text{A.13})$$

so the instrument-side estimation error is $\delta_{\ell,k} - \delta_{\pi,k}$: the generated instrument is formally a fourth cross-fitted nuisance, entering every display of the AR chain exactly where $-\delta_{\pi,k}$ already enters.

Proposition 5 (generated fold-nested cluster-jackknife instrument). *Assume C1–C8 (clustered; court-by-date shift clusters) and G-EX, with the nuisance vector enlarged to $\eta = (\mu, r, \pi, \ell)$: the additional rate $\max_k \|\delta_{\ell,k}\| = o_p(n^{-1/4})$, the product-rate display of C5(ii) augmented by $\|\delta_{\ell,k}\|$ alongside $\|\delta_{\pi,k}\|$, and the fold-nested construction, so that $\hat{\ell}_k$ is $\mathcal{F}_k$ -measurable (C4 for the enlarged vector). Then Theorem 2 holds for the score built from the generated instrument: the cross-fitted AR set for τ_0 retains uniform asymptotic size. At $d_z = 1$, the case of every generated-instrument application in this paper, the same guarantee extends to the score (LM) set, because the score statistic coincides with the AR statistic in the just-identified case (§3.2; the just-identified remark of A.5), so the score conclusion follows from the AR conclusion already proved; for $d_z > 1$ no generated-instrument score claim is made. No lower bound on the limit relevance $\text{Cov}(\tilde{Z}^\ell, \tilde{D})$ is imposed: relevance enters only through the drift class, so identification-robustness is preserved exactly as with an observed instrument.*

Proof. Replace $\tilde{Z}$ by $\tilde{Z}^\ell$ throughout; the feasible residual is (A.13). Lemma 2 applies to the enlarged $(\hat{\gamma}_k, \hat{\eta}_k)$: the fold-nested construction is precisely what makes $\hat{\ell}_k$ $\mathcal{F}_k$ -measurable. Lemma 5's three terms hold with $\delta_{\pi,k}$ replaced by $\delta_{\pi,k} - \delta_{\ell,k}$: Term *A*'s conditional mean-zero step uses G-EX ($\mathbb{E}[\zeta \mid J, W] = 0$), with $\delta_{\ell,k}$ measurable with respect to $\mathcal{F}_k \vee \sigma(J_i, W_i)$; Terms *B* and *C* bound $\|\delta_{\pi,k} - \delta_{\ell,k}\| \leq \|\delta_{\pi,k}\| + \|\delta_{\ell,k}\|$ and apply the augmented product rate. Lemmas 7 and 8 and the four-step closure are untouched: the oracle score uses ℓ_0 , an observed-instrument object. The score clause at $d_z = 1$ needs no separate argument: the statistic coincides with the AR statistic there, so the AR conclusion is the score conclusion. ■

Remark (the score statistic at $d_z > 1$, open). The restriction to $d_z = 1$ is substantive, not notational. For the score statistic the q -block expansion (A.11) must be run with $\hat{\tilde{Z}}$ from (A.13), which replaces the term $\tilde{D} \delta_{\pi,k}$ by $\tilde{D} (\delta_{\pi,k} - \delta_{\ell,k})$; the $\delta_{\ell,k}$ part is a function of (J, W) , so the conditional-centering step would need $\mathbb{E}_P[\tilde{D} \mid J, W] = 0$. No assumption here supplies that display, and a relevant constructed instrument contradicts it, since the first stage makes $\tilde{D}$ covary with $\ell_0(J)$; G-EX restricts the structural error only. A generated-instrument score theorem for $d_z > 1$ therefore needs a joint condition on the construction error and $\tilde{D}$, and is left open.

Remark (what the rate asks of the design). $\|\delta_{\ell,k}\| = o_p(n^{-1/4})$ is a leniency-estimation rate. With bounded cluster sizes and $n \asymp G$, the parametric-in- G rate $G_{\text{est}}^{-1/2} = o(n^{-1/4})$ holds automatically ($G^{-1/2} = o(G^{-1/4})$). The caveat is the judge dimension: this presumes a

fixed or slowly growing number of judges, and with J_n judges the fixed-effect projection error scales like $(J_n/n)^{1/2}$, so the rate condition is $J_n = o(\sqrt{n})$.

Remark (why fold-nesting is a hypothesis). Without fold-nesting, a full-sample jackknife construction is not $\mathcal{F}_k$ -measurable and Lemma 2 fails for the enlarged vector; the alternative route would be an explicit dependence analysis of the full-sample jackknife, which we do not pursue. This is exactly why the implementation rebuilds the construction inside each outer training set (§4.5).

Remark (own-cluster leave-out). The leave-one-cluster-out projection still matters inside the training set: it keeps $\hat{\ell}_k$ well behaved, with no own-cluster treatment data entering the projection for the training clusters used to fit it; for inference clusters the fold structure itself provides the exclusion.

Appendix B. Full result tables

These tables report the per-cell design and nuisance diagnostics, the Monte-Carlo evaluations, and the supplementary panels summarized in the main text; every value is the frozen artifact produced by the replication pipeline.

Table B.1: Per-cell design and nuisance diagnostics (the companion to the headline panel in Table 1 of the article). Each block reports what the cell’s frozen artifact stores: the per-fold nested-selection winners for the outcome nuisance $\hat{\mu}$ and treatment nuisance $\hat{\tau}$, the clustering fields, the weighting status, and, per DML learner, the out-of-fold R^2 of the instrument-residualization fit $\hat{\pi}$, the residualized-variance ratio $\text{Var}(\tilde{Z})/\text{Var}(Z)$, and the SE type. An em-dash marks a field the artifact does not store. The Angrist–Krueger block reports the moment-variance diagnostics for each instrument block: $\text{rank}(\hat{\Omega})$, the condition number of $\hat{\Omega}$, and the fold-cluster count.

DML learner	$R^2_{\text{of}}(\hat{\pi})$	$\text{Var}(\tilde{Z})/\text{Var}(Z)$	SE type
<i>OHIE (lottery, randomized negative control), N = 23,441</i>			
clusters: 20,747; OLS and 2SLS SE type: cluster			
per-fold $\hat{\mu}$ winners: linear, ridge, gb, ridge, linear; per-fold $\hat{\tau}$ winners: linear, gb, gb, gb, gb			
unweighted-panel winners, $\hat{\mu}$: linear, lasso, gb, ridge, linear; $\hat{\tau}$: gb, gb, gb, gb, gb			
weights: primary panel is weighted-responder; nuisance criterion: weighted risk for the weighted panel (weighted-target-law path);			
unweighted risk for the robustness panel			
DML (linear)	0.009	0.991	cluster
DML (ridge)	0.009	0.991	cluster
DML (RF)	-0.053	1.053	cluster
DML (GB)	0.005	0.995	cluster
<i>Examiner (judge leniency, CJIVE) (decoupled), N = 91,282</i>			
clusters (court-by-date bond-hearing shifts): 1,831 shifts, 145 judges; OLS and 2SLS SE type: cluster			
per-fold $\hat{\mu}$ winners: gb, gb, gb, gb, gb; per-fold $\hat{\tau}$ winners: gb, gb, gb, gb, gb			
DML (linear)	-0.001	1.000	cluster
DML (ridge)	0.000	1.000	cluster
DML (RF)	-0.259	1.259	cluster
DML (GB)	-0.030	1.030	cluster
<i>CollegeDistance (distance, decoupled nuisances, cluster-aware on state), N = 4,739</i>			
cluster count: —; OLS SE type: HC0; 2SLS SE type: robust			
per-fold $\hat{\mu}$ winners: —; per-fold $\hat{\tau}$ winners: —			
DML (linear)	-0.027	1.015	cluster
DML (ridge)	-0.024	1.012	cluster
DML (RF)	-0.065	1.052	cluster
DML (GB)	0.022	0.968	cluster
<i>Card 1995 (nearc4, returns to schooling), N = 3,010</i>			
cluster count: —; OLS and 2SLS SE type: robust			
per-fold $\hat{\mu}$ winners: —; per-fold $\hat{\tau}$ winners: —			
DML (linear)	0.242	0.758	HC0
DML (ridge)	0.242	0.758	HC0
DML (RF)	0.235	0.765	HC0
DML (GB)	0.264	0.736	HC0
<i>Angrist–Krueger 1991 (quarter of birth, over-identified), N = 329,509</i>			
cluster: state of birth; SE type on all estimator rows: cluster; per-learner $\hat{\pi}$ diagnostics: —			
$d_z = 3$: $\text{rank}(\hat{\Omega}) = 3$, $\text{cond}(\hat{\Omega}) = 6.6$, rank-deficient: no, fold clusters: 51			
$d_z = 30$: $\text{rank}(\hat{\Omega}) = 30$, $\text{cond}(\hat{\Omega}) = 166.2$, rank-deficient: no, fold clusters: 51			

Table B.2: Monte-Carlo coverage evaluation: cross-fitted Anderson–Rubin vs. Wald coverage across simulation regimes. Each row is one Monte-Carlo operating point with its own independent seed stream; every coverage entry prints its stored Monte-Carlo standard error. In the clustered-DGP rows the AR column is the *cluster*-robust set (what the examiner/OHIE cells use); the naive heteroskedasticity-only AR under-covers there (0.69 (MCSE 0.0146) and 0.51 (MCSE 0.0158)), so the cluster-robust variance is required. In the survey-weighted rows the AR is the *weighted* set covering the survey-population estimand τ_w ; a naive unweighted AR covers τ_w only 0.59 (MCSE 0.0156) and 0.09 (MCSE 0.0091) of the time (it targets the wrong, unweighted estimand τ_u). The combined arm runs all four pipeline components at once (cluster-aware folds, fold-nested constructed instrument, nested learner selection, cluster-robust inference), with an observed-instrument control row isolating the construction-error channel.

Regime	d_z	median eff.- F	AR cov.	AR MCSE	Wald cov.	Wald MCSE	reps
$d_z = 1$ <i>oracle nuisances</i>							
	1	1.11	0.954	0.0038	0.987	0.0021	3,000
	1	2.11	0.950	0.0040	0.988	0.0020	3,000
	1	5.63	0.953	0.0039	0.970	0.0031	3,000
	1	14.12	0.954	0.0038	0.969	0.0032	3,000
	1	41.00	0.943	0.0042	0.954	0.0038	3,000
	1	131.93	0.960	0.0036	0.958	0.0036	3,000
	1	488.44	0.946	0.0041	0.943	0.0042	3,000
$d_z = 1$ <i>estimated (linear) nuisances</i>							
	1	1.99	0.957	0.0064	0.984	0.0040	1,000
	1	5.80	0.955	0.0066	0.970	0.0054	1,000
	1	14.14	0.947	0.0071	0.956	0.0065	1,000
	1	76.98	0.951	0.0068	0.952	0.0068	1,000
$d_z = 1$ <i>estimated (RF, nonlinear) nuisances</i>							
	1	8.39	0.951	0.0068	0.967	0.0056	1,000
	1	29.09	0.949	0.0070	0.959	0.0063	1,000
$d_z = 1$ <i>nonlinear nuisances</i>							
	1	2.99	0.968	0.0111	0.988	0.0069	250
	1	9.43	0.960	0.0124	0.980	0.0089	250
$d_z = 5$ <i>over-identified</i>							
	5	1.11	0.951	0.0048	0.914	0.0063	2,000
	5	1.60	0.955	0.0046	0.912	0.0063	2,000
	5	3.91	0.947	0.0050	0.931	0.0056	2,000
	5	28.59	0.946	0.0051	0.947	0.0050	2,000
$d_z = 3$ <i>over-identified</i>							
	3	4.21	0.946	0.0065	—	—	1,200
	3	14.81	0.963	0.0055	—	—	1,200
	3	36.49	0.938	0.0069	—	—	1,200
$d_z = 30$ <i>over-identified</i>							
	30	4.52	0.958	0.0058	—	—	1,200
	30	15.25	0.947	0.0065	—	—	1,200
	30	37.06	0.950	0.0063	—	—	1,200
$d_z = 3$ <i>over-identified, estimated nuisances</i>							
	3	14.96	0.955	0.0060	—	—	1,200
$d_z = 30$ <i>over-identified, estimated nuisances</i>							
	30	15.21	0.953	0.0061	—	—	1,200
$d_z = 1$ <i>clustered DGP (cluster-robust AR)</i>							
	1	8.52	0.951	0.0068	0.961	0.0061	1,000
	1	6.05	0.948	0.0070	0.946	0.0071	1,000
$d_z = 1$ <i>survey-weighted DGP (weighted AR of τ_w)</i>							
	1	42.89	0.946	0.0071	—	—	1,000
	1	153.94	0.947	0.0071	—	—	1,000
<i>combined: clustered + constructed instrument + nested selection</i>							
constructed, $n = 1,440$	1	2.01	0.912	0.0127	0.924	0.0119	500
observed Z , $n = 1,440$	1	9.58	0.956	0.0092	0.960	0.0088	500
constructed, $n = 1,440$	1	24.84	0.944	0.0103	0.954	0.0094	500
observed Z , $n = 1,440$	1	53.05	0.952	0.0096	0.944	0.0103	500
constructed, $n = 4,800$	1	42.81	0.944	0.0103	0.966	0.0081	500
observed Z , $n = 1,440$	1	2.00	0.936	0.0109	0.962	0.0086	500

Table B.3: Identification-aware selector benchmark (300 paired replications per regime, correctly specified nuisances, common random numbers). For each regime we compare the MSE rule against the identification-aware (IACV) rule. Columns report median effective- F , bias, RMSE, mean Wald CI length, and AR coverage with its Monte-Carlo standard error in parentheses. The paired row reports the paired mean signed-error difference (under bias) and the paired mean absolute-error difference (under RMSE), IACV minus MSE, each with its standard error: every paired difference is within Monte-Carlo uncertainty.

Regime	Rule	median eff.- F	bias	RMSE	Wald CI len.	AR cov.
A (weak)	MSE	3.78	0.0829	1.9305	21.3289	0.967 (0.010)
	IACV	3.64	-0.0379	1.5220	69.4526	0.957 (0.012)
	paired Δ (IACV-MSE)		-0.1208 (0.0882)	-0.0499 (0.0592)		
B (moderate)	MSE	10.68	-0.0066	0.5635	2.1215	0.960 (0.011)
	IACV	10.73	-0.0349	0.4934	1.8019	0.963 (0.011)
	paired Δ (IACV-MSE)		-0.0283 (0.0165)	0.0013 (0.0166)		
C (do-no-harm, low absorption)	MSE	23.10	-0.0248	0.2385	0.8985	0.940 (0.014)
	IACV	22.75	-0.0251	0.2410	0.9028	0.933 (0.014)
	paired Δ (IACV-MSE)		-0.0004 (0.0022)	0.0006 (0.0021)		

Table B.4: Card (1995) control-richness ladder (instrument: `nearc4` ($d_z = 1$), $N = 2,182$). As controls accumulate, the linear and RF first-stage effective- F collapse and the AR set widens until, at the richest rungs, it is the whole line, even though the out-of-fold partial- R^2 and $\text{Var}(\tilde{Z})$ are nearly flat.

Rung	eff.- F (lin. π)	eff.- F (RF π)	f.s. slope	$\text{Var}(\tilde{Z})$	R^2_{of}	AR set
lean	7.50	6.08	0.2601	0.1650	0.23	bounded
+family	5.48	4.16	0.2186	0.1654	0.23	bounded
+parentEd	1.85	1.46	0.1215	0.1657	0.23	whole line
+KWW	3.52	2.29	0.1497	0.1658	0.23	whole line
+poly2	2.51	1.99	0.1235	0.1754	0.18	whole line

Table B.5: OHIE Lee (2009) trimming bounds on the treatment effect under differential survey response. Each panel reports the trimming fraction, the trimmed arm, treatment/control response rates, the first-stage compliance, and the resulting bounds on the intention-to-treat and instrumented (IV) effects. Lee trimming removes mass from the arm with the higher response rate. The weighted panel applies the survey weights of the main specification and is the primary reading; under those weights the control arm responds slightly more and is trimmed. The unweighted panel is a sensitivity reference for the unweighted responder sample. The IV rows divide the trimmed ITT bounds by the untrimmed first-stage compliance; the numerator bounds the ITT for the always-responder population while the denominator is computed on the untrimmed sample, so the two refer to different populations and the ratio constitutes a bound on the instrumented effect only under the assumptions of Lee’s framework. In particular the IV bound is a complier-average object only under its monotonicity condition, which the paper does not otherwise impose (notes to Table 2 of the article).

Quantity	Value
<i>Panel A: weighted (primary)</i>	
Trimming fraction q	0.022
Trimmed arm	control
Response rate, treated	0.495
Response rate, control	0.506
Response rate gap (treated – control)	-0.011
First-stage compliance	0.291
ITT point estimate	0.268
ITT Lee bound $[_{-}, -]$	[0.2247, 0.5957]
Instrumented (IV) effect, point	0.923
Instrumented (IV) effect, Lee bound $[_{-}, -]$	[0.773, 2.050]
<i>Panel B: unweighted (sensitivity)</i>	
Trimming fraction q	0.331
Trimmed arm	selected
Response rate, treated	0.396
Response rate, control	0.265
Response rate gap (treated – control)	0.131
First-stage compliance	0.294
ITT point estimate	0.249
ITT Lee bound $[_{-}, -]$	[−1.1663, 1.2767]
Instrumented (IV) effect, point	0.846
Instrumented (IV) effect, Lee bound $[_{-}, -]$	[−3.961, 4.336]

Table B.6: Angrist–Krueger (1991) over-identified inference. Top panel: OLS, 2SLS, and DML point estimates with weak-IV-robust AR/LM/QLR sets and the J_{CUE} (min-AR) over-identification statistic for each instrument block. Bottom panel: Monte-Carlo AR/LM/QLR coverage and J_{CUE} rejection (size, χ^2_{r-1} strong-identification reference) at $d_z = 3, 30$ from the over-identified Monte-Carlo coverage evaluation. For each instrument block the conservative χ^2_r reference p -value of J_{CUE} , size-valid under weak identification, is 0.4178 at $d_z = 3$ and 0.7149 at $d_z = 30$.

Spec.	Est.	$\hat{\tau}$	SE	eff.- F	AR set	LM set	QLR set	J_{CUE}	$J_{\text{CUE}} p$
<i>Quarter of birth ($d_z = 3$), $N = 329,509$, $\text{df}_J = 2$</i>									
	OLS	0.0673	0.0013	—	—	—	—	2.83	0.2424
	2SLS	0.1077	0.0170	11.41	—	—	—		
	DML (linear)	0.1078	0.0171	11.41	[0.050, 0.148]	[0.057, 0.139]	[0.058, 0.138]		
<i>QOB $\times$ YOB ($d_z = 30$), $N = 329,509$, $\text{df}_J = 29$</i>									
	OLS	0.0673	0.0013	—	—	—	—	25.21	0.6675
	2SLS	0.0869	0.0108	12.05	—	—	—		
	DML (linear)	0.0869	0.0108	12.05	whole line	noncontiguous on grid	noncontiguous on grid		

d_z	median eff.- F	AR cov.	LM cov.	QLR cov.	AR MCSE	LM MCSE	QLR MCSE
3	4.21	0.946	0.950	0.946	0.0065	0.0063	0.0065
3	14.81	0.963	0.952	0.952	0.0055	0.0062	0.0062
3	36.49	0.938	0.948	0.948	0.0069	0.0064	0.0064
30	4.52	0.958	0.941	0.946	0.0058	0.0068	0.0065
30	15.25	0.947	0.954	0.953	0.0065	0.0060	0.0061
30	37.06	0.950	0.966	0.966	0.0063	0.0052	0.0052

J_{CUE} rejection (size, nominal 0.05)

$d_z = 3$ (df = 2): rejection = 0.034 (MCSE 0.0081), reps = 500

$d_z = 30$ (df = 29): rejection = 0.042 (MCSE 0.0090), reps = 500

Table B.7: Adversarial stress panel for the cross-fitted Anderson–Rubin interval: the 11 Monte-Carlo operating points chosen to be unfavorable (flexible random-forest $\hat{\pi}$ at realistic and very weak strength, tiny samples, an estimated $\hat{\pi}$ absorbing most of the instrument, and $d_z = 10$ over-identification). Coverage is for the nominal-95% sets; across all points the stored AR coverage ranges from 0.920 to 0.961. Wald coverage and bias are printed as stored; the artifact stores Monte-Carlo standard errors for the AR column only. The Wald-coverage standard error is recoverable from the printed rate and replication count; the bias column carries no stored dispersion measure. Parenthetical regime tags name the empirical cell whose configuration guided the point’s calibration, not that cell’s realized strength.

Operating point	eff.- F	AR cov.	AR MCSE	Wald cov.	bias	reps
<i>flexible RF π, eff.-$F \approx 10$ (CollegeDistance-RF)</i>						
	9.9	0.958	0.0082	0.963	-0.087	600
<i>flexible RF π, eff.-$F \approx 34$ (examiner)</i>						
	33.5	0.920	0.0111	0.952	-0.016	600
<i>flexible RF π, very weak</i>						
	2.3	0.935	0.0123	0.978	0.233	400
<i>tiny $n = 400$, oracle π</i>						
	1.8	0.952	0.0048	0.983	0.019	2,000
	11.2	0.948	0.0050	0.962	-0.066	2,000
	62.0	0.951	0.0049	0.953	-0.007	2,000
<i>estimated π absorbing 85% of Z</i>						
	5.9	0.957	0.0052	0.977	-0.050	1,500
	29.1	0.949	0.0057	0.961	-0.009	1,500
<i>$d_z = 10$ over-identified, oracle π</i>						
	1.3	0.961	0.0050	0.798	0.289	1,500
	3.1	0.951	0.0056	0.872	0.121	1,500
	20.4	0.957	0.0052	0.933	0.017	1,500

Table B.8: Protocol robustness: the literal inner-CV nested-selection protocol of Çetin (2026a) run in parallel with the three-way LF-NCF primary protocol. Draws and outer partitions are paired by common random numbers (same seed streams), so the comparison isolates the protocol choice. The LF-NCF side is transcribed from the frozen primary artifacts, not recomputed. In the Monte-Carlo block the frozen row fixes linear nuisances with no selection step, so the inner-CV side runs the 2026a selection layer live for $\hat{\mu}$ and $\hat{\tau}$ over a linear-family library with $\hat{\pi}$ held at the same linear residualization; the application blocks recompute each cell’s primary DML panel under the inner-CV protocol. AR coverage differences are within Monte-Carlo noise at every operating point, and the qualitative readings of §4.6–§4.7 are unchanged.

Monte-Carlo point	AR coverage (MCSE)		Wald coverage		median eff.- F	
	inner-CV	LF-NCF	inner-CV	LF-NCF	inner-CV	LF-NCF
$f_s = 0.048$	0.956 (0.0065)	0.957 (0.0064)	0.984	0.984	2.0	2.0
$f_s = 0.078$	0.956 (0.0065)	0.955 (0.0066)	0.972	0.970	5.8	5.8
$f_s = 0.125$	0.946 (0.0071)	0.947 (0.0071)	0.957	0.956	14.1	14.1
$f_s = 0.290$	0.954 (0.0066)	0.951 (0.0068)	0.951	0.952	76.7	77.0
Application, π -learner	$\hat{\tau}$ (Wald SE)		eff.- F		AR 95% set	
	inner-CV	LF-NCF	inner-CV	LF-NCF	inner-CV	LF-NCF
Card, linear	0.122 (0.050)	0.128 (0.053)	17.9	15.8	[0.026, 0.248]	[0.027, 0.269]
Card, ridge	0.122 (0.050)	0.127 (0.052)	18.0	15.9	[0.026, 0.248]	[0.026, 0.266]
Card, rf	0.151 (0.052)	0.152 (0.054)	16.6	15.2	[0.055, 0.294]	[0.054, 0.307]
Card, gb	0.116 (0.047)	0.118 (0.049)	20.1	18.2	[0.025, 0.230]	[0.023, 0.239]
CollegeDistance, linear	-0.008 (0.083)	-0.002 (0.113)	10.1	14.9	[-0.137, 3.688]	[-0.226, 0.782]
CollegeDistance, ridge	-0.008 (0.083)	-0.003 (0.113)	10.0	14.8	[-0.137, 3.332]	[-0.227, 0.766]
CollegeDistance, rf	-0.046 (0.067)	-0.006 (0.106)	7.2	7.9	[-0.205, 0.214]	[-0.263, 0.346]
CollegeDistance, gb	-0.055 (0.077)	0.001 (0.118)	9.4	8.9	[-0.244, 0.173]	[-0.287, 0.381]

Table B.9: Set extraction and informativeness. The AR column is the coverage of τ_0 by the TEST, that is the share of replications with $\text{AR}(\tau_0)$ below its critical value; the set column is the share of replications whose EXTRACTED confidence set contains τ_0 . The two answer different questions, the first about the level of the test and the second about the inversion that turns it into a reported interval, and they agree in every row here, which is the check that a grid inversion warrants. The shape columns give the share of replications whose set is a bounded interval, two rays, the whole line, or empty; the width column is the median total length over the replications whose set is of finite length; the last column is the rejection rate at $\tau_0 + 0.5$. Sets are extracted by the exact analytic inversion at $d_z = 1$ and by the adaptive grid inversion with a bounded expansion ladder at $d_z > 1$ (Online Appendix C.5). The auxiliary clustered, weighted and nonlinear arms of Table B.2 report coverage only: they are $d_z = 1$ designs whose extraction is the same exact inversion already exercised in the strength sweeps above.

eff.-F	AR	set	bounded	two rays	whole line	empty	med. width	rej. at $\tau_0+0.5$
<i>d_z = 1 strength sweep, oracle</i>								
1.11	0.954	0.954	0.163	0.133	0.704	0.000	3.48	0.077
2.11	0.950	0.950	0.310	0.162	0.528	0.000	2.88	0.120
5.63	0.953	0.953	0.667	0.132	0.201	0.000	2.16	0.242
14.12	0.954	0.954	0.960	0.027	0.013	0.000	1.26	0.535
41.00	0.943	0.943	1.000	0.000	0.000	0.000	0.65	0.921
131.93	0.960	0.960	1.000	0.000	0.000	0.000	0.35	1.000
488.44	0.946	0.946	1.000	0.000	0.000	0.000	0.18	1.000
<i>d_z = 1, estimated linear</i>								
1.99	0.957	0.957	0.285	0.170	0.545	0.000	2.65	0.135
5.80	0.955	0.955	0.666	0.146	0.188	0.000	2.11	0.228
14.14	0.947	0.947	0.957	0.025	0.018	0.000	1.24	0.527
76.98	0.951	0.951	1.000	0.000	0.000	0.000	0.46	0.997
<i>d_z = 1, estimated RF, nonlinear π_0</i>								
8.39	0.951	0.951	0.822	0.079	0.099	0.000	1.78	0.284
29.09	0.949	0.949	1.000	0.000	0.000	0.000	0.81	0.764
<i>d_z = 5 over-identified, oracle</i>								
1.11	0.951	0.951	0.115	0.141	0.741	0.003	4.09	0.066
1.60	0.955	0.955	0.262	0.183	0.554	0.001	3.22	0.106
3.91	0.947	0.947	0.847	0.075	0.061	0.017	1.92	0.306
28.59	0.946	0.946	0.973	0.000	0.000	0.027	0.49	1.000
<i>d_z = 3 over-identified, oracle</i>								
4.21	0.946	0.946	0.770	0.114	0.099	0.017	2.03	0.287
14.81	0.963	0.963	0.985	0.000	0.000	0.015	0.83	0.843
36.49	0.938	0.938	0.977	0.000	0.000	0.023	0.50	0.999
<i>d_z = 30 over-identified, oracle</i>								
4.52	0.958	0.958	0.971	0.000	0.000	0.029	0.84	0.907
15.25	0.947	0.947	0.954	0.000	0.000	0.046	0.40	1.000
37.06	0.950	0.950	0.958	0.000	0.000	0.042	0.24	1.000
<i>d_z = 3, estimated linear</i>								
14.96	0.955	0.955	0.982	0.000	0.000	0.018	0.81	0.859
<i>d_z = 30, estimated linear</i>								
15.21	0.953	0.953	0.962	0.000	0.000	0.038	0.40	1.000
<i>combined arm (clusters, generated instrument)</i>								
2.01	0.912	0.912	0.316	0.170	0.514	0.000	2.88	0.100
9.58	0.956	0.956	0.824	0.128	0.048	0.000	2.05	0.280
24.84	0.944	0.944	0.972	0.024	0.004	0.000	1.07	0.678
53.05	0.952	0.952	1.000	0.000	0.000	0.000	0.75	0.880
42.81	0.944	0.944	0.992	0.006	0.002	0.000	0.88	0.794
2.00	0.936	0.936	0.290	0.228	0.482	0.000	3.27	0.130

Table B.10: Structure-matched clustered Angrist–Krueger arm. $G = 51$ unequal clusters whose sizes are the real state-of-birth counts scaled by 0.10 (simulated min, median, max 8, 524, 2,902 against the empirical 78, 5,245, 29,015), $n = 32,954$, the within-state transformation and state-level outer folds of §4.9, every statistic clustered on state. Effective- F is reported cluster-robust with its HC0 counterpart in parentheses; AR (HC0) is the AR coverage a researcher would obtain by ignoring the clustering. $P(\text{whole line})$ is the share of replications whose extracted set is the entire real line, and the last column is the median analytic AR tail limit against the χ^2 critical value it must clear for a bounded set to be possible at all. That limit is $\|P\mathbf{1}_G\|^2$ with P the projector onto the column space of the cluster-sum matrix, so it can never exceed G : at $d_z = 30$ the critical value 43.77 leaves almost no room below $G = 51$, the set is the whole line and its coverage is conservative by construction, while the score statistic still holds its nominal level. The empirical cell’s own tail limit is 33.14. Scale is not matched: this arm reproduces the cell’s dependence structure, not its quarter-of-birth effect sizes, and the clustered conditional-QLR column remains a constructive display carrying no distribution theory.

d_z	strength	eff.- F (HC0)	AR	LM	CLR	AR (HC0)	J size	$P(\text{whole line})$	AR tail / crit
3	weak	5.76 (5.40)	0.967	0.952	0.952	0.952	0.023	0.153	10.6 / 7.8
3	ak	13.12 (14.72)	0.950	0.942	0.947	0.938	0.043	0.002	15.9 / 7.8
3	strong	30.87 (73.49)	0.962	0.963	0.965	0.952	0.043	0.000	21.2 / 7.8
30	weak	7.14 (1.59)	1.000	0.950	0.987	0.967	0.000	1.000	33.4 / 43.8
30	ak	11.93 (3.16)	0.998	0.952	0.978	0.960	0.000	0.997	35.1 / 43.8
30	strong	22.85 (15.07)	1.000	0.962	0.963	0.947	0.000	0.998	37.1 / 43.8

Table B.11: Card (1995) specification sensitivity: the experience block against age. Same sample ($N = 3,010$), same instrument (nearc4 ($d_z = 1$)), same base controls; only the experience or age block changes. Experience is age minus schooling minus six, hence a function of the endogenous treatment, so the classic row is reported for comparability with the literature rather than as a pre-determined specification. Coefficients and standard errors are conventional two-stage least squares with heteroskedasticity-robust (HC0) standard errors; the first-stage F and the AR set come from full-sample partialling of the controls followed by the exact $d_z = 1$ inversion, the same convention as the CollegeDistance row of Table 1 of the article. Replacing experience with age moves the point estimate by about a third and carries the identification-robust set across zero, so the classic magnitude is not robust to this substitution.

Controls	$\hat{\tau}$	HC0 SE	Δ vs classic	first-stage F	AR 95% set
exper+expersq	0.1315	0.0540	0.0000	14.21	[0.028, 0.281]
age	0.0827	0.0446	-0.0488	11.61	[-0.003, 0.217]
age+agesq	0.0827	0.0446	-0.0488	11.71	[-0.003, 0.216]

Table B.12: Examiner cell sensitivity diagnostics ($N = 91,282, 1,831$ court-by-date shifts, 146 judges). *Panel A* varies the cross-fitting partition. Because GroupKFold does not shuffle in the pinned scikit-learn, re-seeding it is a no-op, so the arms resample the shift-level partition explicitly; all of them fix $\hat{\mu}$, $\hat{\tau}$ and $\hat{\pi}$ to the pre-specified ridge and skip the per-fold library selection, which makes them comparable with each other but not with the frozen cell rows of Table 1 of the article. The point estimate ranges over -0.604 to -0.369 and the identification-robust set contains zero in 2 of 6 partitions, so the cell’s exclusion of zero is a property of the reported partition and not of the design alone. *Panel B* reports the in-sample and out-of-fold R^2 of each instrument-residualization learner against the fold-nested instrument: the random forest fits nearly half the variance in sample and none of it out of fold, which is the overfit-dilution signature §4.5 reads off the variance ratio. The fold-nested instrument correlates 0.895 (Spearman 0.905) with the full-sample construction validated against the published implementation, with RMSE 0.016, so the traditional check anchors the construction without certifying the fold-nested chain. Support: $J_n/\sqrt{n} = 0.483$, 2,514 judge-by-shift cells of which 0.271 carry at most two cases.

<i>Panel A: cross-fitting partition (ridge fixed, no per-fold selection)</i>						
Partition	$\hat{\tau}$	SE	eff.- F	AR 95% set	contains 0	
production GroupKFold	-0.5117	0.2051	21.37	$[-1.050, -0.143]$	no	
resampled seed 123	-0.5331	0.2424	15.96	$[-1.212, -0.092]$	no	
resampled seed 7	-0.5785	0.3432	7.04	$[-1.915, 0.005]$	yes	
resampled seed 31	-0.3694	0.2117	19.57	$[-0.889, 0.043]$	yes	
resampled seed 99	-0.6037	0.2746	12.22	$[-1.492, -0.125]$	no	
resampled seed 2024	-0.5439	0.2359	17.48	$[-1.187, -0.116]$	no	

<i>Panel B: instrument-residualization fit, in sample against out of fold</i>			
$\hat{\pi}$ learner	in-sample R^2	out-of-fold R^2	gap
linear	0.0144	-0.0006	0.0151
ridge	0.0144	-0.0005	0.0149
rf	0.4919	-0.2595	0.7513
gb	0.1026	-0.0298	0.1324

Table B.13: Selector interval-length tails and paired post-selection coverage. *Panel A* decomposes the mean Wald interval length of Table B.3: median, ninetieth percentile, the share of the total length carried by the single longest and the ten longest replications, and the mean after dropping those ten. A mean that is several times larger under one rule while the median and the ninetieth percentile agree is a tail statistic, not a systematic widening. *Panel B* reads the post-selection distortion against its paired standard error rather than the one-arm coverage Monte-Carlo standard error: both arms are computed on the same inference observations of the same draws, so only the replications on which they disagree carry information, and the exact McNemar p -value is reported on those replications.

<i>Panel A: Wald interval length by replication</i>							
Regime	rule	mean	median	q_{90}	top-1 share	top-10 share	mean, top-10 dropped
weak	MSE	21.33	2.05	8.74	0.253	0.852	3.26
	IACV	69.45	2.07	9.11	0.770	0.955	3.27
moderate	MSE	2.12	1.19	2.19	0.300	0.403	1.31
	IACV	1.80	1.20	2.17	0.133	0.294	1.32
do-no-harm (low absorption)	MSE	0.90	0.81	1.24	0.009	0.068	0.87
	IACV	0.90	0.83	1.25	0.009	0.068	0.87

<i>Panel B: post-selection coverage distortion, paired</i>						
Regime	rule	in-sample	split	difference	paired SE	McNemar p
weak	MSE	0.975	0.975	0.0000	0.0167	1.000
	IACV	0.975	0.983	-0.0083	0.0083	1.000
moderate	MSE	0.908	0.892	0.0167	0.0117	0.500
	IACV	0.908	0.892	0.0167	0.0167	0.625
do-no-harm (low absorption)	MSE	0.942	0.942	0.0000	0.0118	1.000
	IACV	0.950	0.942	0.0083	0.0083	1.000

Appendix C. Monte-Carlo designs

This appendix records the data-generating processes behind every Monte-Carlo result in §4.2, §4.3, §4.8, and Appendix B: the structural equations, the exact population standardization constants, a parameter table for each simulation arm, and the seeding scheme. The results and their Monte-Carlo standard errors are reported in §4.2–§4.3, §4.8, and in Tables B.2, B.3, B.6, B.7, and B.8. The generators are `dmliv_montecarlo.py`, `mc_coverage.py`, `coverage_clustered.py`, `coverage_weighted.py`, `coverage_nonlinear.py`, `coverage_ak_overid.py`, `run_mc_absorption.py` (the §4.8 known-truth cell), `stress_regime.py`, `protocol_comparison.py` (the Table B.8 protocol comparison), and, for the §4.3 selector benchmark, `selector_regime.py`, `selector_postselect.py`, and `selector_merge.py` in the replication materials, and every artifact carries a stamp of the seed key that produced it.

C.1 The core partially linear IV draw

All sweep arms draw from one partially linear IV design (`make_pliv_data`). Controls are $W \sim N(0, \Sigma_W)$ with $(\Sigma_W)_{ik} = 0.5^{|i-k|}$ (a Toeplitz matrix) and dimension p . Each instrument coordinate $j = 0, \dots, d_z - 1$ carries a base signal $b_j(W)$, the W -explainable part of Z_j , in one of three forms.

Const. $b_j \equiv 0$, a randomized instrument, so the true instrument-residualization nuisance is $\pi_0 \equiv 0$. The estimated-nuisance arms still cross-fit $\hat{\pi}$, so instrument-residualization estimation error is present even where the true nuisance is zero.

Linear. $b_j(W) = W' \beta_j / (\beta_j' \Sigma_W \beta_j)^{1/2}$, where β_j is a standard-normal vector scaled by $p^{-1/2}$, drawn once from the fixed design seed (Appendix C.4) and reused for every replication and sample size; the population variance of $b_j(W)$ is exactly 1 by construction.

Nonlinear. With column indices $(a, b, c, d) = (j, j+1, j+2, j+3) \bmod p$,

$$b_j(W) = \frac{\text{raw}_j(W) - m_j}{s_j}, \quad \text{raw}_j(W) = \sin(W_a) + W_b W_c + \max(W_d, 0),$$

standardized by its exact Gaussian population moments. Writing $S_{xy} = 0.5^{|x-y|}$,

$$m_j = S_{bc} + (2\pi)^{-1/2}, \quad s_j^2 = \frac{1 - e^{-2}}{2} + 1 + S_{bc}^2 + \frac{1}{2} - \frac{1}{2\pi} + S_{ad} e^{-1/2} + \frac{2 S_{bd} S_{cd}}{\sqrt{2\pi}}.$$

The constituent moments are $\text{Var}(\sin W_a) = (1 - e^{-2})/2$, $\text{Var}(W_b W_c) = 1 + S_{bc}^2$, $\text{Var}(\max(W_d, 0)) = 1/2 - (2\pi)^{-1}$, $\text{Cov}(\sin W_a, W_b W_c) = 0$ by sign-flip symmetry, $\text{Cov}(\sin W_a, \max(W_d, 0)) = S_{ad} e^{-1/2}/2$ by Stein's identity applied to the odd part of $\max(u, 0) = (u + |u|)/2$, and $\text{Cov}(W_b W_c, \max(W_d, 0)) = S_{bd} S_{cd} (2\pi)^{-1/2}$. These constants are checked against a 10^6 -draw simulation in the included test suite.

In every form b_j is a fixed population function: its coefficients are deterministic and its centering and scaling use exact population moments, never realized-sample moments, so π_0 is the same function of W in every replication and at every sample size.

The structural system is

$$Z_j = \sqrt{\rho} b_j(W) + \sqrt{1 - \rho} e_{Z,j}, \quad D = W' \beta_r + f_s \sum_j Z_j + \kappa A + \nu, \quad Y = \tau_0 D + W' \beta_g + \kappa A + \zeta,$$

where $\rho \in [0, 1]$ is the W -explainable share of each instrument coordinate, f_s is the design's first-stage loading on the instruments, and $e_{Z,j}$, A , ν , ζ are i.i.d. standard normal, mutually independent and independent of W . The fixed coefficients are $\beta_r = (0.7, -0.5, 0.4, 0, \dots)$, $\beta_g = (1.0, 0.5, -0.5, 0, \dots)$, the confounding loading is $\kappa = 0.8$, and $\tau_0 = 1$ throughout. The true conditional means, stored by the generator and used verbatim as the oracle nuisances, are

$$\begin{aligned}\pi_{0,j}(W) &= \mathbb{E}[Z_j | W] = \sqrt{\rho} b_j(W), \\ r_0(W) &= \mathbb{E}[D | W] = W' \beta_r + f_s \sum_j \pi_{0,j}(W), \\ \mu_0(W) &= \mathbb{E}[Y | W] = \tau_0 r_0(W) + W' \beta_g.\end{aligned}$$

The treatment mean includes the pass-through term $f_s \sum_j \pi_{0,j}$, so r_0 is nonlinear in W whenever the base form is nonlinear, even though the structural part $W' \beta_r$ is linear (§4.8 turns on exactly this term).

Because A is independent of (Z, W) , $\mathbb{E}[\kappa A + \zeta | Z, W] = 0$, so the instrument is valid under full conditioning; endogeneity comes from $\text{Cov}(D, \kappa A + \zeta) = \kappa^2 > 0$. At the true nuisances the residualized instrument is $\tilde{Z}_{0,j} = \sqrt{1 - \rho} e_{Z,j}$, so the residualized concentration scales with $f_s^2(1 - \rho)$ while the estimand τ_0 is untouched. The sweeps therefore move f_s (and, where stated, ρ) to drive the effective- F from about 1 to about 500 (§4.2) without changing the target.

C.2 Auxiliary designs

Clustered arm (dz1_clustered). Two hundred equal clusters of size $M \in \{10, 25\}$, with $p = 20$, $\rho = 0.5$, and the linear base form (fixed design-seed coefficients, exact population scaling). A unit-variance component with cluster share c is built as $\sqrt{c} \eta_{g(i)} + \sqrt{1 - c} \eta_i$ with η_g, η_i i.i.d. standard normal. The instrument's non- W part carries share 0.7, the confounder A share 0.4, and the structural error ζ share 0.4; ν stays i.i.d. Thus $Z = \pi_0(W) + \sqrt{1 - \rho} \varepsilon_Z$ with $\varepsilon_Z = \sqrt{0.7} \eta_{g(i)}^Z + \sqrt{0.3} \eta_i^Z$, and D and Y follow the C.1 equations with the same $\beta_r, \beta_g, \kappa, \tau_0$. W is drawn i.i.d. across observations, so π_0 itself has no cluster component; the within-cluster correlation of the score comes from ε_Z , A , and ζ . The oracle residuals include the $f_s \pi_0$ pass-through. Each row computes the cluster-robust AR and Wald (what the clustered cells use) and their naive heteroskedasticity-only counterparts for the comparison reported in Table B.2.

Survey-weighted arm (dz1_weighted). $n = 6,000$, $\rho = 0.5$, linear base. A binary survey stratum s has sample share 0.5 and is independent of (W, Z, A) ; the design weights are $w = 1.6$ for $s = 0$ and $w = 0.4$ for $s = 1$. The outcome carries a heterogeneous effect, $Y = (\tau_0 + \Delta s)D + W' \beta_g + \kappa A + \zeta$ with $\Delta = 1$. The weighted stratum-1 share is $p_1^w = (0.5 \times 0.4)/(0.5 \times 0.4 + 0.5 \times 1.6) = 0.2$, so the survey-population estimand is $\tau_w = \tau_0 + \Delta p_1^w = 1.2$ while the unweighted-sample estimand is $\tau_u = \tau_0 + \Delta \times 0.5 = 1.5$. The treatment mean r_0 does not involve s ; the oracle outcome mean is estimand-specific, $\mu_w = \tau_w r_0 + W' \beta_g$ for the weighted statistic and $\mu_u = \tau_u r_0 + W' \beta_g$ for the naive unweighted comparison, and both AR versions are evaluated at τ_w .

Nonlinear-nuisance arm (dz1_nonlinear). $n = 1,800$, $\rho = 0.5$. The instrument-residualization nuisance, the treatment structure, and the outcome structure are all nonlinear members of the C.1 family: $\pi_0 = \sqrt{\rho} b_0$, and $D = b_4(W) + f_s Z + \kappa A + \nu$, $Y = \tau_0 D + b_8(W) +$

$\kappa A + \zeta$, each feature standardized by its exact population moments. All three nuisances are cross-fit by random forests (40 trees, minimum leaf size 10) on one common fold partition; nothing is selected.

Angrist–Krueger over-identified arm (ak_overid). The core design with the const base ($\pi_0 \equiv 0$, $\rho = 0$), $n = 12,000$, $p = 20$, $d_z \in \{3, 30\}$, and $f_s \in \{0.022, 0.044, 0.070\}$, calibrated to bracket the Angrist–Krueger cell’s operating strengths (§4.9). Nuisance modes are oracle and estimated-linear (cross-fit linear learners for μ , r , and every π coordinate; run at $f_s = 0.044$). Each replication computes the coverage of τ_0 by the cross-fitted AR ($\chi_{d_z}^2$), the score statistic (χ_1^2), and the conditional QLR, whose conditional null distribution is simulated with 2,000 draws per replication from a dedicated child stream. The J_{CUE} size rows minimize $\text{AR}(\tau)$ over a 61-point grid on $[\tau_0 - 2, \tau_0 + 2]$ and refer the minimum to $\chi_{d_z-1}^2$ (the Stock–Wright S objective); they run at the middle operating point ($f_s = 0.044$) with oracle nuisances. Sampling is i.i.d.; these rows are the i.i.d. analogues of the AK cell’s statistics at its instrument counts and strengths.

Full-stack arm (full_stack). An examiner-style clustered design with a constructed instrument: 30 judges $\times$ 8 shifts per judge, m cases per judge-by-shift cluster, so $G = 240$ clusters and $n = 240m$. Here $W \sim N(0, I_6)$, the latent leniencies ℓ_j are i.i.d. $N(0, 1)$ across judges, and

$$D_i = W_i' \beta_r + f_s \ell_{j(i)} + 0.8 A_i + \nu_i, \quad Y_i = \tau_0 D_i + W_i' \beta_g + 0.8 A_i + \zeta_i,$$

with $\beta_r = (0.6, -0.4, 0.3, 0, 0, 0)$ and $\beta_g = (0.8, 0.4, -0.4, 0, 0, 0)$. The confounder and the structural error carry cluster shares 0.5 and 0.4: $A_i = \sqrt{0.5} \eta_{g(i)} + \sqrt{0.5} \eta_i$ and $\zeta_i = \sqrt{0.4} \xi_{g(i)} + \sqrt{0.6} \xi_i$, while ν is i.i.d. Judges are randomly assigned, so $\mathbb{E}[0.8A + \zeta \mid \text{judge}, W] = 0$. The instrument is constructed within each replication by fold-nested CJIVE on cluster-aware folds (the construction map is fit on training folds only, with leave-one-shift-out values inside the training set, as in §4.5); $\hat{\mu}$ and $\hat{r}$ are selected by leakage-free nested cross-fitting (LF-NCF) from $\{\text{const}, \text{linear}, \text{ridge}\}$; $\hat{\pi}$ is a fold-wise linear regression of the fold’s constructed instrument on W ; AR and Wald are cluster-robust at the judge-by-shift level. The choice $m = 6$ ($n = 1,440$) stresses the leniency-estimation rate condition $J_n = o(\sqrt{n})$ with $J_n/\sqrt{n} \approx 0.79$, and $m = 20$ ($n = 4,800$) relaxes it (≈ 0.43). The observed-instrument control rows replace the constructed instrument by the true leniency $\ell_{j(i)}$ in the same design, isolating the construction-error channel.

Adversarial stress panel (stress). Six regimes probe corners the sweeps do not: a flexible random-forest $\hat{\pi}$ at three strengths on the nonlinear-base design, a tiny sample ($n = 400$), near-total residualization ($\rho = 0.85$ with an estimated $\hat{\pi}$), and heavy over-identification ($d_z = 10$). All six use the C.1 generator; their settings are the stress rows of Table C.1.

C.3 Parameters by arm

Table C.1 lists, for every arm, the dimensions, base form, absorption share ρ , strength ladder f_s , nuisance mode, and replications per row. Learner hyperparameters (forest sizes, ridge grids, fold counts) are fixed in the included scripts and recorded in the artifact stamps. The estimated-nuisance arms use $K = 5$ cross-fitting folds on one common outer partition per

replication, cluster-aware where the design is clustered; the oracle rows use the stored true conditional means directly and need no folds.

Table C.1: Monte-Carlo arms: dimensions, strengths, nuisance modes, and replications per row. “Base form” is the shape of the instrument-residualization nuisance π_0 (Appendix C.1); ρ is the W -explainable share of Z ; f_s is the first-stage loading. “Oracle” means the DGP’s true conditional means are used as nuisances; estimated modes cross-fit them as stated. Random streams derive from the master seed and the arm key (Appendix C.4).

Arm (artifact tag)	n	p	d_z	Base form	ρ	f_s values	Nuisances	Reps
Core sweep, oracle (dz1_oracle)	1,500	20	1	const	0	0.033, 0.048, 0.078, 0.125, 0.21, 0.38, 0.73	oracle	3,000
Core sweep, estimated linear (dz1_est_linear)	1,500	20	1	const	0	0.048, 0.078, 0.125, 0.29	linear $\hat{\mu}, \hat{f}; \#$	1,000
Core sweep, flexible (dz1_est_rf_nonlinear)	1,500	20	1	nonlinear	0.3	0.12, 0.22	RF $\hat{\mu}, \hat{f}; \#$ LF-NCF {linear, RF}	1,000
Over-identified (dz5_overid)	2,000	20	5	const	0	0.015, 0.025, 0.05, 0.15	oracle	2,000
Nonlinear nuisances (dz1_nonlinear)	1,800	20	1	nonlinear ($j = 0, 4, 8$)	0.5	0.07, 0.13	RF $\hat{\mu}, \hat{f}, \#$	250
Clustered (dz1_clustered)	$200 \times M, M \in \{10, 25\}$	20	1	linear	0.5	0.165 ($M=10$), 0.115 ($M=25$)	oracle	1,000
Survey-weighted (dz1_weighted)	6,000	20	1	linear	0.5	0.18, 0.34	oracle (weighted)	1,000
Augstein-Krueger overid. (ak_overid)	12,000	20	3, 30	const	0	0.022, 0.044, 0.070	oracle; est. linear at 0.044	1,200; 500 (<i>Active</i>)
Full stack (full_stack)	240 $m, m \in \{6, 20\}$	6	1	constructed (CHVE); observed- Z controls	—	0.15, 0.35 ($m=20$ at 0.35, constructed only)	LF-NCF $\hat{\mu}, \hat{f}$; linear $\#$	500
Known-truth absorption cell, §4.8 (mc_absorption)	1,500	20	1	nonlinear	0.6	0.455	$\hat{\mu}, \hat{f}$ oracle (correct- r) or cross-fit linear (misspecified- r); $\#$ {linear, ridge, RF, GB}	100
<i>Adversarial stress panel (stress); all rows use the C.1 generator</i>								
flexible $\#$, moderate (flexrf_eff10)	1,500	20	1	nonlinear	0.3	0.12	linear $\hat{\mu}, \hat{f}; \#$ LF-NCF {linear, RF}	600
flexible $\#$, stronger (flexrf_eff25)	1,500	20	1	nonlinear	0.3	0.22	linear $\hat{\mu}, \hat{f}; \#$ LF-NCF {linear, RF}	600
flexible $\#$, very weak (flexrf_weak)	1,500	20	1	nonlinear	0.3	0.06	linear $\hat{\mu}, \hat{f}; \#$ LF-NCF {linear, RF}	400
tiny sample (tiny_n)	400	20	1	linear	0.3	0.10, 0.25, 0.60	oracle	2,000
high absorption (high_absorb_est)	2,000	20	1	linear	0.85	0.18, 0.40	linear $\hat{\mu}, \hat{f}; \#$ LF-NCF {const, linear}	1,500
heavy overid. (dz10)	2,000	20	10	linear	0.3	0.02, 0.05, 0.15	oracle	1,500

C.4 Seeding, common random numbers, and Monte-Carlo standard errors

All Monte-Carlo randomness derives from the single master seed 20260819. Each arm-and-row combination has a key, the arm tag followed by the row’s design parameters; each key component is hashed by SHA-256, the hash truncated to its leading four bytes, and the resulting words appended to the master seed to form a NumPy `SeedSequence`, which is spawned into one independent child stream per replication. The data draw of a replication is seeded by its child; fold shuffles are seeded from the same child’s first generated state, while random-forest fits use a fixed constant seed (the §4.8 cell instead seeds its learner library from the replication’s fold seed). Distinct keys give independently seeded streams, so no two rows share randomness. The keys are (dz1_oracle, f_s), (dz1_est_linear, f_s), (dz1_est_rf_nonlinear, f_s), (dz5_overid, f_s), (dz1_clustered, f_s, M), (dz1_weighted, f_s), (dz1_nonlinear, f_s), (ak_overid, mode, d_z, f_s), (full_stack, f_s, m , instrument), (stress, regime, f_s), and (mc_absorption) (the §4.8 cell’s single row, the arm tag alone); every JSON row prints its master seed, key, and scheme verbatim. The J_{CUE} size rows keep the frozen key (ak_overid_hansen, d_z, f_s): the statistic was renamed J_{CUE} after those draws were included, and the key string is retained so they reproduce exactly. The design seed that fixes the linear base coefficients uses the same master integer with the raw words (j, p); it is used only for the coefficient draw, never to generate data.

Two designs deliberately share randomness within a replication. The selector benchmark (§4.3, Table B.3; key (selector_regime, regime), 300 replications per regime) evaluates both selection rules on the same draw, and the post-selection check (key (selector_postselect, regime), 120 replications) evaluates every rule and both selection modes on the same draw, and its foldwise-aggregation variant (§4.3) reuses the same keys, so its draws are paired with the primary run; the variant’s artifacts carry a `_foldwise` suffix and an explicit `aggregation` field. This pairing is a deliberate common-random-numbers design: it removes between-draw noise from the rule comparison, and the paired differences

are reported with paired standard errors. The pairing is disclosed in the artifacts under a `crn` field; the three regimes still use disjoint streams. A third pairing is cross-protocol: the Table B.8 comparison reruns the (`dz1_est_linear`, f_s) streams under the inner-cross-validation protocol, so the two protocols see identical draws and outer partitions, while the leakage-free side is transcribed from the frozen primary artifacts rather than recomputed.

Replication counts are listed in Table C.1. For a coverage or size proportion $\hat{p}$ over R replications the Monte-Carlo standard error is $\{\hat{p}(1 - \hat{p})/R\}^{1/2}$; for the bias it is the across-replication standard deviation of $\hat{\tau}$ divided by $\sqrt{R}$. Every Anderson–Rubin coverage and size entry in the included artifacts carries its Monte-Carlo standard error, as do the Wald and bias entries of the arms whose artifacts store them; the remaining coverage standard errors are recoverable from the stored counts, while the stress panel’s bias entries carry no stored dispersion measure (Table B.7 notes this). The assembled artifact embeds, for the four core-sweep arms, a per-row parameter stamp (the `dgp` field, written by `dgp_parameters`) recording the dimensions, coefficients, base form, and the exact standardization constants; the other arms’ parameters are fixed in their generator scripts.

C.5 Set extraction, informativeness, and the structure-matched clustered arm

What coverage at the truth does not check. Every coverage entry above is the share of replications in which the test accepts at τ_0 . That establishes the level of the test, and it is silent about the object a reader actually sees, which is the set obtained by inverting the test over a range of candidate values. A grid inversion can in principle miss a narrow accepted region between grid points, so the two can come apart. Table B.9 therefore reports both: the coverage of the test at the truth, and the share of replications whose *extracted* set contains the truth. Sets are extracted by the exact analytic inversion at $d_z = 1$, where the acceptance region is the solution set of a quadratic inequality and no grid is involved, and by the adaptive grid inversion at $d_z > 1$. The Monte-Carlo path bounds the window growth explicitly: each pass runs on a fixed window and a verdict of truncation retries on a window four times wider, at most six times, so the work per replication is bounded and the same for every replication.

Across the forty-eight designs that carry the block, spanning $d_z \in \{1, 3, 5, 10, 30\}$, oracle and estimated nuisances, i.i.d. and clustered sampling, and observed and generated instruments, the two coverages agree at every reported digit. Table B.9 tabulates the strength sweeps, the over-identified arms, and the combined arm; the clustered arm’s columns are in Table B.10, and the stress panel’s are archived with its artifact.

Informativeness. A valid set can still be useless, so Table B.9 also records what the set looks like: the share of replications whose set is a bounded interval, two rays, the whole line, or empty; the median total length among the replications whose set has finite length; and the rejection rate at an alternative half a unit from the truth. Read together with the coverage columns these explain how nominal coverage is attained in the weak arms. At the weakest sweep point the set is the whole line in about seven replications in ten, and rejection at the alternative is near the nominal level, so coverage is achieved by reporting that the design is uninformative rather than by locating τ_0 . As strength rises the whole-line share falls to zero, the bounded share rises to one, the median length contracts, and power at the same

alternative rises. The empty set appears only in the over-identified arms, where the moment conditions can be jointly rejected at every candidate value.

A structure-matched clustered Angrist–Krueger arm. The over-identified arm of §4.2 samples i.i.d. at $n = 12,000$ and matches the Angrist–Krueger cell only in instrument count and strength. Table B.10 reports a second arm that matches the parts of that design the i.i.d. arm cannot reach: $G = 51$ clusters whose sizes are the real state-of-birth counts scaled by one tenth, so the one-to-sixty-seven-to-three-hundred-seventy-two size ratios and the 8.8 percent largest-cluster share are preserved; sparse quarter-of-birth cells counted within state; the within-state transformation and state-level outer folds of §4.9, reusing the same functions as the cell; and cluster-level dependence in both the structural and first-stage errors. Scale is not matched, and neither are the empirical quarter-of-birth effect sizes: τ_0 is a stand-in, the first-stage coefficients are chosen to hit the target effective- F , and the arm makes no claim about the magnitudes in the data. The clustered conditional-QLR column remains a constructive display carrying no distribution theory, as in the cell itself.

At $d_z = 3$ the clustered AR, score, and conditional-QLR sets hold their nominal levels across a weak, a calibrated, and a strong point, and the J_{CUE} size is near nominal, so the machinery the cell uses works at fifty-one unequal clusters. At $d_z = 30$ the picture changes for a structural reason rather than a numerical one. The analytic tail limit of the clustered AR statistic is $\|P\mathbf{1}_G\|^2$, with P the projector onto the column space of the cluster-sum matrix, so it cannot exceed G . A bounded set therefore requires the χ_r^2 critical value to fall below that ceiling, and at $G = 51$ with thirty instruments the critical value leaves almost no room: the simulated tail limit never clears it, the set is the whole line in essentially every replication, and its coverage is conservative by construction. The score statistic, read against χ_1^2 , keeps its nominal level in the same replications. The cell’s own tail limit sits in the same place, which is why §4.9 reads its thirty-instrument set as uninformative about τ rather than as evidence against the instruments.

Appendix D. Cell-level and arm-level detail

This appendix records the cell-level and arm-level detail that supports Section 4 of the article. Each subsection is referenced from the matching subsection of the article. Table D.1 reports the descriptive statistics and covariate balance for the four observational cells of §4.1. Table D.2 reports the residualized first-stage strength by nuisance learner for the same cells (§4.1 of the article).

Table D.1: Descriptive statistics and covariate balance

Cell	N	clusters	Y mean	D mean	Z mean	balance
OHIE	23,441	20,747	1.95	0.279	0.50	by design
Examiner	91,282	1,831	0.585	0.327	0.03 (0.03)	by design
CollegeDistance	4,739	43	2.24	13.81	1.80	—
Card	3,010	robust	6.26	13.26	0.68	—

Notes. Y : OHIE primary-care visits, examiner conviction, CollegeDistance log state wage, Card individual log wage; D : ever-Medicaid, pretrial release, years of education; Z : lottery, cluster-jackknife leniency, distance, growing up near a four-year college. Clusters: households, court-by-date (bond-hearing shift) assignments, state proxies (Card uses heteroskedasticity-robust standard errors). The Z -mean SD is archived only for the examiner cell and shown there in parentheses; the other cells’ frozen artifacts record no instrument SD. “Balance: by design” indicates that the (quasi-)randomization is established in the source study rather than recomputed here: OHIE by lottery within household-size strata (Finkelstein et al. 2012), examiner leniency by quasi-random assignment within court-by-time with the covariate-balance test reported by Chyn, Frandsen, and Leslie (2025). CollegeDistance and Card rely on conditional-on-observables identification and carry no design-based balance test.

Table D.2: Residualized first-stage strength (effective- F) by nuisance learner

Cell	ridge	RF	GB
OHIE	2093	2024	2094
Card	15.9	15.2	18.2
Examiner	20.1	14.2	20.6
CollegeDistance	14.8	7.9	8.9

Notes. Effective- F on the cross-fitted, **cluster-aware, decoupled-nuisance** residualized first stage. OHIE uses the survey-weighted responder primary specification (§4.4); the other three cells are unweighted. The examiner cell shows a clear dilution drop under its random-forest $\hat{\pi}$, and CollegeDistance under both flexible learners; OHIE and Card stay flat. CollegeDistance’s earlier flexible-learner “collapse” was fold leakage (§4.6). *Sample construction.* OHIE: 74,922 lottery records merged to the state-program and twelve-month survey files, restricted to the twelve-month survey responders with non-missing outcome, treatment, ever-Medicaid, household size, survey weight, and household identifier, leaving 23,441 weighted responders in 20,747 households. Examiner: 94,210 Miami-Dade cases restricted to judges with at least 200 cases, leaving 91,282 cases in 1,831 shifts. CollegeDistance: 4,739 complete cases with positive wage.

D.1 Monte-Carlo arm detail

Adversarial stress panel (§4.2). The panel probes six corners not covered by the primary sweep: tiny $n = 400$; near-total residualization with π explaining 85% of Z ; $d_z = 10$; and

the very-weak-and-flexible worst case. The per-point results are Table B.7. In the headline coverage range of §4.2, the floor (0.938) is the strongest $d_z = 3$ over-identified point and the ceiling (0.968) is the nonlinear-nuisance arm.

Nonlinear-nuisance arm (§4.2). This arm replaces the linear structural functions g_0 and r_0 with nonlinear ones recovered by random forests (all three nuisances (μ, r, π) then carry estimation error) and finds the cross-fitted AR coverage at or above nominal (0.968 and 0.960 at effective- F of about 3.0 and 9.4), confirming that this Monte-Carlo evidence does not rest on linear structural nuisances.

Clustered arm (§4.2). This arm makes the data clustered: shared cluster effects enter the structural error and a cluster-level component enters the instrument, as in the examiner-shift and OHIE-household cells. It confirms that the cluster-robust cross-fitted AR retains nominal coverage (0.948 and 0.951 at effective- F of about 6.1 and 8.5, over 200 clusters of size 25 and 10) exactly where a naive heteroskedasticity-only variance under-covers badly (0.51 and 0.69); the cluster-robust inference the clustered cells rely on is covered asymptotically by Theorem 2 (Appendix A.4), and this arm shows that it also holds its nominal level at these finite cluster counts.

Survey-weighted arm (§4.2). A survey stratum carries a heterogeneous effect, so that the survey-population effect τ_w differs from the unweighted-sample effect τ_u , as in OHIE, where weighting moves the estimand. The survey-weighted cross-fitted AR covers τ_w near nominal (0.946 and 0.947 at effective- F of about 43 and 154), whereas a naive unweighted AR, centered on the wrong estimand τ_u , covers it only 0.59 and 0.09 of the time, the gap widening as the instrument strengthens; survey weighting is therefore necessary, the weighted statistic is the one Proposition 4 (Appendix A.6) covers, and the weighted inference holds nominal coverage in this design.

Calibrated flexible-regime design points (§4.2). The two calibrated design points of the estimated flexible random-forest regime have median effective- F 8.4 and 29.1, chosen to bracket the flexible-learner strengths the empirical cells occupy, with 1,000 replications each and a Monte-Carlo standard error of about 0.007.

Just-identified Wald instability (§4.2). In the just-identified design the Wald point estimate was wildly unstable at weak instruments: at effective- $F \approx 2.0$ under an estimated-linear $\hat{\pi}$ the estimator’s standard deviation across replications is about 15, so large that even its mean bias (0.37, Monte-Carlo standard error 0.47) is not pinned down by 1,000 replications, while the enormously inflated Wald standard error (averaging above 100) mechanically covered (0.984) without conveying information.

Angrist–Krueger regimes (§4.2, §4.9). Across the Angrist–Krueger regimes ($d_z = 3$ and $d_z = 30$ at effective- F from about four to thirty-seven; 1,200 replications per coverage point and 500 for the J_{CUE} sizes), the cross-fitted AR, score, and conditional-QLR coverage rates all lie between 0.938 and 0.966 (Monte-Carlo standard errors ≈ 0.006), and the J_{CUE} over-identification statistic has size 0.034 at $d_z = 3$ (2 d.f.) and 0.042 at $d_z = 30$ (29 d.f.) against its χ^2_{r-1} reference, so the i.i.d. analogues of the sets displayed in §4.9 and Appendix D.6 operate at their nominal levels in these designs. The real-data effective- F values reported in §4 lie in the range where the simulated AR coverage remains close to nominal; in the weak part of that range, where Wald departs, AR is far closer in the primary sweep, and §4.2 records the two arms where it is not.

Two-specification absorption contrast (§4.8). In the known-truth contrast of §4.8, the

random forest recovers the nonlinear π_0 out of fold ($R_{\text{out}}^2(\hat{\pi}; Z) = 0.50$ against a linear 0.18; $\text{Var}(\tilde{Z})/\text{Var}(Z) = 0.50$ against 0.82). Both estimators are essentially unbiased over 100 replications ($|\text{mean bias}| \leq 0.01$ under either nuisance specification, within about one Monte-Carlo standard error of zero); under the correct r this is Proposition 1’s prediction, and under the misspecified r , where Proposition 1’s hypotheses fail, the flexible $\hat{\pi}$ is consistent and keeps the residualized moment centered (Appendix A.3, Remark). The linear- π arm under the misspecified r is covered by neither Proposition 1 nor the Appendix A.3 Remark; its near-zero bias there is an empirical feature of that design, whose linear g_0 makes the outcome- and treatment- nuisance misspecifications cancel in the bracket, and it would not be guaranteed under a nonlinear g_0 .

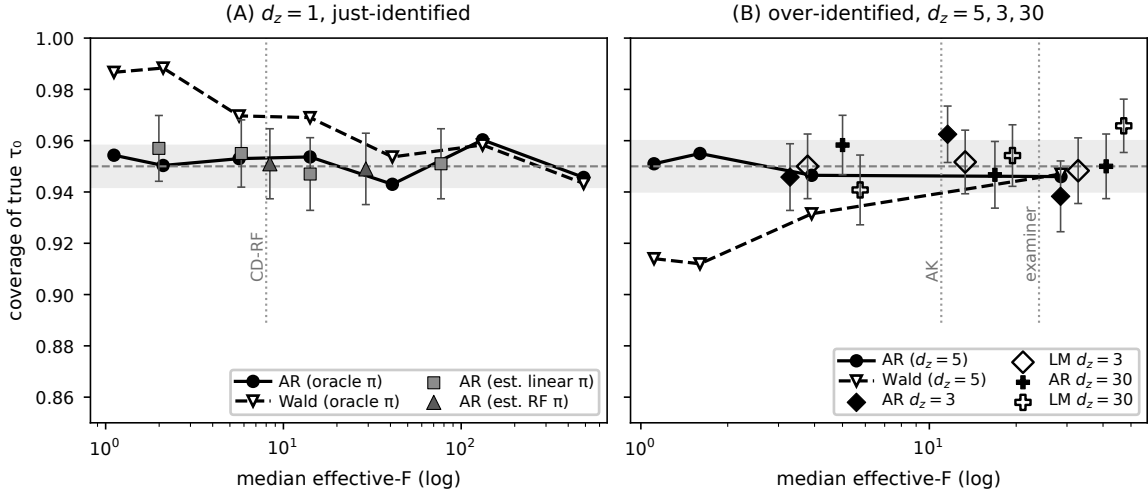


Figure S.1: Monte-Carlo coverage evaluation for the cross-fitted Anderson–Rubin set. Panel (A): just-identified ($d_z = 1$) AR vs. Wald coverage of τ_0 across the concentration sweep under oracle, estimated-linear, and estimated-flexible instrument residualizations. Panel (B): over-identified regimes ($d_z = 5, 3, 30$), where AR and the score statistic hold; the Wald curve is drawn at $d_z = 5$, the arm that stores it. Dashed line: nominal 0.95; band: ± 2 Monte-Carlo standard errors of the oracle curve in (A) (3,000 replications) and of the $d_z = 5$ curves in (B) (2,000); the error bars on the estimated- π points in (A) and on the $d_z = 3, 30$ points in (B) are ± 2 of their own stored Monte-Carlo standard errors (1,000 and 1,200 replications).

D.2 OHIE weighting and Lee-bound detail

The primary weighted-responder sample comprises the 23,441 twelve-month survey responders with complete analysis variables, out of the 23,741 responders to that survey. The weights define the estimand: the weighted moment targets τ_w , the PLIV coefficient under the survey-weighted responder law, which the study’s nonresponse-adjusted weights tie back toward the eligible population. Because the weights are not a function of the controls alone, the weighted target law is carried through the whole pipeline: every nuisance fit, selection loss, and diagnostic in the primary specification is computed under the weights, not only the final moment. This is what the hypotheses of regime (b) of Proposition 4 require when the weights depend on variables beyond the controls. The weight-on- W regression is the regime-(a) diagnostic: residual variation near zero would have licensed the unweighted- nuisance shortcut.

In the raw, unweighted sample, the lottery raises twelve-month survey response (39.6% among the selected versus 26.5% among the non-selected), so the survey sample is itself selected on treatment. Under the primary specification’s survey weights, the response shares are nearly equal: 49.5% among the selected versus 50.6% among the non-selected. This is why the Lee (2009) trimming bounds of \$4.4 are computed in the weighted frame, where the implied trim is the control arm’s 2.2%, rather than in the unweighted frame, where the 13.1-percentage-point response gap forces a 33% trim of the lottery arm.

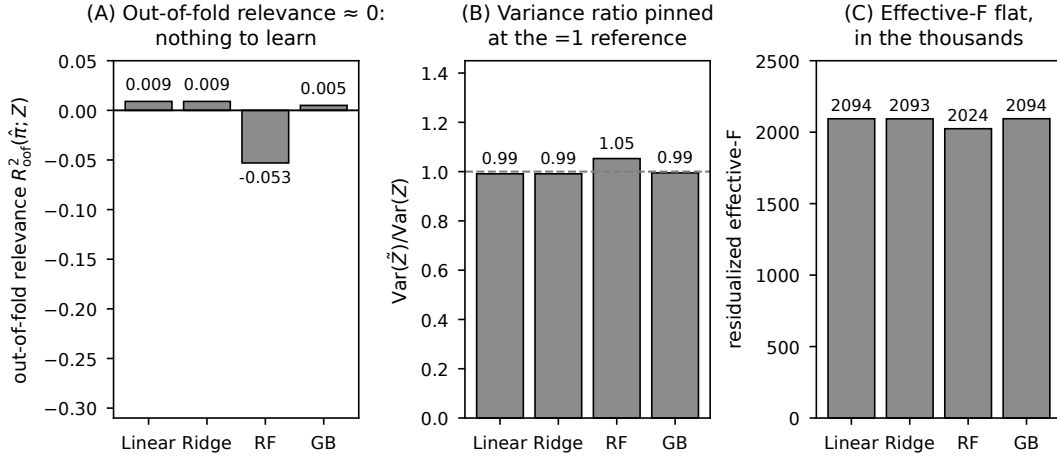


Figure S.2: OHIE randomized negative control, decomposed by nuisance learner. Because the lottery fixes the assignment shares within randomization strata, a flexible $\hat{\pi}$ has only those known strata shares to recover. Panel (A): the out-of-fold relevance $R_{\text{oof}}^2(\hat{\pi}; Z)$ is ≈ 0 for every learner (the random forest reaches only -0.053). Panel (B): the residualized-variance ratio $\text{Var}(\tilde{Z})/\text{Var}(Z)$ stays at the dashed $= 1$ reference (at most 1.053), so residualization neither removes nor adds instrument variance. Panel (C): the residualized effective- F stays flat (≈ 2100) across the library. The population absorption channel is closed.

D.3 Examiner design detail

The leniency instrument is a generated, cross-fitted object: within each outer training set the leave-one-shift-out leniency map is rebuilt from that training set’s clusters alone and evaluated on the held-out fold’s cases, so the instrument enters the protocol as a fourth trained nuisance with the same out-of-fold measurability as $(\hat{\mu}, \hat{r}, \hat{\pi})$.

In the examiner-design estimator literature, leave-one-out and cluster-jackknife estimators correct the many-instrument bias of two-stage least squares (Angrist, Imbens, and Krueger 1999; Frandsen, Leslie, and McIntyre 2025); Angrist and Frandsen (2022) bring machine-learning first stages to this design class and document the biases that machine-generated instruments can carry. We embed the design as the portfolio cell where a flexible residualization makes overfitting of the instrument an economically real concern.

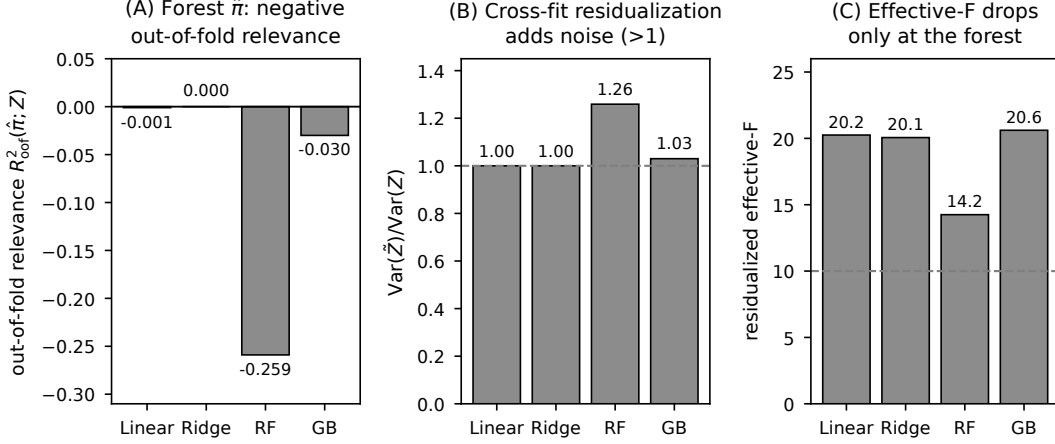


Figure S.3: Examiner cell (fold-nested CJIVE, cluster-aware folds), the dilution mechanism decomposed by nuisance learner. Panel (A): the random-forest $\hat{\pi}$ cannot predict the orthogonalized leniency instrument out of fold ($R^2_{\text{oof}}(\hat{\pi}; Z) = -0.26$, against ≈ 0 for the other learners). Panel (B): cross-fitting therefore turns that fit into added noise, $\text{Var}(\tilde{Z})/\text{Var}(Z) = 1.26$, above the dashed = 1 reference. Panel (C): the residualized effective- F falls from 20.1 (ridge) to 14.2 (random forest). Overfit dilution, not genuine absorption.

Sensitivity diagnostics. Table B.12 reports three checks that the cell’s headline rows do not carry. First, the fold-nested instrument is compared with the full-sample construction that was validated against the published implementation: the two correlate 0.90, with a root-mean-square difference of 0.016, so the traditional check anchors the construction without certifying the fold-nested chain that the reported rows use. Second, the cross-fitting partition is varied. Because GroupKFold does not shuffle in the pinned scikit-learn version, re-seeding it leaves the partition unchanged, so the arms resample the shift-level partition explicitly, holding the learner fixed at the pre-specified ridge. The point estimate ranges from -0.37 to -0.60 across six partitions and the identification-robust set contains zero in two of them, a wider spread than the one across the learner library, so the cell’s exclusion of zero is a property of the reported partition and not of the design alone. Third, the in-sample and out-of-fold fits of each instrument-residualization learner are recorded. The random forest attains $R^2 = 0.49$ in sample against -0.26 out of fold, while ridge and the linear learner move by less than 0.02; the dilution reading of §4.5 is the cross-fitted consequence of that gap.

D.4 CollegeDistance detail

The leakage diagnosis (§4.6). The outcome is a state-level average hourly manufacturing wage (logged) that takes only 41 distinct values and is determined by the state-level controls. The state-level controls that determine the state-level average wage are region, unemployment, and tuition. Under ordinary shuffled cross-validation the same state appears in training and held-out folds, so a flexible $\hat{\pi}$ “predicts” the instrument by memorizing the state: the random forest then shows out-of-fold $R^2_{\text{oof}}(\hat{\pi}; Z) = 0.78$ and the residualized instrument appears strongly absorbed, with the effective- F apparently collapsing. The state proxy used for fold assignment is the 43-group (region, tuition) identifier, within which the state wage is constant in every group on this sample, so whole proxy-states never straddle a fold boundary.

With cluster-aware folds on this proxy the diagnosis reverses: R_{oof}^2 falls to -0.07 with a large in-sample overfit gap (0.94), the residualized-variance ratio moves only slightly above one ($\text{Var}(\tilde{Z})/\text{Var}(Z) = 1.05$), and the effective- F does not collapse. The apparent genuine absorption was geographic fold leakage; with the leakage removed the cell is simply weak, diagnosed by the overfit gap rather than by the variance ratio, whose mild excess over one falls well short of the examiner cell’s dilution signature, and it is neither a clean absorption nor a clean dilution exemplar.

The cautionary features (§4.6). Two further features make this a cautionary illustration rather than a clean mechanism cell. First, the outcome is a state-level index rather than an individual wage, and under cluster-aware folds no outcome learner predicts a held-out group’s wage, so it is not a clean economic estimand. The outcome is constant within the 43 proxy groups, and every out-of-fold outcome R^2 is negative. Second, because each nuisance is fit by the same learner, every coupled specification pairs $\hat{\pi}$ with one of these badly fitting outcome models, biasing the estimates through the coupled nuisances rather than $\hat{\pi}$: the coupled linear and random-forest runs disagree ($\hat{\tau} = -0.04$ versus -0.16 , with erratic AR sets $[-0.18, 10.39]$ versus $[-1.19, 0.29]$) because the nuisances, not the instrument, differ. On these coupled runs: once a nuisance is misspecified, the invariance of Proposition 1 no longer applies, and different specifications legitimately produce different estimates. Indeed, once the outcome and treatment nuisances are fit by their own MSE-best learners rather than coupled to the π -learner, the ridge and random-forest residualizations agree: every π -learner yields $\hat{\tau}$ between -0.006 and 0.001 , with wide, overlapping Anderson–Rubin sets that all contain zero (linear: $[-0.23, 0.78]$); the original disagreement came from the coupled nuisances, not the instrument. The known-truth Monte Carlo can isolate the treatment-nuisance specification effect because there π_0 is a designed nonlinear function, τ_0 is known, and the nuisance specification can be controlled (§4.8).

D.5 Card ladder and literature detail

The standard controls of the full-sample Card specification are experience and its square, race, region, and urban-residence indicators. The control-richness ladder of §4.7 has five rungs: lean controls; adding family background; parental education; the knowledge-of-the-working-world (KWW) score; and finally a degree-two expansion. The upper rungs mix predetermined and 1976-measured controls (experience is age minus schooling minus six), so the rungs are concentration-formation stages, not competing causal specifications. On this ladder, the effective- F path is non-monotonic with an uptick at the +KWW rung, and the random-forest ladder traces the same shape, from about 6.1 to about 2.0, with the same uptick (Table B.4). Across the ladder the flat diagnostics of §4.7 are $\text{Var}(\tilde{Z}) \approx 0.165$ – 0.175 and $R_{\text{oof}}^2(\hat{\pi}; Z) \approx 0.18$ – 0.23 , and while the linear- π point estimate drifts only within about $[0.04, 0.08]$, the random-forest ladder wanders more widely, within $[0.02, 0.13]$. The background controls absorb the part of `nearc4`’s relevance mediated by family background and region.

The rich specification of §4.7, the one under which Card is most often judged weak in the returns-to-schooling literature, uses two proximity instruments (`nearc4` and `nearc2`), the full background battery, and a degree-two expansion, on the restricted 2,182-man sample. The gap between the rich-specification null and the canonical 0.13 tracks the ladder: restricting

to the background-complete subsample roughly halves the effective- F (from about 16 to 7.5), and the richer controls then cut it to about a third (single-instrument effective- F about 2.5, already below the two-instrument 4.69 and well into the uninformative-null regime). In the rich specification the DML estimate, a π -library and residualization choice apart from the published implementations, is a similar uninformative null.

Specification sensitivity of the experience block. Because experience is age minus schooling minus six, the classic control set is not predetermined with respect to the treatment. Table B.11 holds the sample, the instrument, and every other control fixed and swaps that block for age, and for age and its square. The point estimate falls by about a third and the identification-robust set crosses zero, while the first-stage F is nearly unchanged. The canonical Card magnitude is therefore a property of the canonical control set rather than of the instrument, which is why §4.7 reads the ladder as concentration formation and not as a menu of competing causal specifications.

D.6 Angrist–Krueger cell and state-bundle detail

Design (§4.9). On the full $N = 329,509$ men in the Census extract, Y is log weekly wage, D is years of schooling, and standard errors are clustered on state of birth. The controls W enter as an *additive* specification of year-of-birth and state-of-birth dummies, not the saturated state-by-year cell means; the nuisances are fit within that maintained additive form, and flexible learners add nothing to it, so the cell isolates the over-identified *inference*, not the absorption mechanism. Cross-fitting is cluster-aware here as everywhere: outer folds partition states of birth, with the state effects swept by a within transformation that never crosses a fold boundary (training-side means for fitting; each held-out fold demeaned by its own state means); Lemma 9 (Appendix A.4) gives the formal bridge. We run two instrument sets: the three quarter-of-birth dummies ($d_z = 3$) and their interactions with year of birth ($d_z = 30$), the Bound, Jaeger, and Baker (1995) many-weak specification.

The $d_z = 3$ results (§4.9). With the three quarter-of-birth dummies the cell behaves as a clean over-identified design. Two-stage least squares and the cross-fitted DML estimate agree at $\hat{\tau} = 0.108$ (Wald SE 0.017 for both), the canonical Angrist–Krueger return to schooling and well above the ordinary-least-squares 0.067; the effective- F is 11.4; and the three identification-robust sets (distinct objects at $d_z > 1$) are bounded and mutually close (Table B.6): AR [0.050, 0.148], score [0.057, 0.139], conditional QLR [0.058, 0.138]. The J_{CUE} statistic does not reject ($J = 2.83$, 2 d.f., $p = 0.24$): at this moderate strength the over-identifying restrictions are not refuted.

The $d_z = 30$ results (§4.9). Saturating the instrument to the thirty quarter-of-birth $\times$ year-of-birth interactions turns the design into the many-weak case. Two-stage least squares biases toward ordinary least squares ($\hat{\tau} = 0.087$, between the 0.108 of the clean design and the 0.067 of OLS), with a *tighter* Wald standard error (0.011) that reads as spurious precision, exactly the Bound–Jaeger–Baker pathology. The identification-robust sets, by contrast, reflect the weakness: the AR set is the whole line and the score and conditional-QLR sets are noncontiguous across the search window, so none of the three localizes τ . The J_{CUE} statistic still does not reject ($J = 25.2$, 29 d.f., $p = 0.67$), but here nonrejection is weak evidence: under a design this weak the over-identification statistic has little power, so the nonrejection does not establish that the instruments are valid; it says only that the over-identifying

restrictions cannot be refuted at this strength. The per-instrument effective- F (about twelve) sits above the rule of ten even as the thirty-dimensional design cannot inform τ . The cell thus supplies the real-data counterpart to the over-identified failure simulated in §4.2: the effective- F screens strength while inference rests on the identification-robust set, not on an F -threshold.

Theorem coverage on the transformed bundles (§4.9). The within transformation preserves independence across states and the fold-measurability of the training fits; Lemma 9 (Appendix A.4) records this and states the price. On the transformed bundles, under a state-bundle exogeneity condition (stated below), the clustered AR statistic at a tested value is covered by Theorem 2 via Lemma 9, and the clustered score statistic by Theorem 3 (Appendix A.5) under the conditions K1 and K3 read on the transformed score. The two theorems are $G \rightarrow \infty$ statements; with $G = 51$ state-of-birth clusters this cell sits at the boundary of that regime, aggravated at $d_z = 30$ by a 30×30 cluster meat built from 51 cluster scores, so we do not read the theorems as finite- G evidence here (§3.2, Appendix A.4–A.5), and the clustered conditional-QLR variant carries no distributional claim and is reported as a construction. The finite-sample complement is the matched coverage simulation of §4.2 (i.i.d. sampling; Appendix D.1): at both $d_z = 3$ and $d_z = 30$ the cross-fitted AR, score, and conditional-QLR sets hold near-nominal coverage (0.938–0.966) and the J_{CUE} statistic’s size is 0.034 and 0.042 against its strong-identification reference.

The within transformation of §4.9 is a fixed linear map inside each state bundle. The state-bundle exogeneity condition is the within-estimator’s strict exogeneity in this setting: each individual’s structural error must be mean-independent of the full state bundle of instruments and year-of-birth controls, while the instrument- and treatment-equation errors are conditionally centered given the control bundle and its size. The moment and eigenvalue conditions are restated on the transformed score. Under random within-state sampling and the quarter-of-birth exclusion this is the natural reading of the design. For the clustered score statistic, the stacked-moment condition K1 and the same-aggregation coherence condition K3 are read on the transformed score; K1’s non-degeneracy bound rules out the collinear score-and-Jacobian corner (§3.2).

A plain individual-fold comparison run is archived alongside the state-partitioned cross-fitting run and shifts the point estimates only in the third decimal place.

The Mikusheva–Sun (2022) pre-test finds the quarter-of-birth design strongly identified at $K = 180$ interacted instruments ($\tilde{F} = 13.42$ against a first-stage F of 2.43), whereas our $d_z = 30$, state-clustered specification makes clustering rather than instrument count the binding constraint (§4.9).

A structure-matched clustered Monte-Carlo arm. Appendix C.5 and Table B.10 report an arm built to this cell’s dependence structure rather than to its scale: fifty-one clusters carrying the real state-of-birth size profile, sparse quarter-of-birth cells counted within state, the within transformation, and state-level folds. At $d_z = 3$ the clustered AR, score, and conditional-QLR sets hold their nominal levels at a weak, a calibrated, and a strong point, and the J_{CUE} size is near nominal, so the inference this cell reports at three instruments is exercised at its own cluster count rather than only at its instrument count. At $d_z = 30$ the simulated AR set is the whole line in essentially every replication, for a reason that is structural rather than numerical. The analytic tail limit of the clustered AR statistic equals $\|P\mathbf{1}_G\|^2$, where P projects onto the column space of the cluster-sum matrix, so it is bounded

above by G . A bounded set requires that limit to clear the χ_r^2 critical value, and at $G = 51$ with thirty instruments the critical value of 43.77 leaves almost no room below fifty-one. The cell’s own tail limit is 33.14, below that critical value, which is why its thirty-instrument set is the whole line: at this cluster count and instrument count a bounded clustered AR set is not attainable, whatever the strength of the instruments. The score statistic, read against χ_1^2 , keeps its nominal level in the same simulated replications, which is the reading behind reporting all three sets rather than the AR set alone.

Appendix E. Selector and post-selection detail

E.1 The identification-aware selector: motivation, construction, and justification

Although under correctly specified nuisances the prediction-optimal residualization is also concentration-optimal in population (Proposition 3), the two need not coincide at finite n : the relevance g_{ZD} is invariant only in the limit, so among learners that predict Z comparably well out of fold the *realized* residualized strength still varies. This is the finite-sample gap the identification-aware π -selector of §4.3 addresses.

In full, the selection step runs as follows. Within each outer training set, candidate learners are fit on an estimation subsample; the learners whose out-of-fold predictive risk on a disjoint selection subsample lies within one estimated standard error of the best form the tie set; and within the tie set the selector maximizes a residualized-strength proxy computed on the selection subsample alone. The predictive screen is additive, so the tolerance shrinks with the selection-sample size rather than sitting at a fixed ratio. The residualized-strength proxy is the ridge-stabilized proxy of Çetin (2026b), run at zero ridge here since $d_z = 1$; its treatment residualization uses a linear fit on the estimation subsample, so the correct-nuisance headline of the §4.3 benchmark refers to the estimator’s residuals, not to the selector’s internal proxy.

The population justification is supplied by Çetin (2026b, Propositions 3–5), stated there for selection over full nuisance pipelines and applied here to the π slot: prediction-only tuning can select π -learners whose residualized strength is asymptotically negligible, the screened strength proxy attains a near-oracle bound, and the tie-break has no first-order cost under regular identification. Theorem 3 of that paper gives the Anderson–Rubin set uniform asymptotic validity under i.i.d. sampling when the nuisances are selected this way and the library satisfies its rate conditions (imported as Theorem 1, Appendix A.4; the clustered counterpart, with folds and selection at cluster level, is Theorem 2).

In the paired benchmark of §4.3, the tie-break does change selections mechanically: the random-forest share of fold-level winners rises from about a quarter to about 45 percent in the two absorbing regimes.

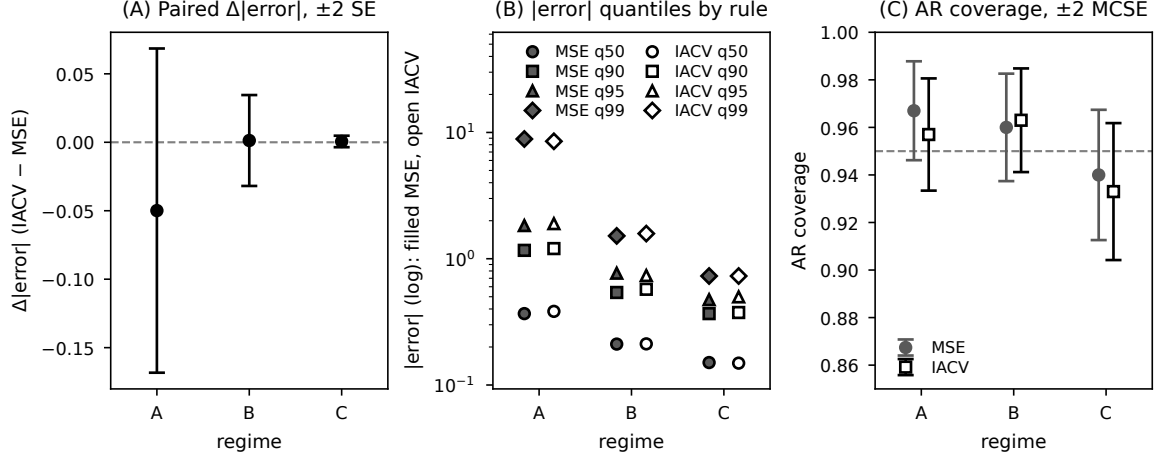


Figure S.4: Identification-aware (IACV) versus MSE π -selection across three regimes (300 paired replications, correctly specified nuisances, common random numbers). Panel (A): paired absolute-error differences (IACV minus MSE) with ± 2 standard-error bars; every interval contains zero. Panel (B): upper tail of the absolute error by rule (median, 0.90, 0.95, and 0.99 quantiles, log scale); through the 0.95 quantile the two rules are nearly indistinguishable in every regime, and at the 0.99 quantile the weak regime favors the identification-aware rule slightly, so the weak-regime RMSE gap is carried by the extreme tail. Panel (C): Anderson–Rubin coverage by rule with ± 2 Monte-Carlo standard errors against nominal 0.95.

The mean interval length is a tail statistic. Table B.3 reports a mean Wald interval length in the weak regime that is about three times larger under the identification-aware rule, which would suggest systematically wider intervals. The per-replication length quantiles now archived with the regime artifacts show that it is not: the two rules have nearly identical median lengths, agree through the ninetieth percentile, and coincide once the ten longest replications are dropped, while a single replication contributes more than three quarters of the identification-aware mean (Table B.13). The gap is a handful of draws with a near-zero realized first stage, the regime in which the Wald interval is not the object we report anyway, rather than a systematic widening.

E.2 The two post-selection control experiments

The matched split-sample experiment of §4.3 is constructed as follows: on a held-out inference half we compute the AR coverage of τ_0 both when the π -learner is selected on that same half (in-sample) and when it is selected on a disjoint half (which yields a post-selection-valid interval). One design caveat applies: the split-sample arm collapses the fold-specific winners into a single majority-vote learner before computing coverage, while the in-sample arm retains the per-fold winners. The contrast therefore varies selection reuse and winner aggregation together, and this first experiment does not separate the two.

A pure control isolates the first factor: a foldwise variant retains the per-fold winner structure in both arms, so the only remaining difference is whether the selection and inference observations overlap, and it shares its draws with the original run through the same common-random-numbers scheme. Its selection distortions are the -0.008 to $+0.017$ range reported in §4.3.

The mild departure from nominal that both modes inherit in places (coverage between 0.89 and 0.99 across regimes, in the original contrast and the pure control alike) is the same finite-sample generated-instrument effect recorded in §4.2, not a selection effect.

The right yardstick for the distortion. Both arms are computed on the same inference observations of the same draws, so the in-sample-minus-split difference is a paired contrast and its paired standard error, not the one-arm coverage Monte-Carlo standard error, is the scale against which it should be read. The per-replication coverage indicators archived with the post-selection artifacts give that paired standard error directly, together with an exact McNemar test on the replications where the two arms disagree (Table B.13). The paired standard errors are smaller than the marginal ones, so this is the stricter reading, and the distortions remain inside it.

Appendix F. Protocol comparison: literal inner cross-validation and LF-NCF

Section 2.1 of the article takes as given the leakage-safe nested cross-fitting of Çetin (2026a,b), run in the three-way LF-NCF form of Çetin (2026b, Algorithm 1), and notes that Çetin (2026a, Theorem 2) is stated for the inner-cross-validation form of the protocol in that paper. That literal inner-cross-validation variant is also run in parallel as a robustness check: on a paired Monte-Carlo arm and on the Card and CollegeDistance cells it reproduces the primary protocol’s coverage within Monte-Carlo noise and leaves every qualitative reading unchanged (Table B.8). Table B.8 (Appendix B) reports the full comparison, with the pairing by common random numbers described in its caption and in Appendix C.

The fold mechanics of the three-way form, as implemented: within each outer fold, the outer-training sample is split into an estimation and a selection subsample; candidate learners for (μ, r, π) (hyperparameters fixed within each library entry or tuned internally on the estimation subsample) are fit on the estimation subsample, selected on the selection subsample, and the selected fits evaluate the held-out fold, with no refit on the union.

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